# Particles and Nuclei 2 Final Assignment

Physics 7502

Instructor: Prof. Justin Frantz

Spring 2022

The final assignment is a reading assignment on several closely related topics plus some basic questions just to verify you actually did read it critically..

The reading assignment is completely contained in this document. When links are given, they are just the references to some material used or for your reference in case you want more information.

# Hadrons/Hadronization/Jets

- What follows is several reading selections taken from several sources
  - The biggest fraction is taken from CY Wong, Introduction to High Energy Heavy Ion Collisions however, none of this is specific to RHIC physics
- Rather the focus is on non-perturbative and some perturbative aspects of models of hadrons and especially their formation.
- Since jets are one of the only ways we can see creation of hadrons "from scratch" including at the first part of the process, the quarks/color charges themselves, it is one of the most important applications of such models.
- And since they are themselves another important QCD object almost equivalent to the bound states themselves we will learn a little about their properties
- Just FyI QGP formation has developed into its own "higher level" model of hadron formation and along with similar statistical/thermal type approaches may apply even to simpler e.g. p+p collisions--and I don't just mean the very rare p+p events where us RHI physicists think may actually create the QGP.
  - However these descriptions never include the microscopic-level processes they are only statements about the thermal/statistical properties of the many body interactions of many particle-systems.

# Hadron Models

- Besides the issues I was critical of it before for [or perhaps partially this was one of the issues], the Gell-man Quark Model in Ch 2/4/5 of Griff is not even close to a holistic model of hadronic structure--~never meant to be. As we already discussed, it ONLY captures some leading contributions (the "valence quark") to the expected quantum numbers of the known hadrons—and as we see with the spin problem it doesn't really even do that perfectly.
- In these readings are presented a few other approximation models for hadron structure/formation. Lattice QCD is currently the only tool we can be fairly certain provides the answer of the "FULL" model. But we still can't use it for many important calculations (grid size too small/computing resources, limited in the technique to "steady state" type results…)
- Thus a range of models for different situations have been developed.
- Some of these "models", like the Reggeon model are more like "classification models" like the Eightfold way.
- Some are just concepts/guiding principles (e.g. "parton-hadron duality)
  - This one is at least mentioned in passing in these readings, but it is essentially a model of formation that in some cases you can just convert some number of quark and gluons including their four vectors into hadrons (ie keeping same 3- or 4-vectors)

# Start with Jets – Event Generators

- We start with the jets:
- I give some slides about the experimental definition of jets which also already introduces the concepts of hadronization.
- Remember jets start as a parton/QCD (quark or gluon) but they can only be measured after the step of hadronization
- https://www.hep.ucl.ac.uk/postgrad/teaching/lhc/Jets_2017.pdf
- Something quick as background, since the slides can't help mentioning them, the study of jets since it is defined by "jet finding " algorithms that are implemented ~ONLY as computer programs, relies heavily on **"Event Generator"** simulations which simulate the conversion of partons to hadronic jets. These of course are "monte carlo" (stochastic/random number sampling is involved) simulations.
- The most common of these is called PYTHIA– the PYTHIA computer program/package. A little more info about this.

# High Energy Event Generators

- **PYTHIA and HERWIG – LO** Leading order generators -- calculate the tree-level cross sections of the partons, use "global fit" PDF's of incoming hadrons, and use some tuned "empirical" models to convert partons to hadrons for final state.
  - PYTHIA https://pythia.org/ became the most widely used by about 2005, 5 years before LHC started. Most successful. Dial up any combination of particles to collide (e, p, n,.. not NUCLEI though), specify any special reaction you want to study (though not needed) and it generates very realistic event samples of particles (lists of 4-vectors, particle ID codes) for as many events as you want.
    - Uses "String model" of hadrons/hadronization to provide framework but constants needed in this framework are "tuned" empirically on other data.
  - HERWIG. No longer used but was an alternative using "Clustering" hadronization model.

- NLO Generators. SHERPA (most like Pythia) "MC@NLO", DiPhoX for photons

- HIJING Heavy Ion collisions (superimposing PYTHIA N+N events ie uses PYTHIA underneath) but adds a little bit of known nuclear effects
  - does NOT include QGP effects usually (newer versions can).
  - https://arxiv.org/abs/1901.04220 https://arxiv.org/abs/nucl-th/9502021

# Jet physics at the LHC
# an introduction

**UoL intercollegiate postgraudate course**

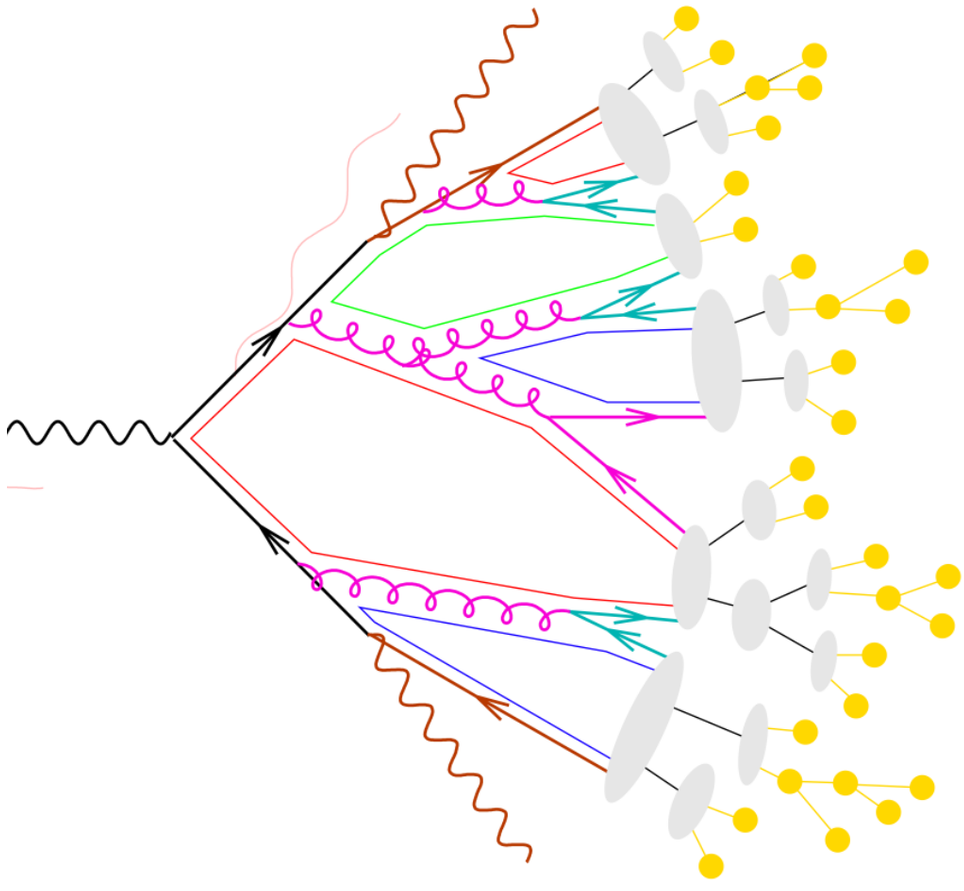
Amal Vaidya

# Why do we need jets?

## Consequence of QCD

- Quarks and gluons are produced at high energies (perturbative QCD)
- Will radiate more partons as they propagate
- At lower energies they will form colourless hadrons

**parton (or something decaying to partons)**



**Mainly pions, with additional hadrons, photons and neutrinos**

Reconstructing a jet gives us a proxy for the kinematics of the parent particle (and more!)

# Why do we need jets?

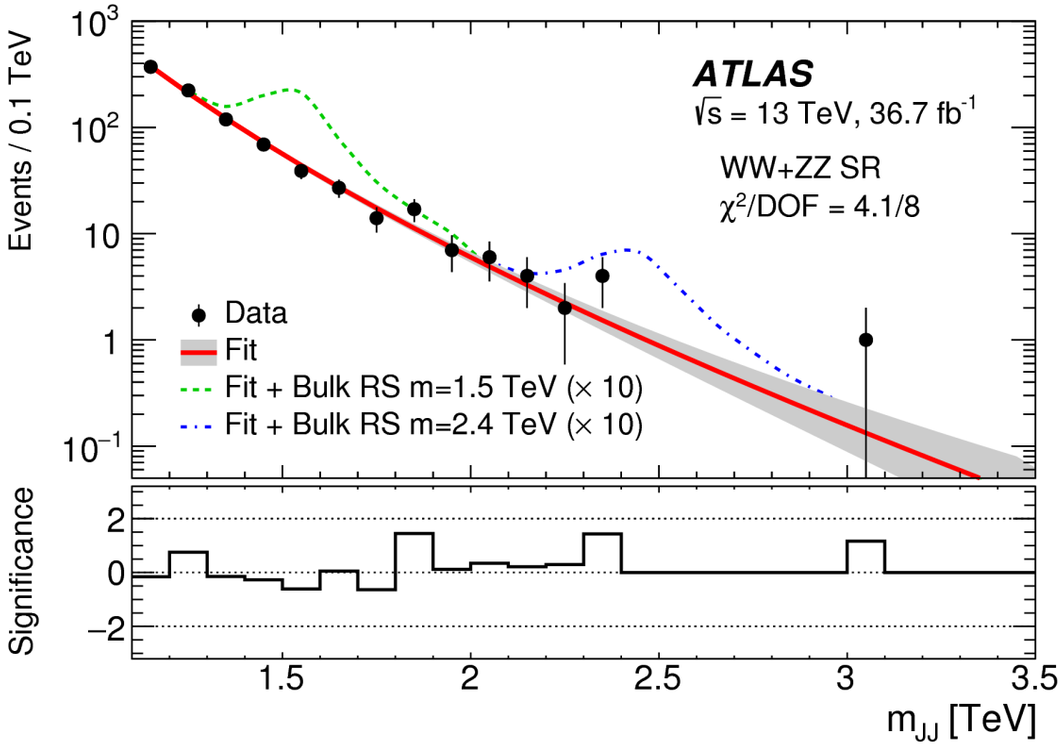
**Even though they are less well defined than leptons or muons they are essential for understanding LHC physics**

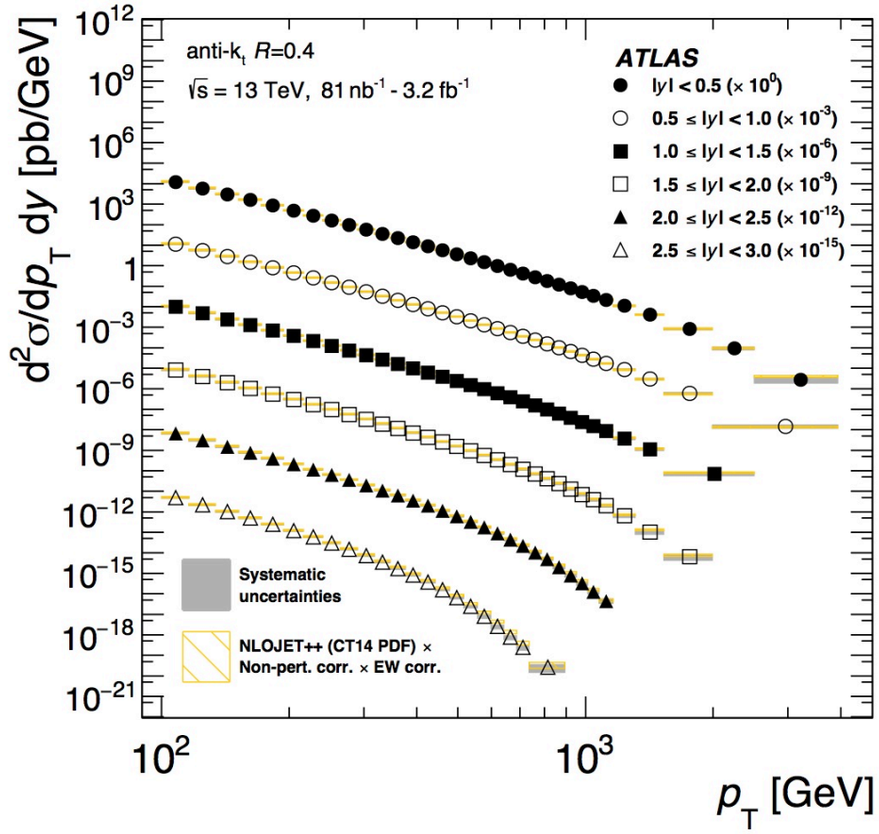
**Standard Model physics:**

- Many standard model processes produce jets, are sensitive to **α** strong
- Multijet cross section measurements test QCD
- Hadronic decays of heavy particles

**New physics searches:**

- Many searches looking for final states with jets, or in regions of phase space with high jet multiplicities



# What is a jet?

## Different kinds of jets

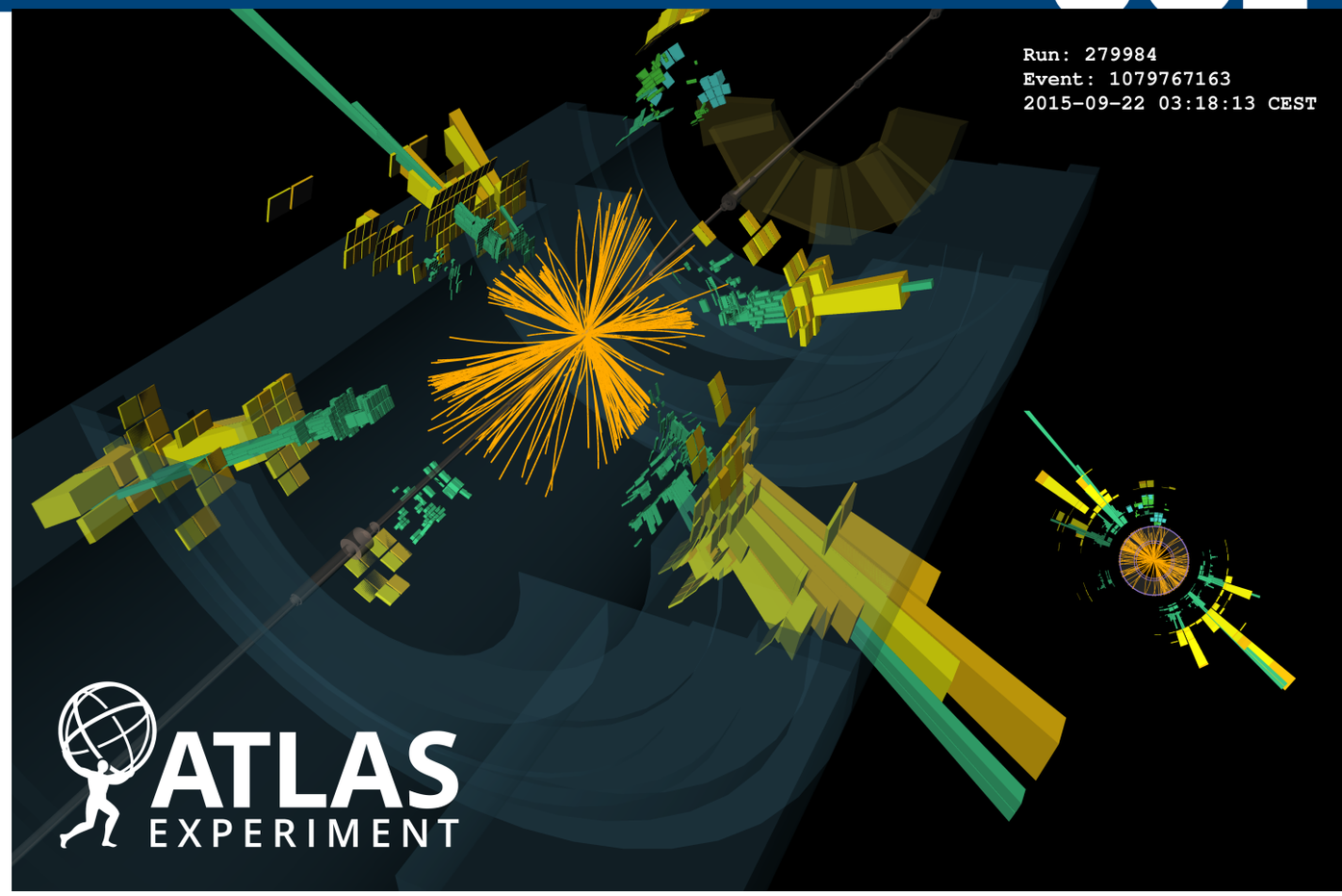
**Truth/Particle level:**

The constituents of the jet are final state (visible) particles

**detector level:**

Can be constructed from a number of detector objects

- calorimeter clusters
- charged tracks
- some combination thereof (particle flow)




**Jet have typical kinematics and a number of other properties:**

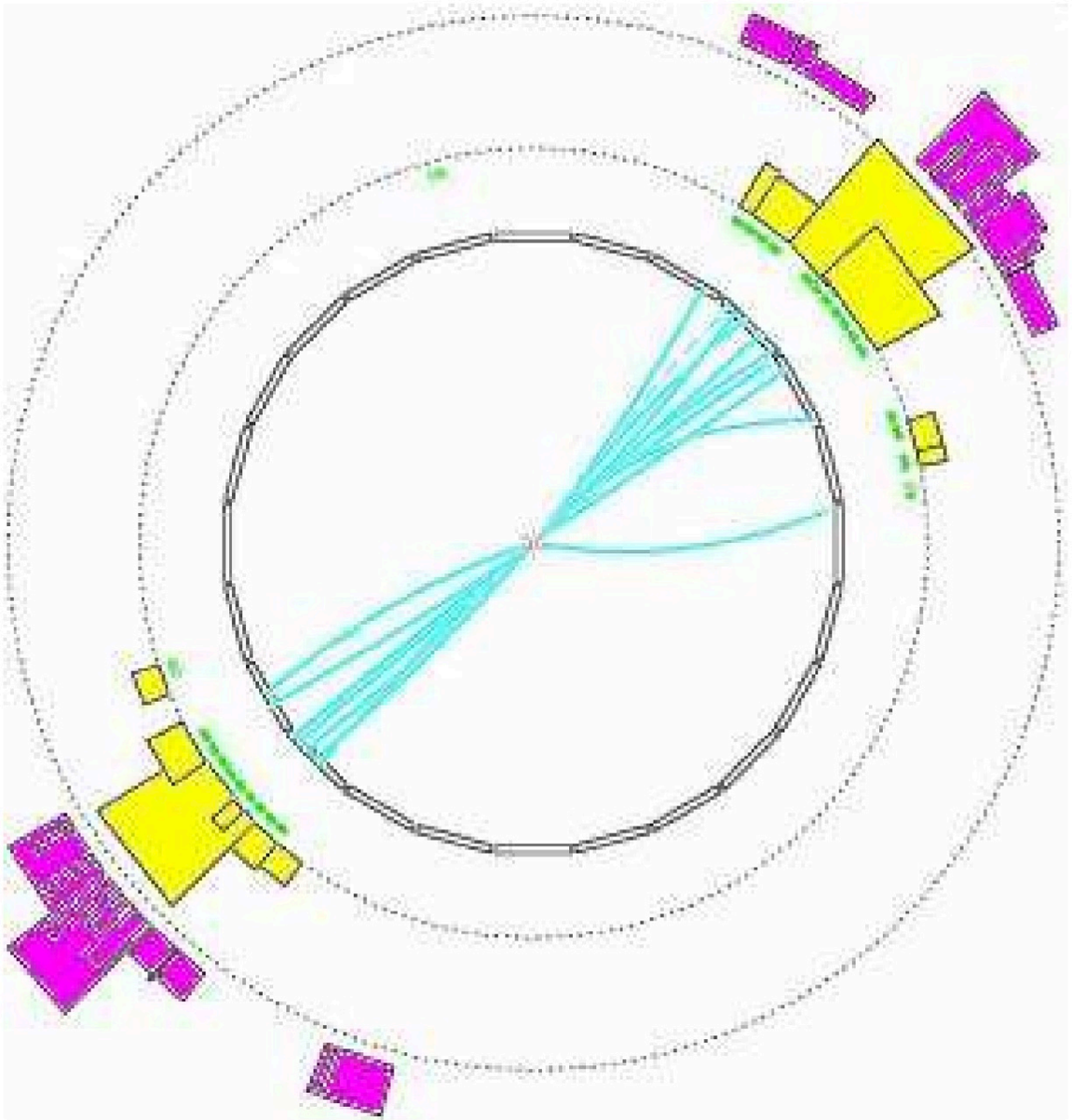
- Cone size
- Jet finding algorithm
- Substructure
- Charge fraction
- Active area….

**But first, how do we define what is and isn't a jet?**

# Jet finding

**You have your constituents, now find jets! How many are there….**

**in this event?**



# Jet finding

**You have your constituents, now find jets! How many are there….**

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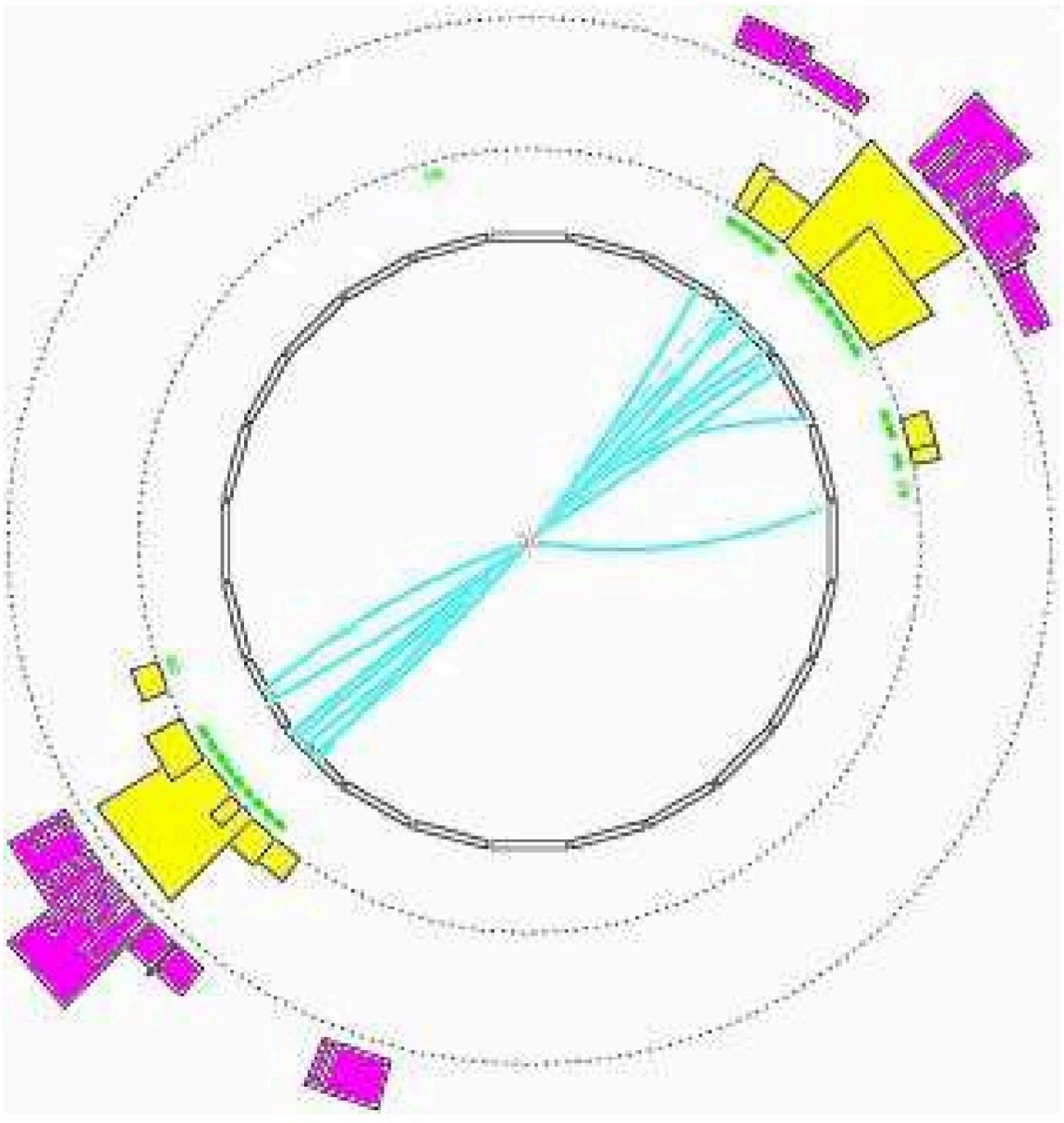
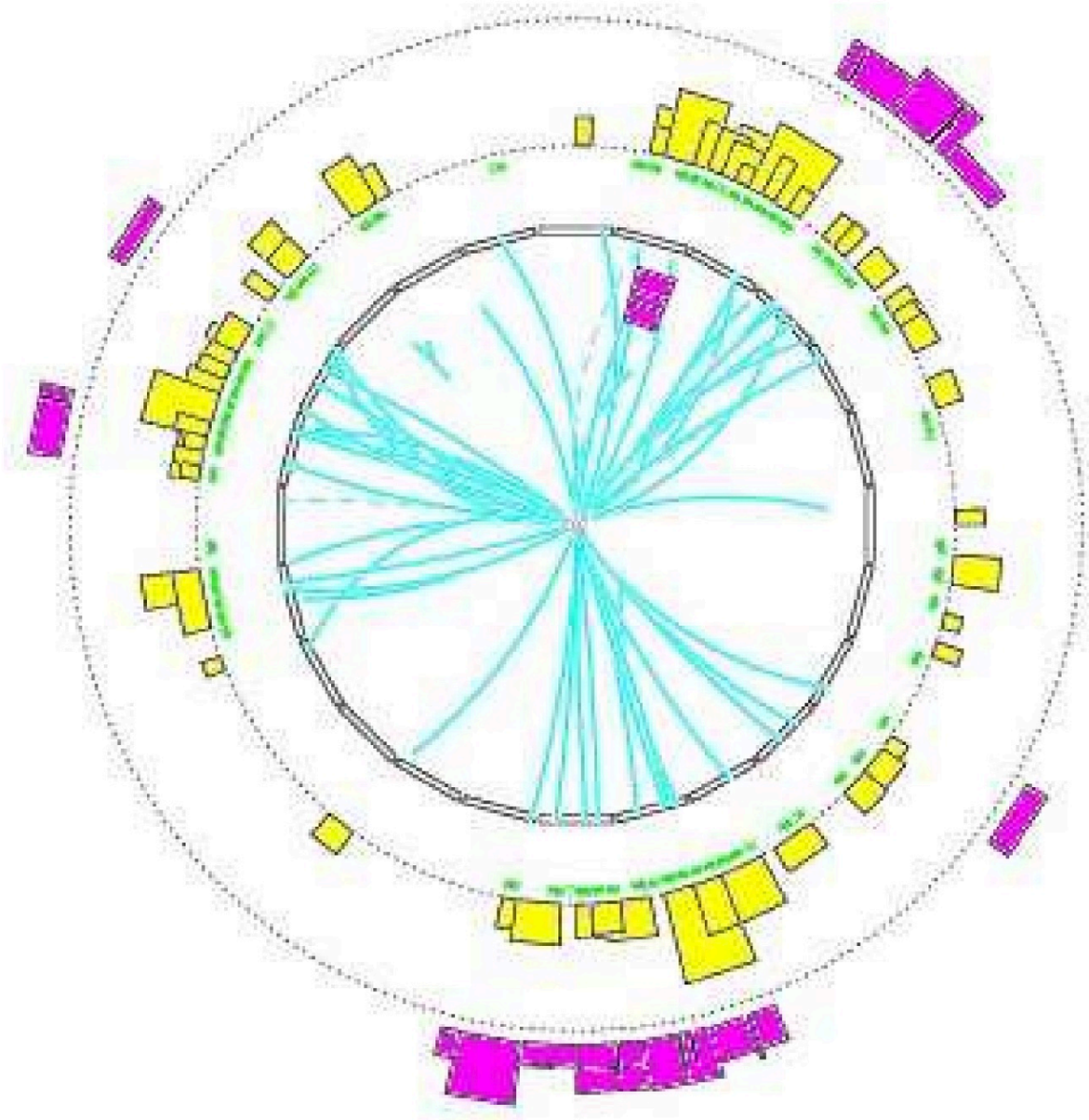
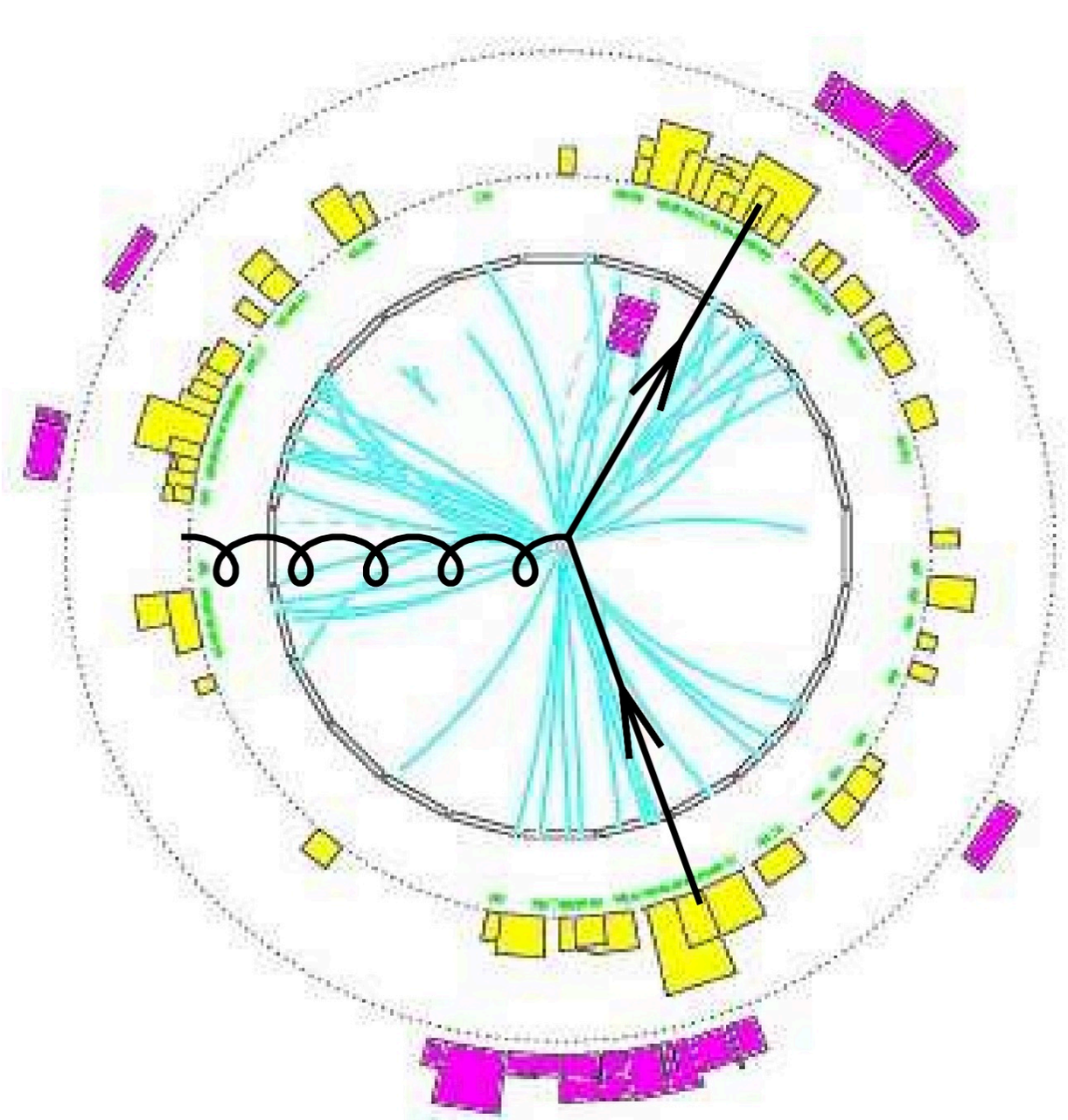


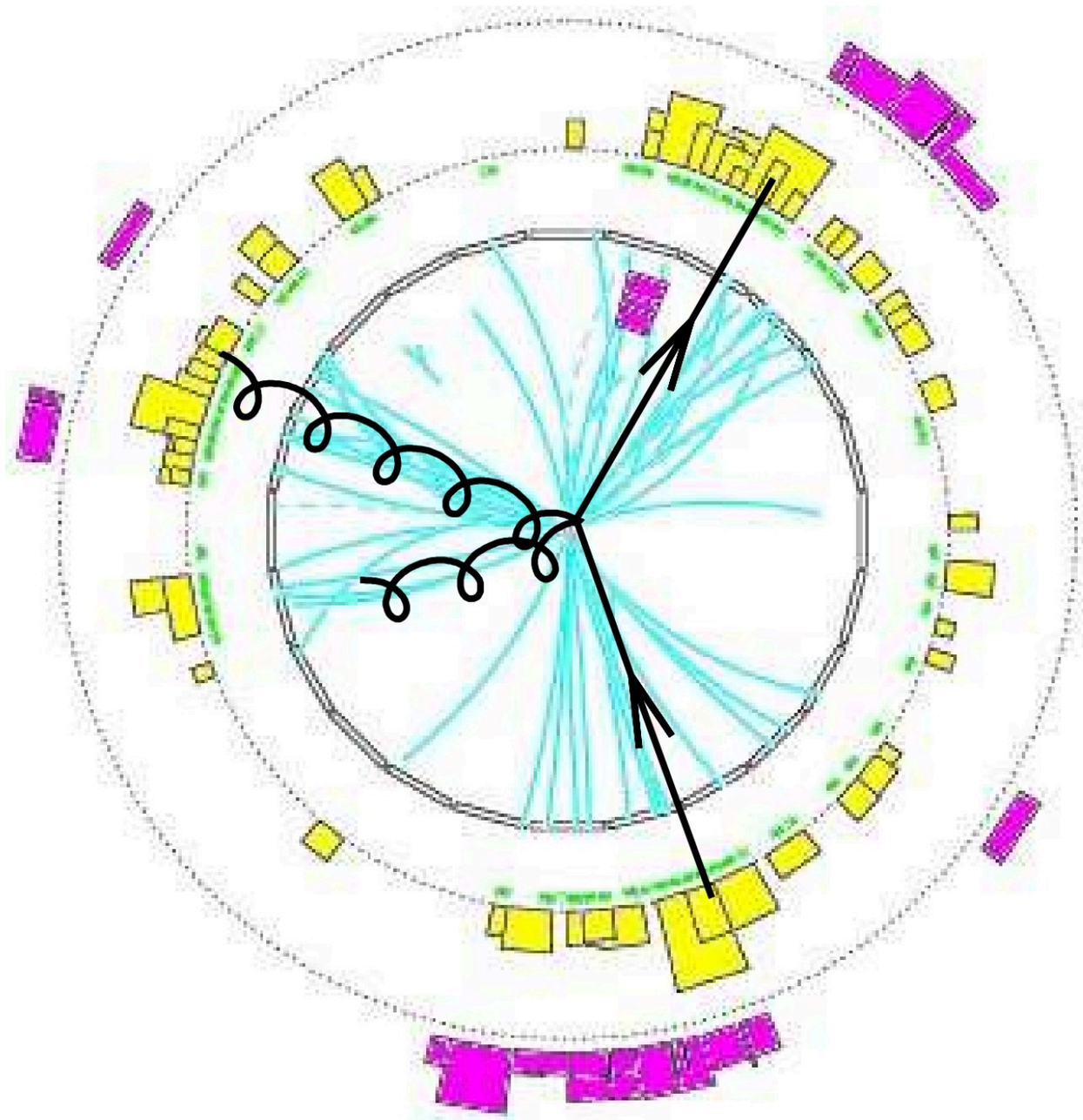
**and this one?**



# Jet finding

**We need a robust, unambiguous definition of a jet**



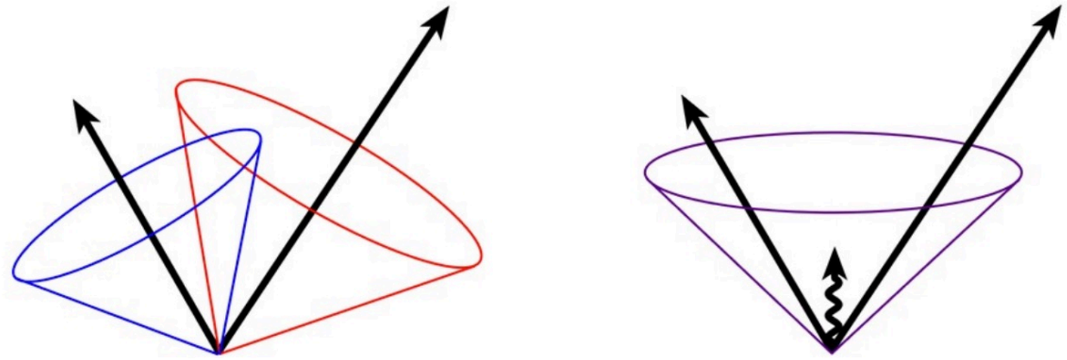


# Jet Algorithms

**What properties should a good jet finding algorithm have?**

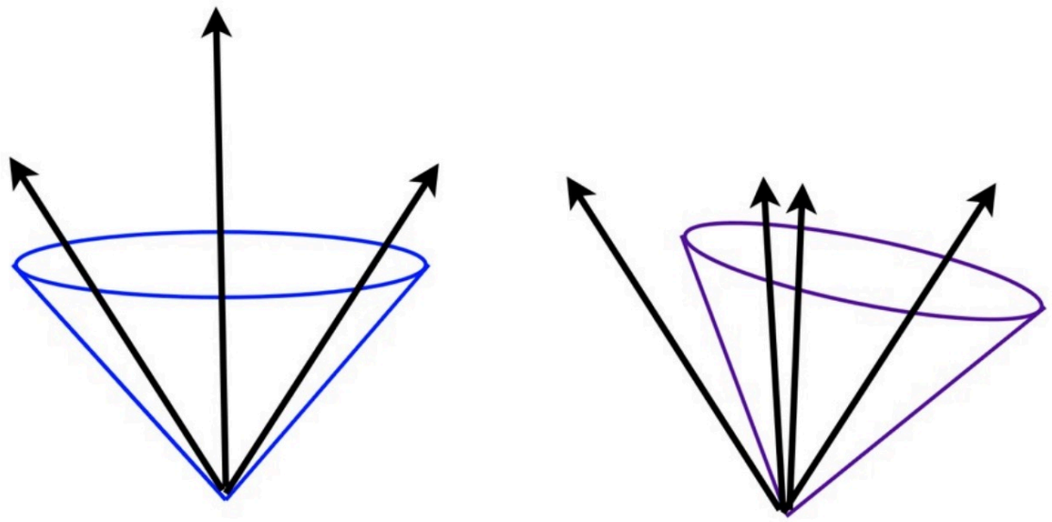
**parton and jet correspondence**

- find all physically interesting jets from high energy partons



**Infrared safety**

- soft radiation should not effect jet configuration
- Only observables that are IR safe can be calculated in pQCD



**collinear safety**

- Collinear splittings should not bias jet finding

**Other things to consider**

- should be independent of detector technology (works at particle level)
- computationally fast
- Easy to calibrate and stable in noisy, pileup filled detector environments

# Jet Algorithms

**Cone algorithms (no one uses these anymore)**

**Iterative cone**

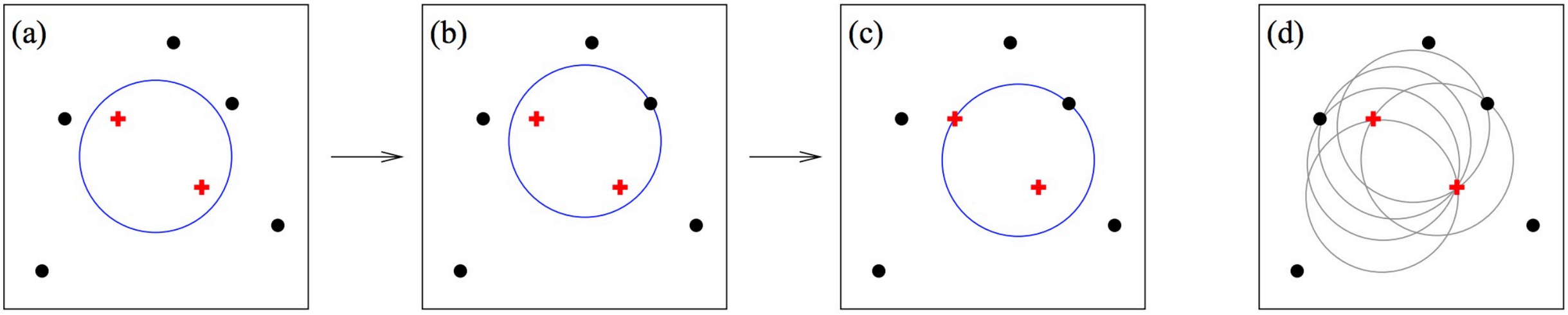
- select the most energetic particle as a seed
- all constituents within cone of radius R are considered part of the jet
- jet axis re-calculated, if it's stable, w.r.t seed axis. STABLE CONE

**Not IR safe**

**SIScone (seedless infrared-safe cone) algorithm**

**scales as $N^2 \ln(N)$ :(**

- find all stable cones as above as "protojets"
- remove constituents from those cones and repeat until new no cones are found
- merge overlapping protojets into final jets

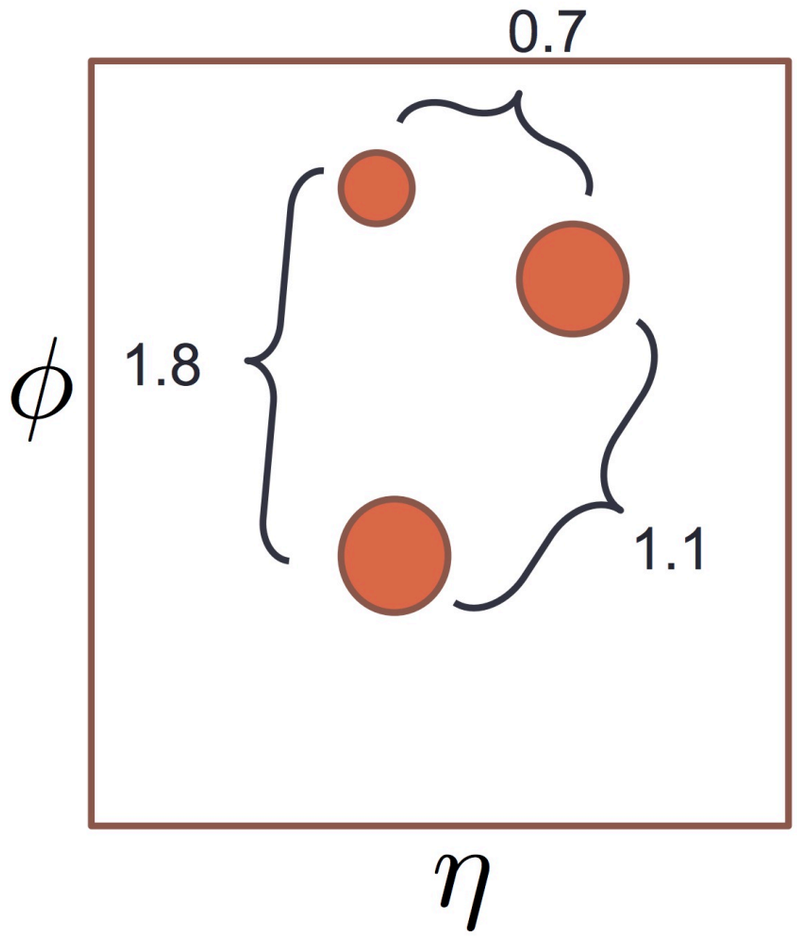


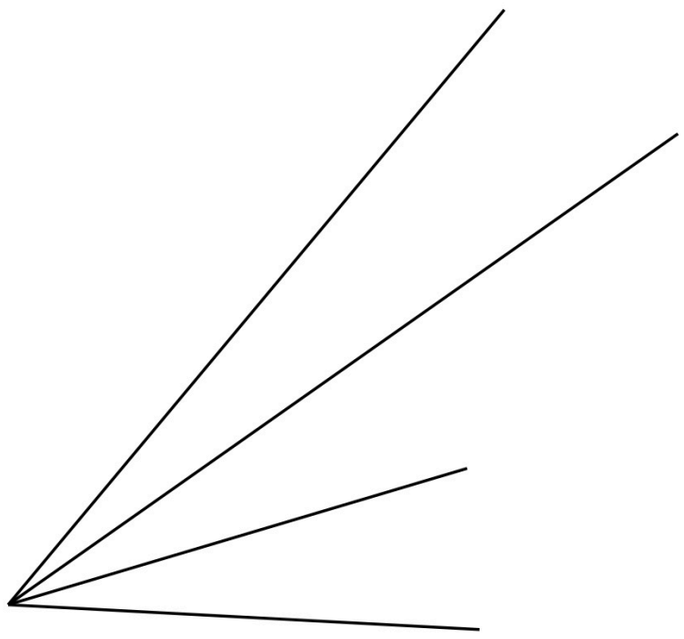
# Jet Algorithms

**Sequential Recombination (clustering) algorithms**

Can intuitively think of clustering algorithms as working their way back through the parton splittings

Define a distance measure based on the constituent angular separation and their energy/pT and combined particles which are closest



ON ALL THESE SLIDES THE VARIABLE R or Delta R (Rij means R difference between i and j) IS USED TO REPRESENT THE SIZES BETWEEN the PARTICLES/JETS in the ABOVE FIGURE

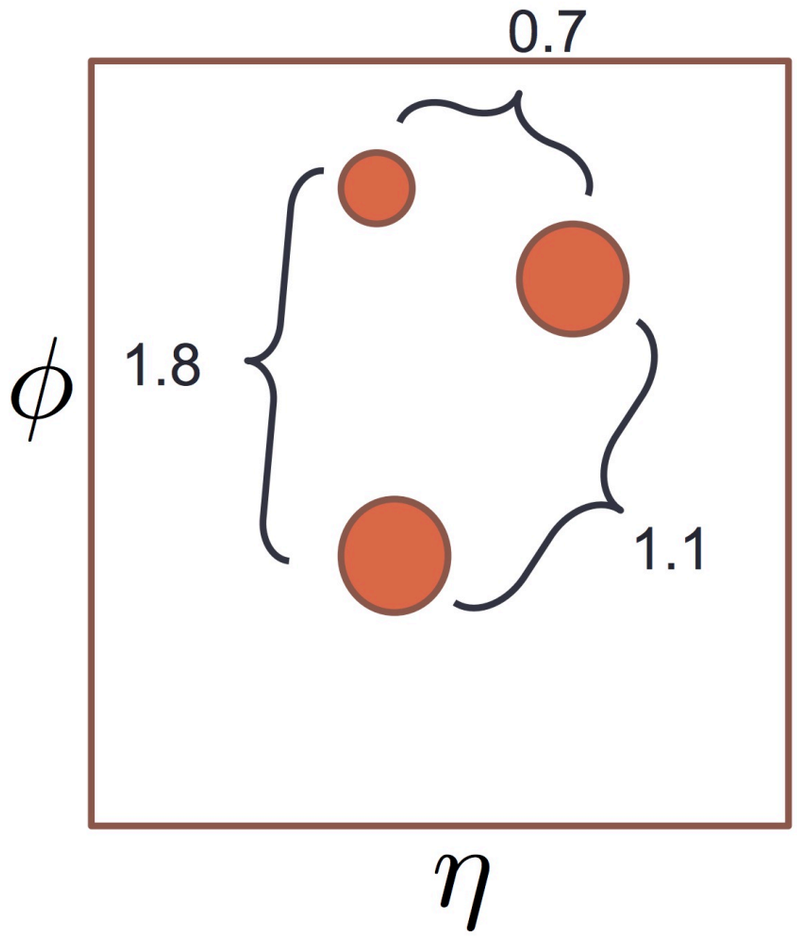
THIS IS THE SAME R DEFINED BACK WHEN WE INTRODUCED RAPIDITY in terms of PSEUDORAPDITY (eta, the x-axis in the above plot)
dR = sqrt( dEta^2 + dPhi^2)

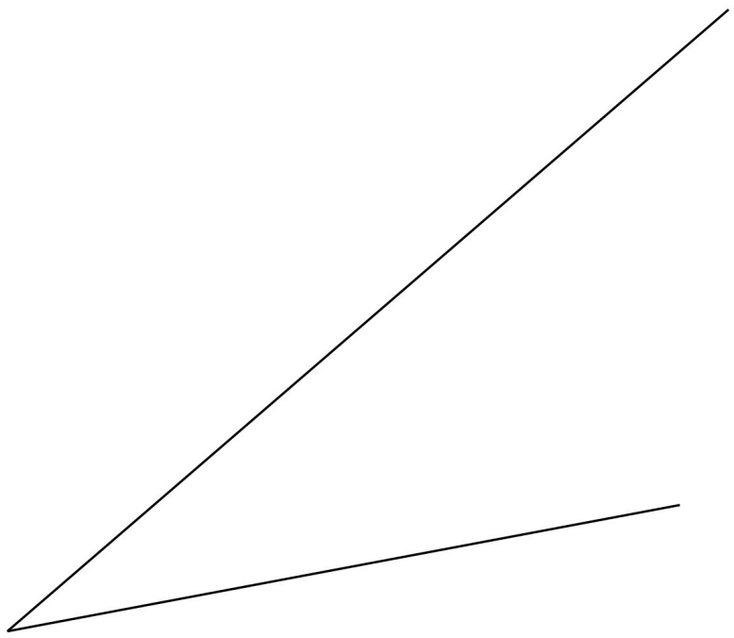
# Jet Algorithms

**Sequential Recombination (clustering) algorithms**

Can intuitively think of clustering algorithms as working their way back through the parton splittings

Define a distance measure based on the constituent angular separation and their energy/pT and combined particles which are closest



The JADE algorithm was the first clustering algorithm.

IR and collinear safe

Could sometimes cluster soft, back to back particles together…

# Jet Algorithms

## Modern ("second generation") Jet clustering algorithms

3 jet algorithms are currently used for various purposes at both ATLAS and CMS (AFAIK!)

**All can be defined using a set of generalised distance parameters**

**constituent pT** — **angular separation**

$$d_{ij} = min(k_{ti}^{2p}, k_{tj}^{2p}) \frac{\Delta_{ij}^2}{R^2}$$

$$d_{iB} = k_{ti}^{2p}$$

**Radius parameter**

**"Beam distance"**

**indices *i* and *j* run over all candidate jet constituents**

*p* = 1 : $k_t$ algorithm

*p* = 0 : Cambridge/Aachen algorithm

*p* = -1: anti-$k_t$ algorithm

**Cluster as follows**

- work out all of the $d_{ij}$ and $d_{iB}$
- Find the minimum of the $d_{ij}$ and $d_{iB}$
- If it is a $d_{ij}$ the combine *i* and *j*, if not, *i* is considered a final state jet and removed
- repeat until now particles are left

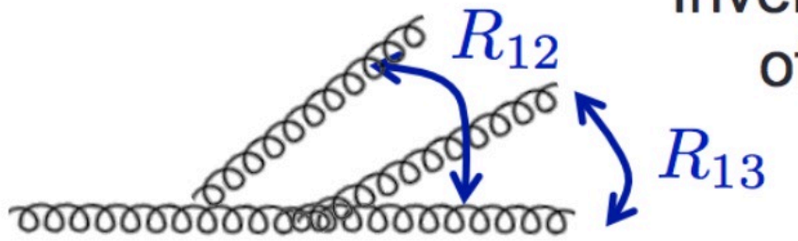
# Jet Algorithms

**(Shameless slide theft)**

## Cambride/Aachen algorithm

$$d_{ij} = \left(\frac{R_{ij}}{R_0}\right)^2$$

- clusters closest radiation first

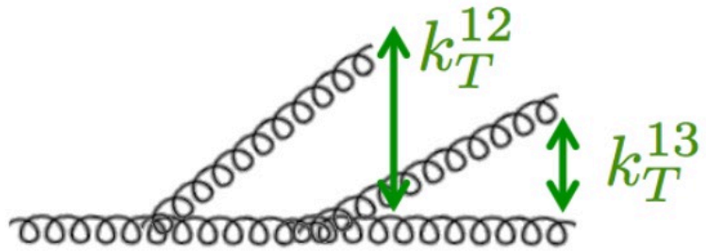



Inversion of **Herwig shower**

## $k_T$ algorithm

$$d_{ij} = \min(p_{Ti}^2, p_{Tj}^2)\left(\frac{R_{ij}}{R_0}\right)^2$$

- clusters hard collinear radiation first




Inversion of **Pythia shower**

## anti $k_T$ algorithm

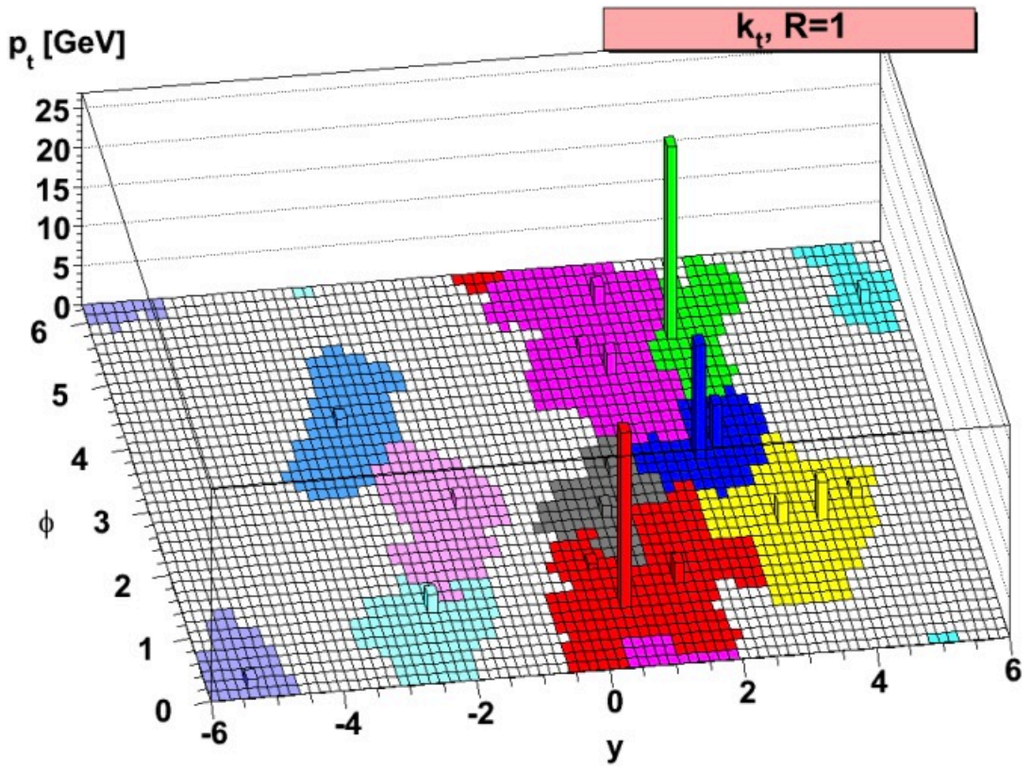
$$d_{ij} = \min(p_{Ti}^{-2}, p_{Tj}^{-2})\left(\frac{R_{ij}}{R_0}\right)^2$$

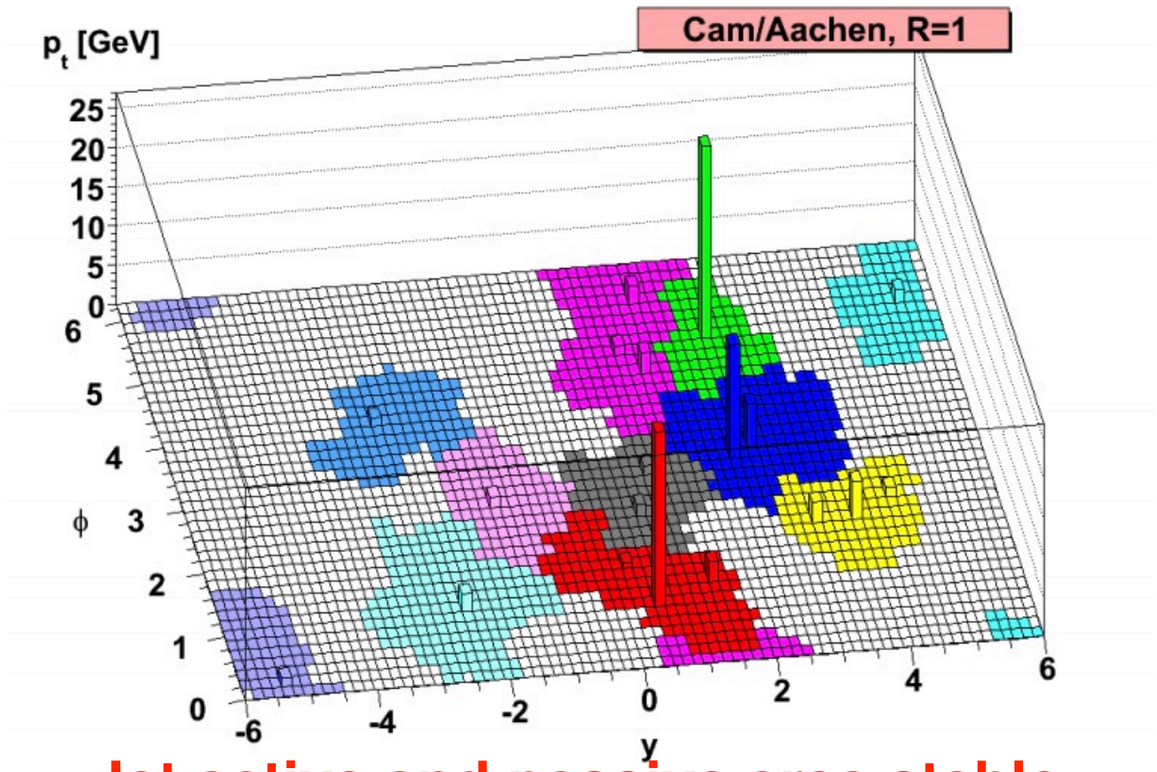
- Clusters farthest first
- No inverse parton-shower interpretation

**Other have their niche uses too (later)**

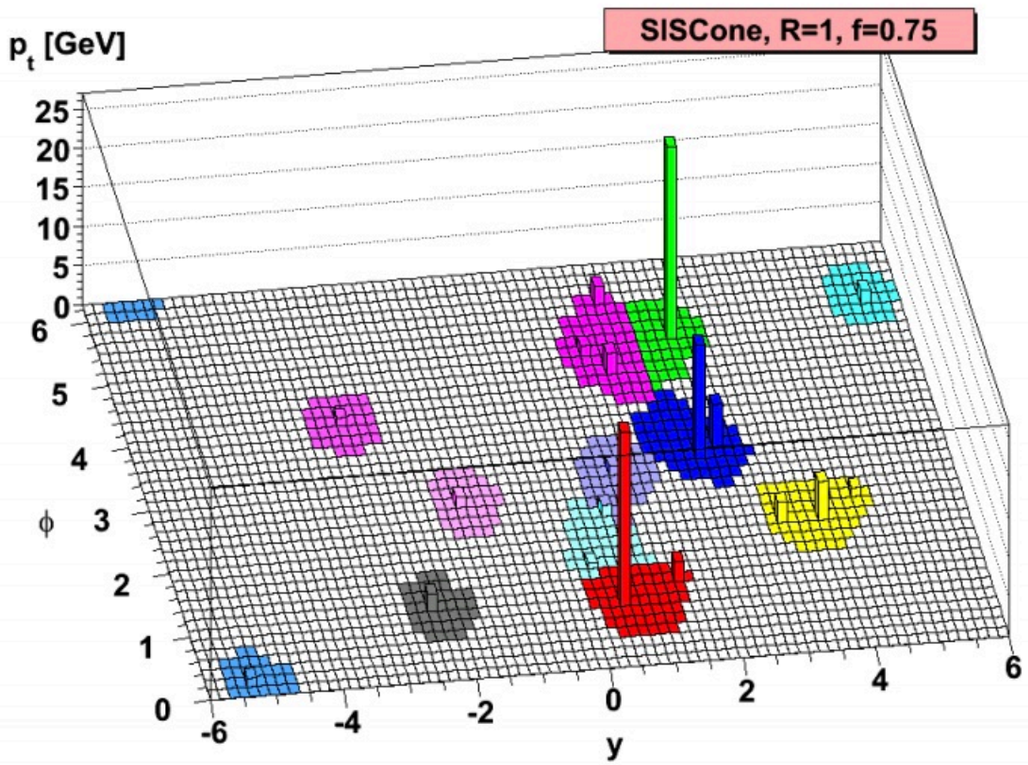
- Produces round jets
- Almost exclusively used by ATLAS and CMS

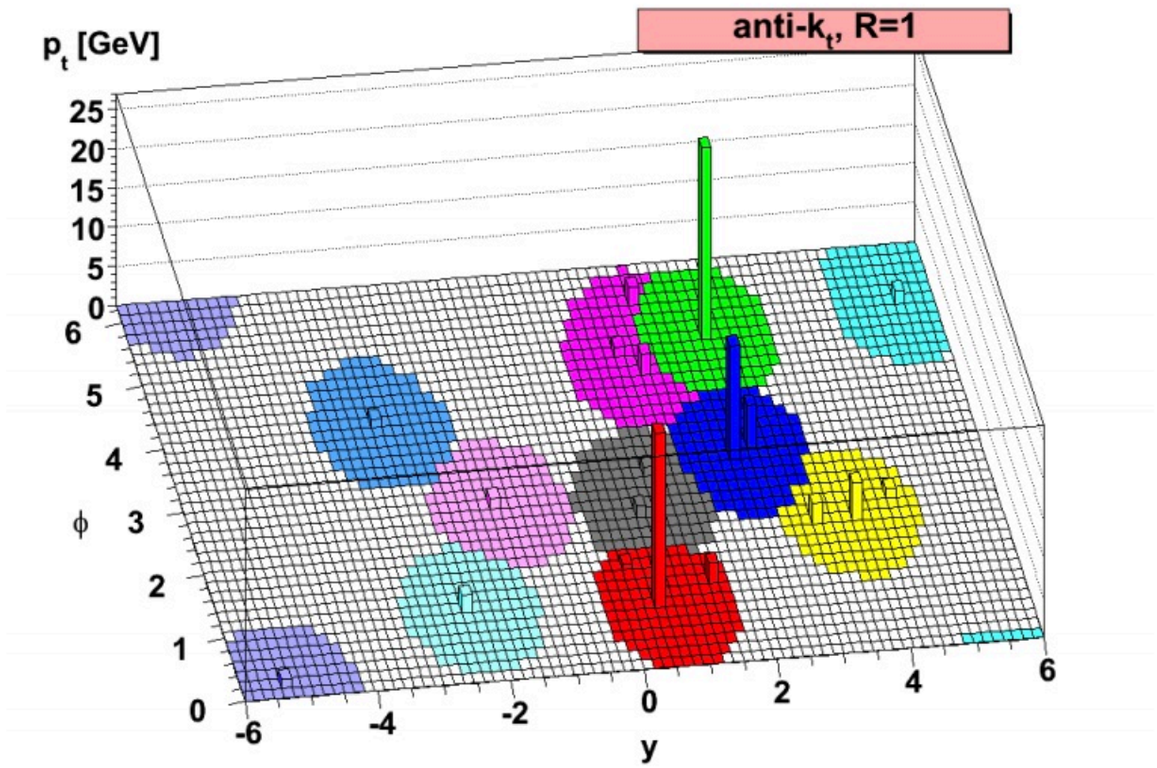
# Jet Algorithms



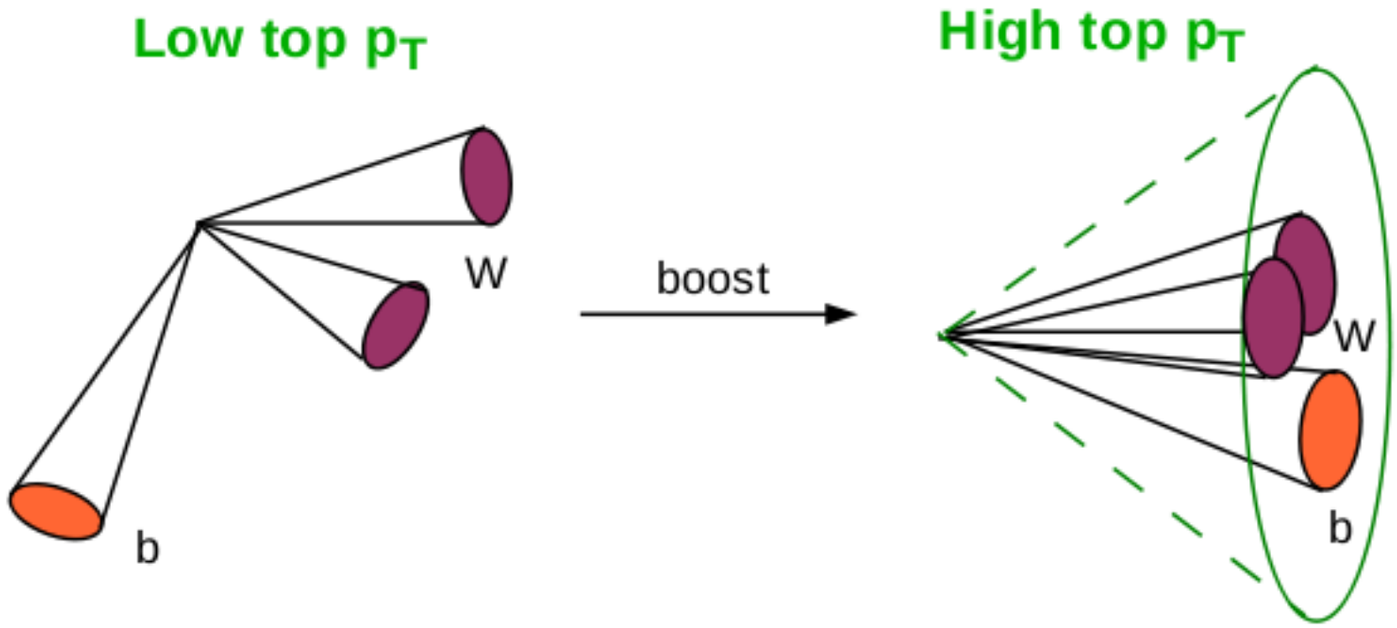

**Jet active and passive area stable**



# Boosted jets and substructure

**How do we reconstruct heavy, hadronically decaying particles?**




YOU MAY SKIM THIS SECTION: the state of the art for jet algorithms

At high pT can typically reconstruct a heavy object within an R=1.0 jet

Rule of thumb: angular separation of decay products of a massive particle in a 1 to 2 decay is

$$R = \frac{2m}{p_T}$$

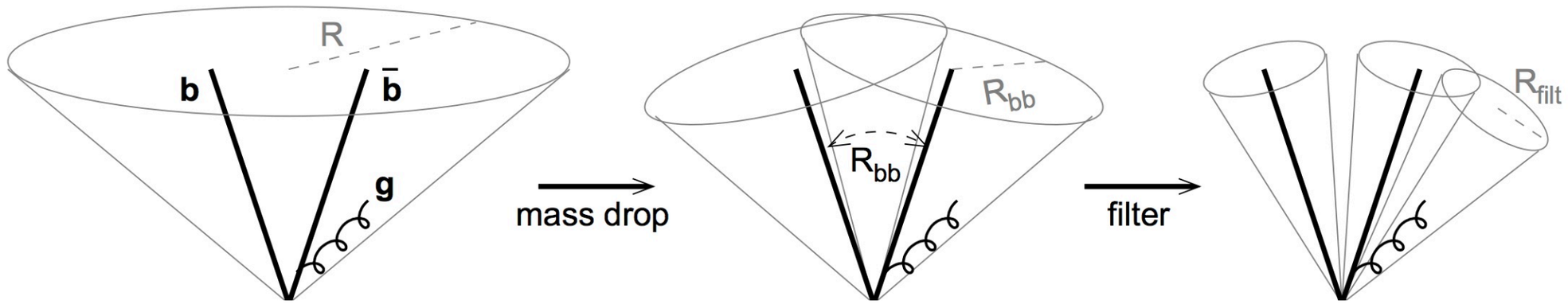
Jets from quarks and gluons typically have a single, hard core

**Other challenges**

- Have to deal with pileup, now at a constituent level rather than a jet level
- Finite resolution of the calorimeter: angular separation of constituents matters more

# Substructure origins

## BDRS tagger: Higgs tagging with split filtering



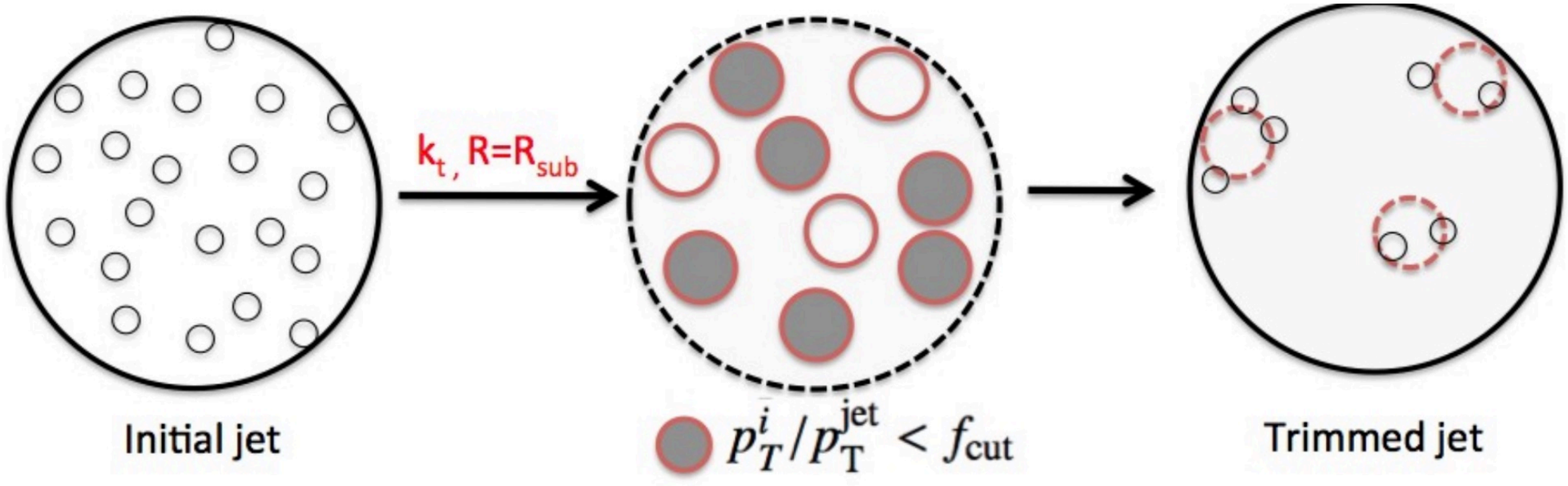

$$\mu = \frac{m_{j1}}{m_j} \qquad y = \frac{\min(p_{T1}, p_{T2}) \cdot (\Delta R_{12})^2}{m_{jet}^2}$$

- Cluster jet with C/A algorithm
- Undo the clustering history and at each step evaluate mass drop and subjet asymmetry
- If mass drop is small and asymmetry large, discard the subheading jet and repeat

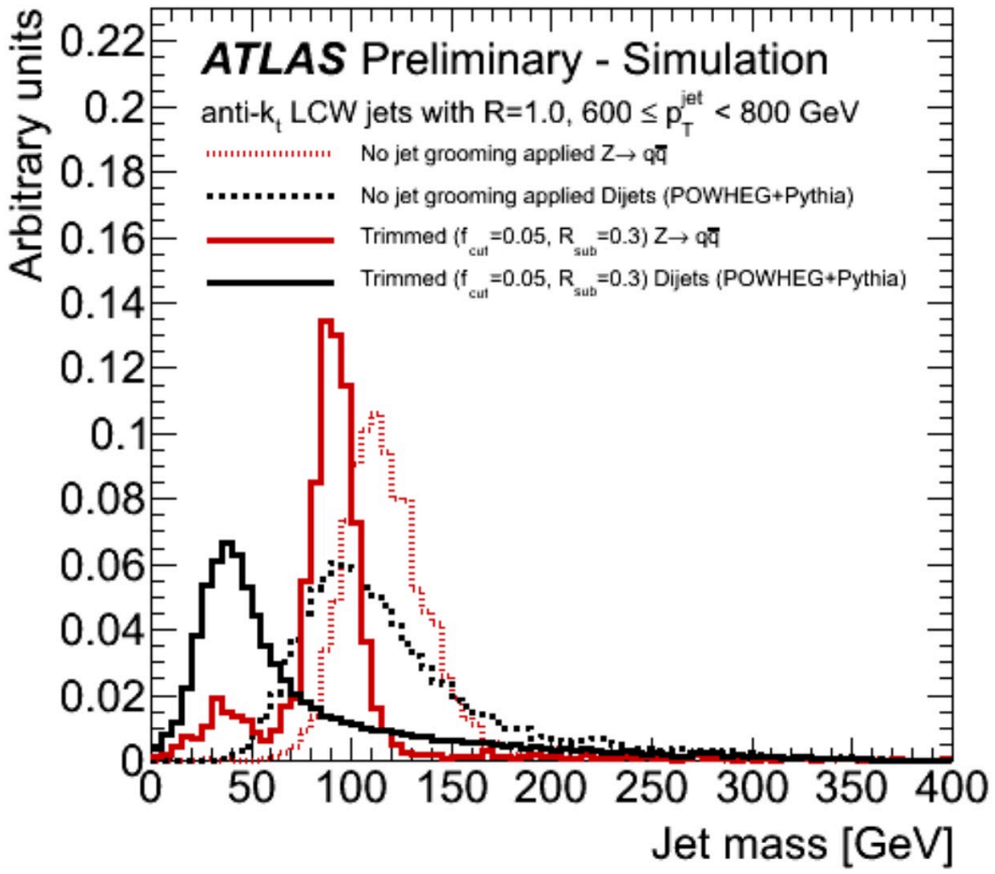
**This will pick out the "hard splitting" and help identify the mass peak**
Showed that more could be learnt about the physics of a jet by looking inside

# Jet Grooming

## The trimming algorithm




**anti-kt R=1.0, (before/after trimming)**




- JSS variables are smeared by soft radiation from ISR, pileup sources
- Grooming attempts to remove this while preserving substructure information
- Can be too aggressive

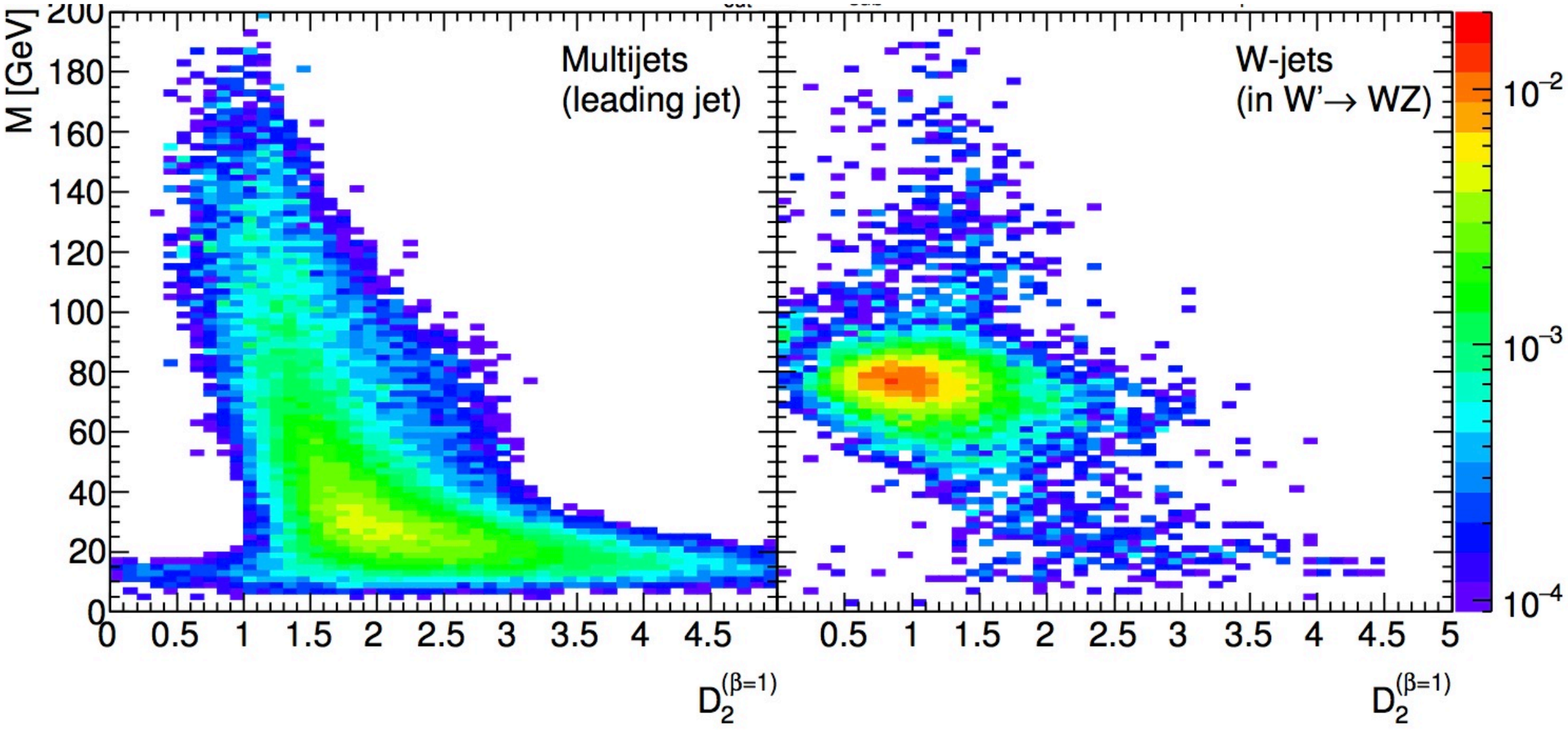
Trimming is currently used by ATLAS

softdrop is likey to replace it, and has interesting theoretical properties

# Making a W tagger

**Compare different combinations of variables and cuts**

Apply cuts optimise signal selection and background rejection



# Fragmentation Functions

- Next we have more of a theoretical perturbative, take on the jet formation part– Fragmentation Functions
- This was already shown in the slides by me as part of the main "factorization" equation—e.g. from my thesis plot. It most universally is represented by the function D(x) or D(z) where x or z is supposed to represent the fraction of the OUTGOING parton's (q/gluon) momentum the jet's hadron ends up with.
  - If x is used therefore this x is different than the x in the Structure Functions / PDF's in Griffiths Edition 1 handout, that one represented the fraction of the INCOMING hadron's momentum that the incoming parton scatters with initially.
  - Most people use z and D(z)...this notation in this paper is still compatible with that because the x applies only in e+e–.
  - But just as with the structure fn's /pdfs many "partially integrated"/ integrated /un-integrated versions exists and it's not always clear which one it is from the notation D

$$\frac{d\sigma_{pp\to C+X}}{d^3p_C} = \int dx_{q_i} dx_g dz_q G_{q_i/p}(x_{q_i}, M_\gamma^2) G_{g/p}(x_g, \mu_F^2) D_{C/c}(z_C, \mu_F a^2) \times \frac{\hat{s}}{z_C^2 \pi} \frac{d\sigma_{qg\to qc}}{d\hat{t}} \delta(\hat{s} + \hat{t} + \hat{u}) \quad (2.11)$$

We've suppressed the $f$ on $q_f$, but it hardly matters since we've integrated out the momentum of this final state quark (the $X$). One example of the application of this

from my Ph.D dissertation as before it's in BB folder too

# Reading Sources

- The main ~~slides~~ are from Bryan Webber:
  - https://www.slac.stanford.edu/econf/C990809/docs/webber.pdf

- But first to give you a broader brief intro that uses more familiar notation:
  - arXiv:1607.02521v2 https://arxiv.org/abs/1607.02521

- **In this reading don't get bogged down, just scan** pay particular attention though to the end sections about the difference btw quark and gluon FF's, differentiating q vs g these has become pretty commonplace, even at the "jet finding" procedure level.

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# 1 Introduction

Quantum chromodynamics (QCD) is the generally accepted fundamental theory of the strong interaction. Thanks to the asymptotic freedom of QCD [1, 2] many high-energy scattering processes can be analyzed using perturbation theory. In most cases such analyses are in the form of factorization theorems, which separate the perturbatively calculable part of the cross section from the non-perturbative part [3]. If specific particles are identified in the final state, parton fragmentation functions (FFs) appear frequently as non-perturbative ingredient of QCD factorization formulas. (In the older literatue the term parton decay functions instead of FFs was often used.) Generally, FFs describe how the color-carrying quarks and gluons transform into color-neutral particles such as hadrons or photons.

The concept of FFs was already used shortly after the parton model [4, 5, 6] had been introduced [7]. This was in the pre-QCD era, at a time when key characteristic features of the partons, like their interactions, were heavily under debate. Early on FFs were considered as counterpart of parton distribution functions (PDFs). While PDFs were understood as probability densities for finding partons, with a given momentum, inside color-neutral particles, FFs were understood as probability densities for finding color-neutral particles inside partons [7, 6].

FFs are intimately connected with another type of non-perturbative objects, the so-called (time-like) cut-vertices [8, 9]. More precisely, cut-vertices correspond to certain moments of FFs [10]. They were introduced in order to obtain a formulation of processes with identified hadrons that is similar to, for instance, Wilson's operator product expansion [11]. Nowadays FFs are more frequently used than cut-vertices.

The best studied FF is what we denote by $D_1^{h/i}(z)$ in this paper. It describes the fragmentation of an unpolarized parton of type $i$ into an unpolarized hadron of type $h$, where the hadron carries the fraction $z$ of the parton momentum. Here one has in mind the longitudinal momentum of the hadron, that is, the component of the momentum along the direction of motion of the parton. Therefore, $D_1^{h/i}(z)$ is often called a collinear FF or also an integrated FF, since the transverse momentum of the hadron $\vec{P}_{hT}$ relative to the parton is integrated over. To be now more specific about the meaning of this function we note that the quantity $D_1^{h/i}(z)\,dz$ is the number of hadrons $h$ inside parton $i$ in the momentum fraction range $[z, z+dz]$. When taking into account higher-order QCD effects this parton model interpretation of FFs gets distorted [12].

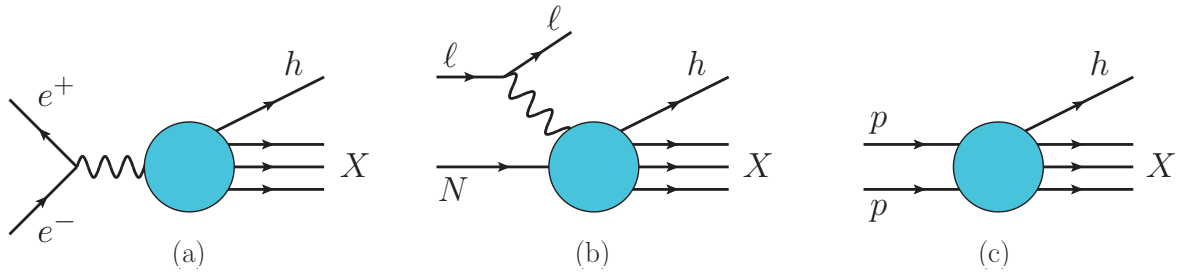


Figure 1: Representation of the scattering amplitude for processes whose QCD description involves FFs: Single-inclusive hadron production in (a) $e^+e^-$ annihilation, (b) deep-inelastic lepton-nucleon scattering, (c) proton-proton scattering.

In general, the following processes have played and continue to play a crucial role in studies of FFs:

- single-inclusive hadron production in electron-positron annihilation, $e^+ + e^- \to h + X$. Often this process is simply denoted as single-inclusive annihilation (SIA).
- semi-inclusive deep-inelastic lepton-nucleon scattering (SIDIS), $\ell + N \to \ell + h + X$.
- single-inclusive hadron production in proton-proton collisions, $p + p \to h + X$. Related processes like proton-antiproton ($p\bar{p}$) collisions have been studied as well.

The scattering amplitudes for these reactions are displayed in Fig. 1. In these cases QCD factorization theorems schematically read [3, 13]

$$\sigma^{e^+e^-\to hX} = \hat{\sigma} \otimes FF\,, \tag{1}$$

Skim here:

$$\sigma^{\ell N\to \ell hX} = \hat{\sigma} \otimes PDF \otimes FF\,, \tag{2}$$

$$\sigma^{pp\to hX} = \hat{\sigma} \otimes PDF \otimes PDF \otimes FF\,, \tag{3}$$

where $\hat{\sigma}$ indicates the respective process-dependent partonic cross section that can be computed in perturbation theory. The parton-model representation of the cross section for the three processes is shown in Fig. 2. Using the parton model, or in other words leading order perturbative QCD (pQCD), it is often straight forward to write down a factorization formula. However, in full QCD it is typically challenging to analyze and factorize radiative corrections to arbitrary order in the strong coupling [3, 13]. Factorization theorems only hold if specific kinematic conditions are satisfied, where the minimum requirement is the presence of a hard scale that allows one to use pQCD. For SIA that scale is provided by the center-of-mass ($cm$) energy $\sqrt{s}$. For SIDIS it is the momentum transfer between the leptonic and the hadronic part of the process, while in the case of hadronic collisions it is the transverse momentum of the final state hadron relative to the collision axis. The specific form of the factorization theorem, including the precise meaning of the "multiplication" $\otimes$, also depends on the kinematics of the process. In addition, it can depend on the polarization state of one or more of the involved particles. More information on this point will be given later in this paper and the references quoted there. We also mention that the factorization theorems in Eqs. (1)-(3) hold in the sense of a Taylor expansion in powers of $1/Q$, where here $Q$ denotes the hard scale of a process. The term on the r.h.s. of these equations then represents the leading contribution. Factorization theorems have been written down for certain subleading terms as well, but in most such cases all-order proofs do not exist.

An interesting and important early application of FFs in the 1970s was for the production of large-transverse-momentum hadrons in hadronic collisions, where FFs are needed according to (3). Data for this process had been obtained in $pp$ collisions at the ISR (Intersecting Storage Ring) collider at

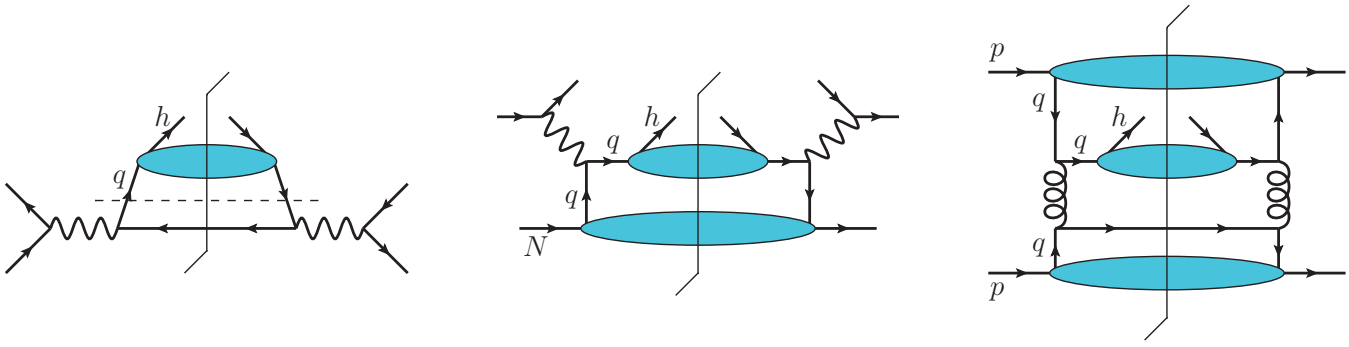


Figure 2: Parton-model representation of the cross section for single-inclusive hadron production in (a) $e^+e^-$ annihilation, (b) deep-inelastic lepton-nucleon scattering, (c) proton-proton scattering. In the case of SIA the factorization of the cross section into the hard process $e^+e^- \to q\bar{q}$ and the quark fragmentation part is indicated by the horizontal dashed line. Processes (b) and (c) depend on both FFs and PDFs. For proton-proton scattering only one of the leading-order diagrams is shown.

CERN [14, 15, 16], and in fixed-target proton-tungsten collisions at Fermi-Lab [17]. The data could not be explained unless one takes subprocesses involving gluons into account [18, 19, 20, 21, 22]. This observation can be considered the first phenomenological indication for the existence of gluons, as it was made before the discovery of the 3-jet events by the TASSO, PLUTO, MARK-J and JADE collaborations at DESY [23, 24, 25, 26] which are often quoted in this context.

In addition to $D_1^{h/i}(z)$, one can consider a number of other FFs by including (i) the spin of the parton and/or hadron, (ii) the transverse momentum of the hadron relative to the parton, (iii) higher-twist effects, and (iv) fragmentation into more than one hadron. All the FFs that emerge when making such generalizations are important in their own as they contain a lot of information about non-perturbative QCD dynamics in general, and the hadronization process in particular. In a number of cases they are simply needed to describe existing data. Very often they are also considered and used just as a tool for studying PDFs of hadrons, most notably of the nucleon. Especially in that regard generalizations of $D_1^{h/i}(z)$ received a lot of attention over the past two decades. In this article we will discuss the following four types of FFs:

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- **Integrated FFs:** The major focus for this type of FFs will be on $D_1^{h/i}(z)$. There exists an enormous body of experimental and theoretical work for this function. Its phenomenology is very rich, not the least due to the large variety of hadrons that has been studied. But we also discuss the corresponding spin-dependent integrated FFs $G_1^{h/i}(z)$ and $H_1^{h/i}(z)$, which are needed when the hadron is polarized parallel or perpendicular to its momentum, respectively. The three functions listed here are leading-twist (twist-2) objects.

- **Transverse-momentum dependent (TMD) FFs:** One can also study FFs that depend on the hadron's transverse momentum $\vec{P}_{hT}$, in addition to their dependence on $z$. Such TMD FFs allow one, for instance, to probe the transverse-momentum dependence of PDFs in a process like SIDIS. By keeping the transverse momentum of the fragmentation process one also finds new FFs. Two of them already became important in high-energy spin physics, with the Collins function for quarks $H_1^{\perp\, h/q}$ representing the most prominent example [27]. At present people are interested in this function mainly because in SIDIS it couples to the transversity PDF [27], which is one of the three leading-twist collinear quark distributions of the nucleon [28, 29, 30]. Because of its chiral-odd nature transversity cannot be measured in inclusive DIS. As a result the present knowledge about this function is still poor compared to the other leading-twist PDFs, that is, the unpolarized distribution and the helicity distribution of partons.

# Fragmentation and Hadronization

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## 1 Introduction

Hadronic jets are amongst the most striking phenomena in high-energy physics, and their importance is sure to persist as searching for new physics at hadron colliders becomes the main activity in our field. Signatures involving jets almost always have the largest cross sections, but are the most difficult to interpret and to distinguish from background. Insight into the properties of jets is therefore doubly valuable: both as a test of our understanding of strong interaction dynamics and as a tool for extracting new physics signals in multi-jet channels.

In the present talk, I shall concentrate on jet fragmentation and hadronization, the topic of jet production having been covered admirably elsewhere [1, 2]. The terms fragmentation and hadronization are often used interchangeably, but I shall interpret the former strictly as referring to inclusive hadron spectra, for which factorization 'theorems'[2] are available. These allow predictions to be made without any detailed assumptions concerning hadron formation. A brief review of the relevant theory is given in Section 2.

Hadronization, on the other hand, will be taken here to refer specifically to the mechanism by which quarks and gluons produced in hard processes form the hadrons that are observed in the final state. This is an intrinsically non-perturbative process, for which we only have models at present. The main models are reviewed in Section 3. In Section 4 their predictions, together with other less model-dependent expectations, are compared with the latest data on single-particle yields and spectra.

One of the most important objectives of jet studies is to understand the differences between jets initiated by different types of partons, especially quarks versus gluons, which could also be valuable in new physics searches. The wealth of recent data on quark-gluon jet differences is discussed in Section 5. Another

[1] Permanent address.

[2] I put this term in inverted commas because proofs of factorization in fragmentation do not really extend beyond perturbation theory [3].

goal is to compare jets produced in different processes such as $e^+e^-$ annihilation and deep inelastic scattering (DIS). This is being done by the H1 and ZEUS collaborations at HERA; some of their results are discussed in Section 6.

A good understanding of heavy quark jets is especially important because these are expected to be copiously produced in new processes such as Higgs boson decay. Section 7 discusses some recent results on heavy quark fragmentation.

Finally, Sections 8 and 9 deal with correlation effects, first those of Bose-Einstein origin and then those expected in fully hadronic final states from $e^+e^- \to W^+W^-$. A summary of the main points is given in Section 10.

Regrettably, I have no space to discuss many other relevant and interesting topics, such as: fragmentation function parametrizations [4]; photon fragmentation function; polarization in hadronization; event shapes and power corrections [5]; fluctuations and intermittency; dynamical two-particle correlations; heavy quark production in jets ($g \to c\bar{c}, b\bar{b}$); transverse energy flow in deep inelastic scattering; underlying event in DIS and hadron-hadron collisions; identified particle production in DIS; tests of QCD coherence; jet profiles and substructure [1]. Apologies also to all those whose work I have omitted or mentioned only superficially.

In citing the latest experimental data (mostly still preliminary) I have relied on the reference system of the EPS-HEP99 conference in Tampere, Finland [6], because few experimental papers were explicitly submitted to LP99. In the case of the large collaborations, papers submitted to EPS-HEP99 and/or LP99 can usually be found easily via the collaboration web pages [7]–[14].

## 2 Jet fragmentation—theory

Let us start by recalling the basic factorization structure of the single-particle inclusive distribution, *e.g.* in $e^+e^- \to hX$ (Fig. 1):

$$F^h(x,s) = \sum_i \int_x^1 \frac{dz}{z} C_i(z, \alpha_s(s)) D_i^h(x/z, s)$$

$$s = q^2 , \qquad x = 2p_h \cdot q/q^2 = 2E_h/E_{cm}$$

where $C_i$ are the coefficient functions for this particular process (including all selection cuts etc.) and $D_i^h$ is the universal fragmentation function for parton $i \to$ hadron $h$.

The fragmentation functions are not perturbatively calculable, but their $s$-

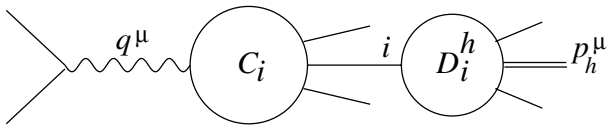



Figure 1: Factorization structure of $e^+e^- \to hX$.

dependence (scaling violation) is given by the DGLAP equation:

$$s\frac{\partial}{\partial s}D_i^h(x,s) = \sum_j \int_x^1 \frac{dz}{z} P_{ji}(z,\alpha_s(s)) D_j^h(x/z,s) \; .$$

Thus, they can be parameterized at some fixed scale $s_0$ and then predicted at other energies [4].

In certain kinematic regions, higher-order corrections are enhanced by large logarithms, which need to be resummed. At small $x$, $\log x$ enhanced terms can be resummed by changing the DGLAP equation to

$$s\frac{\partial}{\partial s}D_i^h(x,s) = \sum_j \int_x^1 \frac{dz}{z} P_{ji}(z,\alpha_s(s)) D_j^h(x/z,z^2 s) \; .$$

This is commonly known as the modified leading-logarithmic approximation (MLLA) [15, 16, 17]. The effect of resummation is to generate a characteristic humpbacked shape in the variable $\xi = \ln(1/x)$, with a peak at $\xi_p \sim \frac{1}{4}\ln s$.

Large logarithms of ratios of invariants may also appear inside the coefficient functions $C_i$, for example in three-jet events when the angles between jets become small. In some cases, these can be absorbed into a change of scale in the fragmentation functions. Examples will be encountered in Sections 4 and 5.

Although universal, fragmentation functions are factorization scheme dependent. The splitting functions $P_{ji}$ are also scheme dependent in higher orders. To specify the scheme requires calculation of the coefficient functions to (at least) next-to-leading order. This has only been done in a few cases. Thus, there is need for theoretical work to make full use of the data on fragmentation functions.

# 3 Hadronization Models

## 3.1 General ideas

*Local parton-hadron duality* [18]. Hadronization is long-distance process, involving only small momentum transfers. Hence, the flows of energy-momentum and

flavor quantum numbers at hadron level should follow those at parton level. Results on inclusive spectra and multiplicities support this hypothesis.
*Universal low-scale* $\alpha_s$ [19, 20, 21]. Perturbation theory works well down to low scales, $Q \sim 1$ GeV. Assume therefore that $\alpha_s(Q^2)$ can be defined non-perturbatively for all $Q$, and use it in the evaluation of Feynman graphs. This approach gives a good description of heavy quark spectra and event shapes.

## 3.2 Specific models

The above general ideas do not try to describe the mechanism of hadron formation. For this we must so far resort to models. The main current models are *cluster* and *string* hadronization (Fig. 2). We describe briefly the versions used in the HERWIG and JETSET event generators, respectively.

1. *Cluster model* [22]-[26]. The model starts by splitting gluons non-perturbatively, $g \to q\overline{q}$, after the parton shower. Color-singlet $q\overline{q}$ combinations have lower masses and a universal spectrum due to the *preconfinement* [27, 28] property of the shower (Fig. 3 [29]). These color-singlet combinations are assumed to form clusters, which mostly undergo simple isotropic decay into pairs of hadrons, chosen according to the density of states with appropriate quantum numbers [23]. This model has few parameters and a natural mechanism for generating transverse momenta and suppressing heavy particle production in hadronization. However, it has problems in dealing with the decay of very massive clusters, and in adequately suppressing baryon and heavy quark production.

2. *String model* [30]-[34]. This model is based on the dynamics of a relativistic string, representing the color flux stretched between the initial $q\overline{q}$. The string produces a linear confinement potential and an area law for matrix elements:
$$|M(q\overline{q} \to h_1 \cdots h_n)|^2 \propto e^{-bA}$$
where $A$ is the space-time area swept out (Fig. 4). The string breaks up into hadrons via $q\overline{q}$ pair production in its intense color field. Gluons produced in the parton shower give rise to 'kinks' on the string. The model has extra parameters for the transverse momentum distribution and heavy particle suppression. It has some problems describing baryon production, but less than the cluster model.

3. *UCLA model* [35]-[37] is a variant of the JETSET string model which takes the above area law for matrix elements more seriously, using it to determine the relative rates of production of different hadron species. This results in

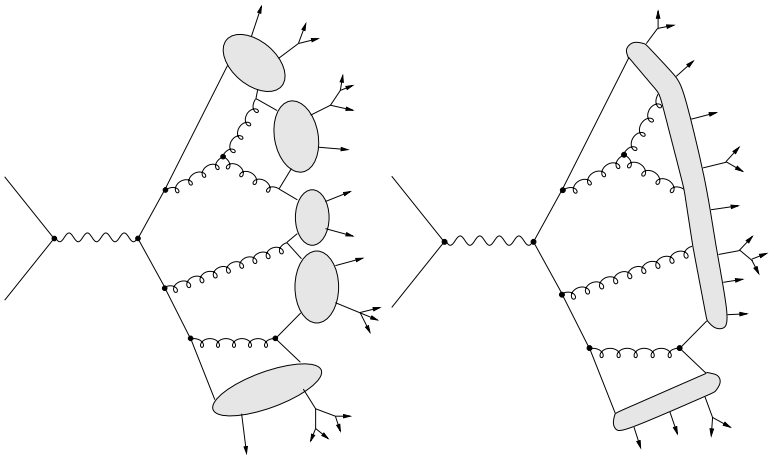


Figure 2: Cluster and string hadronization models.

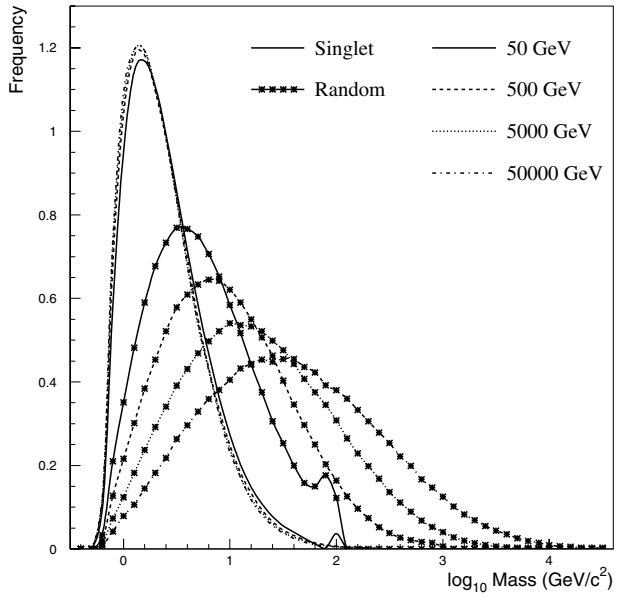
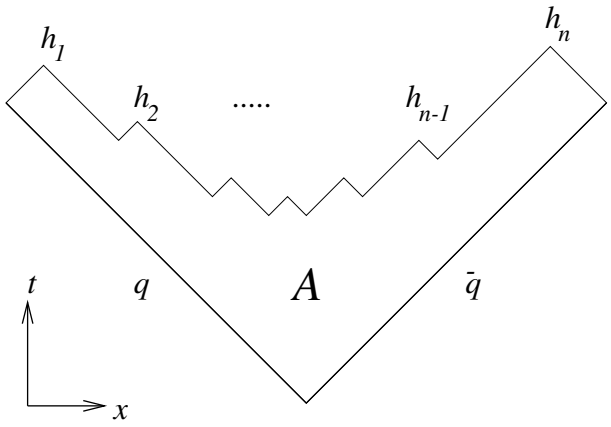



Figure 3: Cluster model: mass distribution of $q\overline{q}$ pairs.

heavy particle suppression without extra parameters, the mass-squared of a hadron being proportional to its space-time area. At present, the model still uses extra parameters for $p_T$ spectra, and again has some problems describing baryon production.




You will see more about diagrams like this one later, don't bother trying to understand things here like this in detail yet.

Figure 4: String model: space-time picture.

## 4 Single-particle yields and spectra

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Tables 1 and 2 compare predictions of the above models[3] with data on $Z^0$ decay from LEP and SLC.[4] Of course, the models have tunable parameters, but the overall agreement is encouraging. As stated earlier, the main problems are in the baryon sector, especially for HERWIG.

It is remarkable that most measured yields (except for the $0^-$ mesons, which have special status as Goldstone bosons) lie on the family of curves

$$\langle n \rangle = a(2J+1)e^{-M/T}$$

where $M$ is the mass and $T \simeq 100$ MeV (Fig. 5 [39]). This suggests that mass, rather than quantum numbers, is the primary factor in determining production rates. Note that, surprisingly, the orbitally-excited $J = \frac{3}{2}$ baryon $\Lambda(1520)$ (not yet included in models) is produced almost as much as the unexcited $J = \frac{3}{2}$ baryon $\Sigma(1385)$ [40, 41].

At other energies, model predictions for identified particle yields are in broad agreement with $e^+e^-$ data (Fig. 6 [42]), but statistics are of course poorer. Charged particle spectra at low $x$ agree well with the resummed (MLLA) predictions [15, 16, 17] over a wide energy range, as illustrated in Fig. 7 [43].

In $p\overline{p} \to$ dijets [44] the relevant scale is taken to be $Q = M_{JJ} \sin\theta$ where $M_{JJ}$ is the dijet mass and $\theta$ is the jet cone angle (Fig. 8). Results are then in striking agreement with theory and with data from $e^+e^-$ annihilation at $Q = \sqrt{s}$ (Fig. 9).

[3] Using the recent ALEPH HERWIG tuning with strangeness suppression 0.8 [38].
[4] Updated from Ref. [29].

| Particle | Multiplicity | HERWIG 5.9 | JETSET 7.4 | UCLA 7.4 | Expts |
|---|---|---|---|---|---|
| Charged | 20.96(18) | 20.95 | 20.95 | 20.88 | ADLMO |
| $\pi^{\pm}$ | 17.06(24) | 17.41 | 16.95 | 17.04 | ADO |
| $\pi^0$ | 9.43(38) | 9.97 | 9.59 | 9.61 | ADLO |
| $\boldsymbol{\eta}$ | 0.99(4) | 1.02 | 1.00 | 0.78 | ALO |
| $\rho(770)^0$ | 1.24(10) | 1.18 | 1.50 | 1.17 | AD |
| $\omega(782)$ | 1.09(9) | 1.17 | 1.35 | 1.01 | ALO |
| $\boldsymbol{\eta'(958)}$ | 0.159(26) | 0.097 | 0.155 | 0.121 | ALO |
| $\mathbf{f_0(980)}$ | 0.155(8) | 0.111 | ~0.1 | — | ADO |
| $a_0(980)^{\pm}$ | 0.14(6) | 0.240 | — | — | O |
| $\phi(1020)$ | 0.097(7) | 0.104 | 0.194 | 0.132 | ADO |
| $\mathbf{f_2(1270)}$ | 0.188(14) | 0.186 | ~ 0.2 | — | ADO |
| $f_2'(1525)$ | 0.012(6) | 0.021 | — | — | D |
| $K^{\pm}$ | 2.26(6) | 2.16 | 2.30 | 2.24 | ADO |
| $\mathbf{K^0}$ | 2.074(14) | 2.05 | 2.07 | 2.06 | ADLO |
| $K^*(892)^{\pm}$ | 0.718(44) | 0.670 | 1.10 | 0.779 | ADO |
| $K^*(892)^0$ | 0.759(32) | 0.676 | 1.10 | 0.760 | ADO |
| $K_2^*(1430)^0$ | 0.084(40) | 0.111 | — | — | DO |
| $D^{\pm}$ | 0.187(14) | 0.276 | 0.174 | 0.196 | ADO |
| $D^0$ | 0.462(26) | 0.506 | 0.490 | 0.497 | ADO |
| $D^*(2010)^{\pm}$ | 0.181(10) | 0.161 | 0.242 | 0.227 | ADO |
| $D_s^{\pm}$ | 0.131(20) | 0.115 | 0.129 | 0.130 | O |
| $B^*$ | 0.28(3) | 0.201 | 0.260 | 0.254 | D |
| $B_{u,d}^{**}$ | 0.118(24) | 0.013 | — | — | D |
| $J/\psi$ | 0.0054(4) | 0.0018 | 0.0050 | 0.0050 | ADLO |
| $\psi(3685)$ | 0.0023(5) | 0.0009 | 0.0019 | 0.0019 | DO |
| $\chi_{c1}$ | 0.0086(27) | 0.0001 | — | — | DL |

Table 1: Meson yields in $Z^0$ decay. Experiments: A=Aleph, D=Delphi, L=L3, M=Mark II, O=Opal. Bold: new data this year. Underlined: disagreement with data by more than $3\sigma$.

New SLD data include hadron spectra in light quark (distinguished from antiquark) fragmentation, selected by hemisphere using the SLC beam polarization [45]. One sees strong particle–antiparticle differences in the expected directions (Fig. 10), bearing in mind the predominance of down-type quarks in $Z^0$ decay.

| Particle | Multiplicity | HERWIG 5.9 | JETSET 7.4 | UCLA 7.4 | Expts |
|---|---|---|---|---|---|
| p | 1.04(4) | 0.863 | 1.19 | 1.09 | ADO |
| $\Delta^{++}$ | 0.079(15) | 0.156 | 0.189 | 0.139 | D |
| | 0.22(6) | 0.156 | 0.189 | 0.139 | O |
| $\Lambda$ | 0.399(8) | 0.387 | 0.385 | 0.382 | ADLO |
| $\Lambda(1520)$ | 0.0229(25) | — | — | — | DO |
| $\Sigma^{\pm}$ | 0.174(16) | 0.154 | 0.140 | 0.118 | DO |
| $\Sigma^0$ | 0.074(9) | 0.068 | 0.073 | 0.074 | ADO |
| $\Sigma^{\star\pm}$ | 0.0474(44) | 0.111 | 0.074 | 0.074 | ADO |
| $\Xi^-$ | 0.0265(9) | 0.0493 | 0.0271 | 0.0220 | ADO |
| $\Xi(1530)^0$ | 0.0058(10) | 0.0205 | 0.0053 | 0.0081 | ADO |
| $\Omega^-$ | 0.0012(2) | 0.0056 | 0.00072 | 0.0011 | ADO |
| $\Lambda_c^+$ | 0.078(17) | 0.0123 | 0.059 | 0.026 | O |

Table 2: Baryon yields in $Z^0$ decay. Legend as in Table 1.

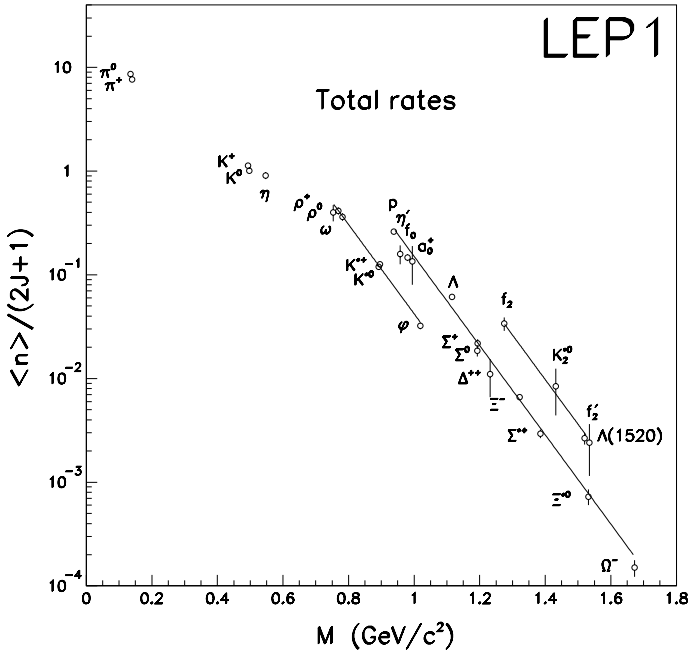


Figure 5: Particle yields in $Z^0$ decay.

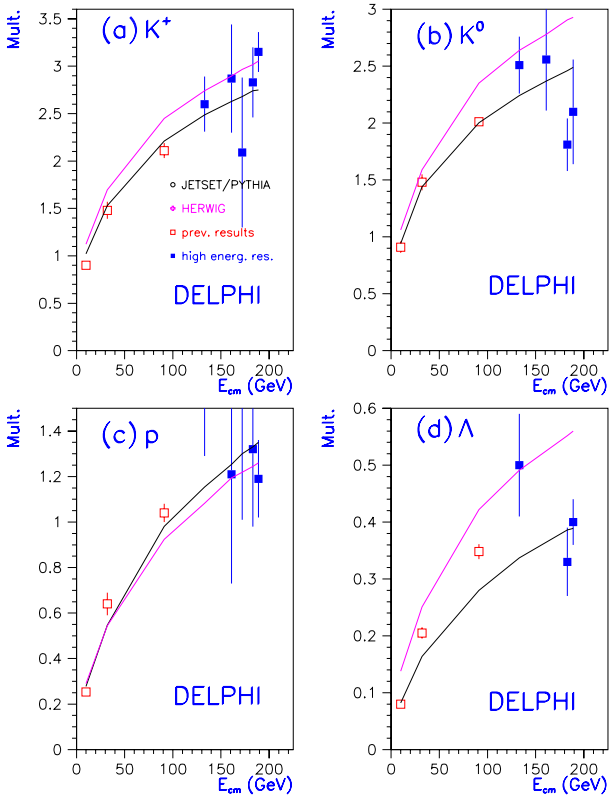


Figure 6: Particle yields in $e^+e^-$ annihilation.

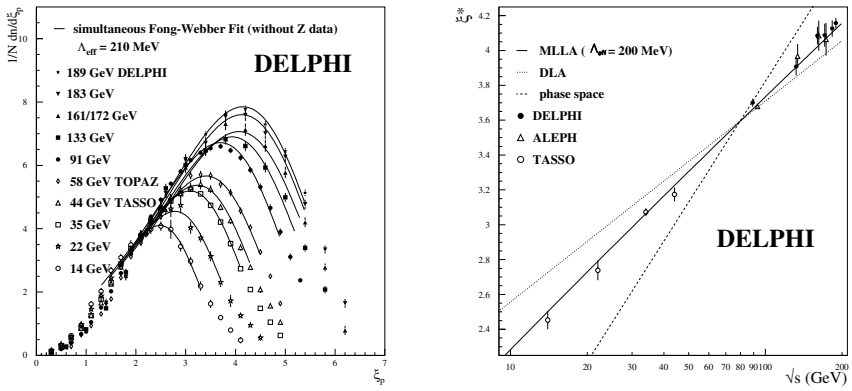


Figure 7: Low-$x$ fragmentation in $e^+e^-$ annihilation.

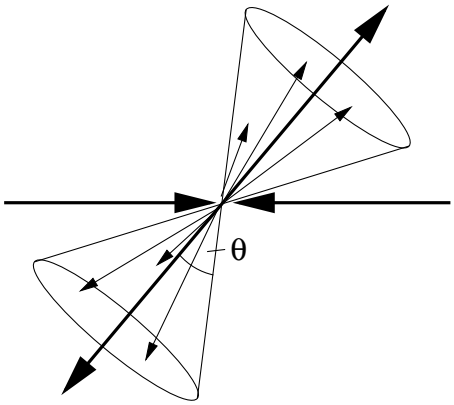


Figure 8: Cone angle in $p\overline{p} \to$ dijets

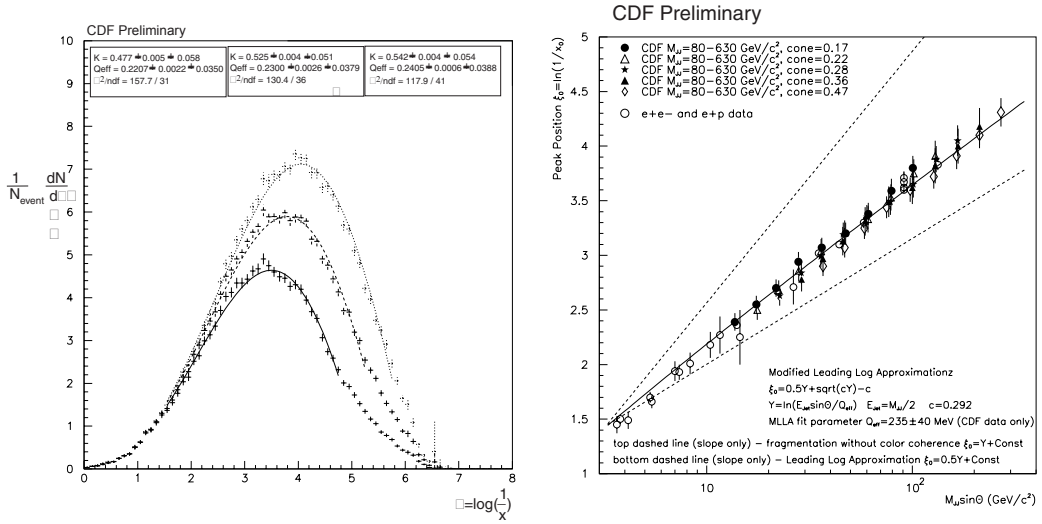


Figure 9: Low-$x$ fragmentation in $p\overline{p} \to$ dijets.

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## 5 Quark and gluon jets

DELPHI [46] selects gluon jets by anti-tagging heavy quark jets in Y and Mercedes three-jet events (Fig. 11). As expected, the higher color charge of the gluon ($C_A = 3$ vs. $C_F = 4/3$) leads to a softer spectrum and higher overall multiplicity (Fig. 12). In general, the relative multiplicities of identified particles are consistent with those of all charged, with no clear excess of any species in gluon jets (Fig. 13). In particular there is no enhanced $\phi(1020)$ or $\eta$ production:

$$\text{DELPHI [46]}: \quad N_g(\phi)/N_q(\phi) \quad = \quad 0.7 \pm 0.3$$

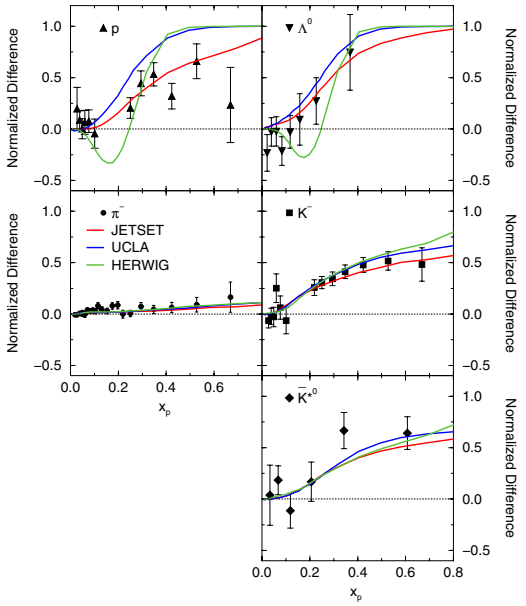



Figure 10: Normalized particle–antiparticle differences in quark jet fragmentation. Results are from SLD [45].

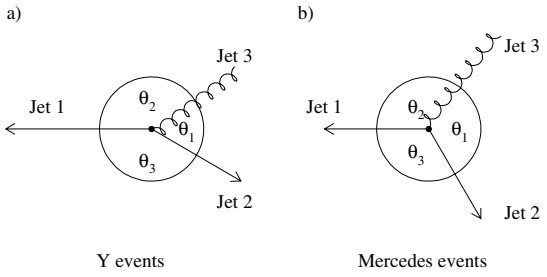
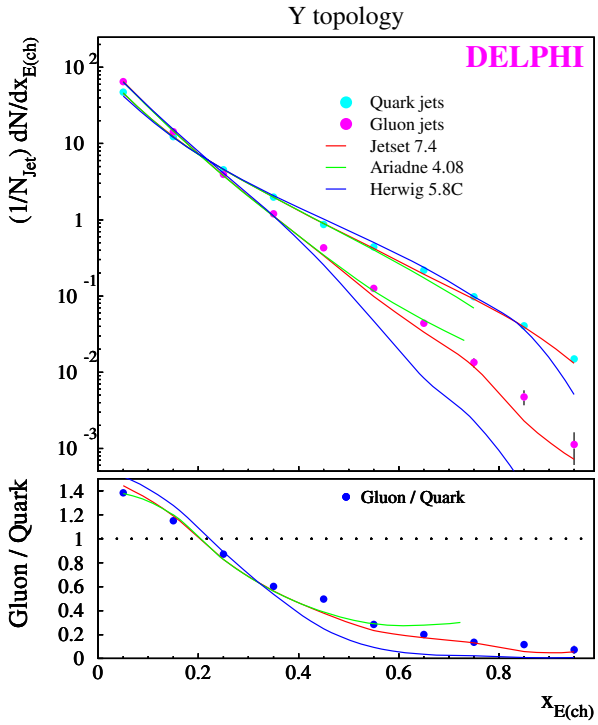



Figure 11: Selection of gluon jets by DELPHI.

$$\text{OPAL [47]}: \; N_g(\eta)/N_q(\eta) \;\; = \;\; 1.29 \pm 0.11 \; .$$

OPAL [48] selects gluon jets recoiling against two tagged $b$-jets in the same hemisphere. Monte Carlo studies indicate that such jets should be similar to those emitted by a point source of gluon pairs. The qualitative message from the data is again clear (Fig. 14): Gluon jets have softer fragmentation than light quark jets, and higher multiplicity. The precision of the data is now such that next-to-leading order calculations of the relevant coefficient functions, taking into account the experimental selection procedures, are needed to check the universality of the extracted gluon fragmentation function.




Note, this quantity measured here is essentially (probably identically) the same as the FF, D(x or z). Experimentalists like to use this notation instead often but sometimes they use D(z).

Figure 12: Charged particle spectra in quark and gluon jets.

The ratio of gluon/quark multiplicities at low rapidity (large angle) is close to the ratio of color charges $r \equiv C_A/C_F = 2.25$, in agreement with local parton-hadron duality:

$$\text{OPAL: } r_{ch}(|y| < 1) = 1.919 \pm 0.047 \pm 0.095 \ .$$

Monte Carlo studies [48] suggest that a better measure of $C_A/C_F$ is obtained by selecting low-momentum hadrons with relatively large $p_T$ (*i.e.* low rapidity). This gives

$$\text{OPAL: } r_{ch}(p < 4,\ 0.8 < p_T < 3 \text{ GeV}) = 2.29 \pm 0.09 \pm 0.015 \ .$$

DELPHI [49] has observed scaling violation in quark and gluon jet fragmenta-

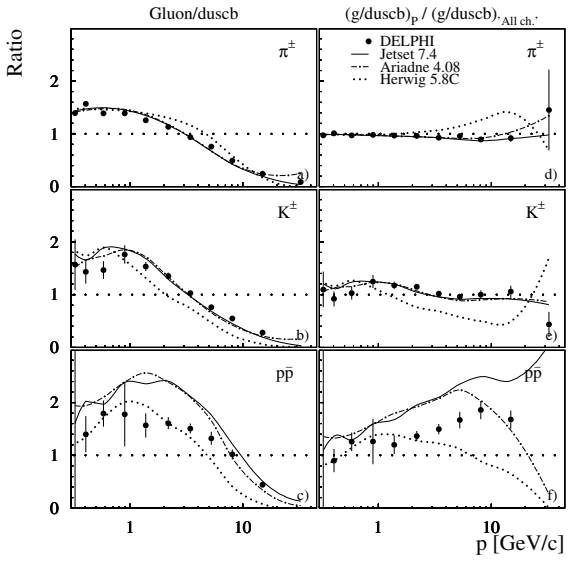



Figure 13: Comparisons of particle spectra in quark and gluon jets.

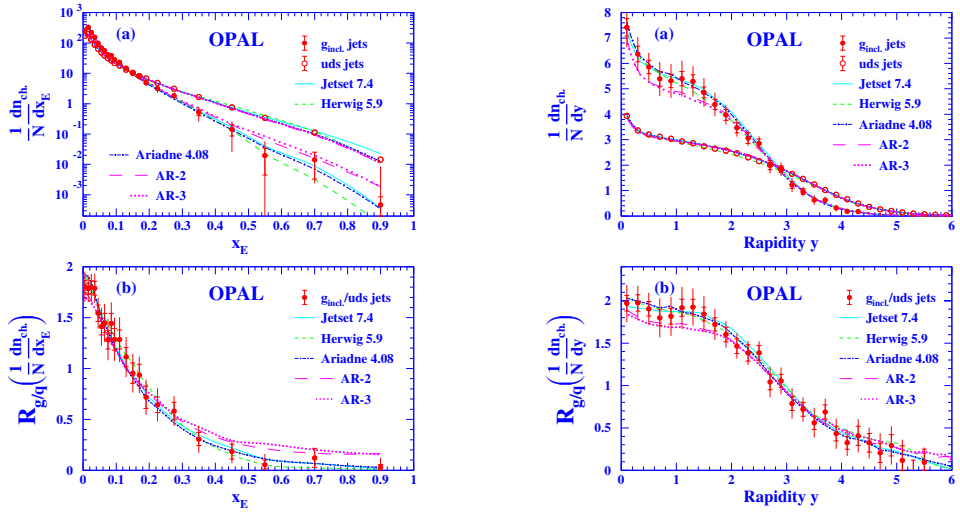



Figure 14: Momentum fraction and rapidity distributions in quark and gluon jets.

# Now the Hadronization Models Details

- First we start with a Wong section about the String Model.

- Just before that I include a bit about a precursor called the Schwinger Mechanism, you can mostly skip that part, but read at least the first couple pages of that, it kind of sets up the idea of color flux tubes which is what we think of the strings as being.
  - because of the gluon-gluon field field interaction i.e. non-Abelian behavior, the field forms "flux tubes" -- this has been shown in Lattice QCD

- This not only describes the **String Model (Lund string model**, see below) used in PYTHIA previously mentioned but it sets up the beginnings of what turns into in a later section (the DPM section) the explanation of the Reggeon/ Pomeron descriptions of the interactions.
  - Lund (U) is a big university in Stockholm Sweden, the main PYTHIA developers and maintainers have been there since its beginning, ~40-ish years ago)

We therefore have

$$(1-x_b)\frac{\tilde{\phi}_B^2(b)}{(\beta^2-m_\beta^2)^2} = x_b\frac{x_b}{(1-x_b)}\frac{\tilde{\phi}_B^2(b)}{(b^2-m_b^2)^2}.$$

The results suggest the usefulness of introducing the structure function

$$\tilde{G}_{b/B}(x_b,\boldsymbol{b}_T) = \frac{1}{(2\pi)^3}\frac{x_b}{2(1-x_b)}\frac{\tilde{\phi}_B^2(b)}{(b^2-m_b^2)^2},$$

which differs with $G_{b/B}$ of Eq. (5) of Supplement 4.1 only in the vertex function. The vertex function is evaluated when $b$ is on the mass shell for $G_{b/B}(x_b,\boldsymbol{b}_T)$ and is evaluated when $b$ is off shell for $\tilde{G}_{b/B}(x_b,\boldsymbol{b}_T)$. In terms of $\tilde{G}$, the invariant cross section is

$$E_b\frac{d^3\sigma}{db^3}\bigg|_{BA\to bX} = x_b\tilde{G}_{b/B}(x_b,\boldsymbol{b}_T)\sum_i\int\frac{d^3i}{E_i}E_i\frac{d^3\sigma}{di^3}(\beta+A\to i+X'),$$

which is Eq. (4.35).

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## 5. Particle Production in a Strong Field

We saw in Chapter 3 that the production of soft particles is a very important process in high-energy nucleon-nucleon collisions. How are particles produced? What is the physical basis for their production? While the answers to these questions should be found in the theory of quantum chromodynamics (QCD), the quantitative analysis of the particle production process cannot yet be described from first principles of QCD. On the other hand, many qualitative aspects of the production process can be explained in terms of simple and analogous models. This chapter and several following chapters will discuss different models of particle production.

### §5.1 Schwinger Particle Production Mechanism

The Schwinger particle production mechanism, first put forth to examine the production of electron-positron pairs in a strong and uniform electromagnetic field, has been applied to many problems in contemporary physics [1]. In particular, it has been invoked to study particle production in QCD [2]. As applied to nucleon-nucleon collisions or $e^+e^-$ annihilations, the field between a quark and an antiquark is represented, as an approximation, by an Abelian gauge field in the same form as a constant electric field between two condenser plates in quantum electrodynamics. A particle-antiparticle pair is produced when a particle tunnels from the negative energy continuum to the positive energy continuum.

We can study how this approximate description arises. The interaction between a quark and an antiquark can be represented phenomenologically by a linear potential proportional to the separation between the two particles [3], which is an essential feature for quark confinement. The constant of proportionality is called the *string tension* $\kappa$, which has been estimated from the Regge slope parameter in hadron spectroscopy to be about 1 GeV/fm [3]. The linear potential suggests the usefulness of describing a $q\bar{q}$ pair in terms of the model of a confined color flux tube. In this model, one considers a quark $q_0$ with a color charge $q$ at $z=0$ and an antiquark $\bar{q}_0$ with a color charge $-q$ at $z=L$ (Fig. 5.1). The color electric field $\mathcal{E}$ inside the tube has a constant magnitude $\mathcal{E}$ and is directed in the positive $z$ direction. The color flux is confined within a tube of length $L$ and a cross sectional area $A$. The cross section of the tube is approximately a circle with

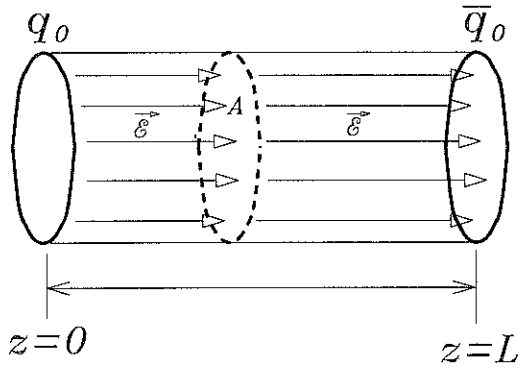


**Fig. 5.1** The color electric field $\vec{E}$ between a quark $q_0$ at $z = 0$ and an antiquark $\bar{q}_0$ at $z = L$ in the color flux tube model.

a radius of about 0.5 fm [3]. Outside the flux tube, the color electric field is zero. The energy contained in the color flux tube is

$$\frac{1}{2}\mathcal{E}^2 \cdot A \cdot L ,$$

which also gives the interaction energy between the quark $q_0$ and the antiquark $\bar{q}_0$ at a separation $L$. In this model, the interaction energy between a quark and an antiquark is linearly proportional to their separation $L$ with a constant of proportionality $\mathcal{E}^2 A/2$. The model can be used to study interesting qualitative features involving the dynamics of quarks. One can identify the constant of proportionality of the linear interaction in the color flux tube model with the phenomenological string tension $\kappa$,

$$\frac{1}{2}\mathcal{E}^2 \, A = \kappa . \tag{5.1}$$

Because the color electric field is confined inside the tube and vanishes outside, Gauss's law can be used to relate $\mathcal{E}A$ to the color charge $q$ of the quark $q_0$ at $z = 0$ by

$$\mathcal{E}A = q . \tag{5.2}$$

Using (5.1) and (5.2), one finds that the relation between the product $q\mathcal{E}$ and the string tension $\kappa$ is

$$q\mathcal{E} = 2\kappa . \tag{5.3}$$



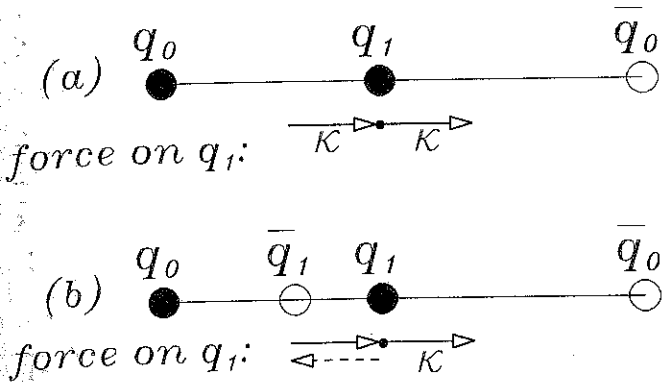


**Fig. 5.2** (*a*) The vector diagram shows the two forces acting on a single quark $q_1$ coming from $q_0$ and $\bar{q}_0$. (*b*) The production of the quark $q_1$ is accompanied by the production of $\bar{q}_1$. Here the vector diagram includes, in addition, the force acting on $q_1$ coming from the produced $\bar{q}_1$, shown as the dashed arrow.

When the separation $L$ between the quark $q_0$ and the antiquark $\bar{q}_0$ exceeds a threshold value, a quark $q_1$ and an antiquark $\bar{q}_1$ can be produced in the flux tube. To study the production of $q_1$ and $\bar{q}_1$ in the tube, we treat the quark $q_0$ at $z = 0$ and the antiquark $\bar{q}_0$ at $z = L$ as static sources which generate an 'external' color electric field $\boldsymbol{\mathcal{E}}$ for the production of $q_1$ and $\bar{q}_1$. For this picture, a $q_0\bar{q}_0$ pair acts as a pair of condenser plates placed at different potential levels with the potential difference being due to the color electric field from $q_0$ to $\bar{q}_0$. If a single quark $q_1$ with a color charge $q$ were produced in the flux tube (Fig. 5.2(*a*)), it would experience a force $q\boldsymbol{\mathcal{E}}$ and therefore a potential

$$\begin{aligned} V(z) &= -q\mathcal{E}z \\ &= -2\kappa z \quad (\text{ single } q_1 \text{ produced, unphysical}) . \end{aligned} \tag{5.4}$$

The force $q\boldsymbol{\mathcal{E}}$ acting on $q_1$ is equal to $2\kappa$ pointing in the positive $z$ direction. This force can be divided into two parts, as illustrated in Fig. 5.2(*a*). One part $\kappa$ is the repulsive 'push' force from the quark $q_0$ at $z = 0$; another part $\kappa$ is from the attractive 'pull' force from the antiquark $\bar{q}_0$ at $z = L$.

However, a single quark is never produced all by itself, as this would violate charge conservation. The production of the quark $q_1$

is always accompanied by the production of an adjacent antiquark $\bar{q}_1$ [see Fig. 5.2($b$)]. In consequence, the force acting on the quark $q_1$ due to the quark $q_0$ at $z = 0$ is cancelled by the force arising from the interaction due to the newly produced antiquark $\bar{q}_1$. The additional force coming from the additional interaction is shown by the dashed arrow in Fig. 5.2($b$). It has a magnitude $\kappa$ but points in the negative $z$ direction. The cancellation of the two forces acting on $q_1$ due to $q_0$ and due to the the newly produced $\bar{q}_1$ is another manifestation of the phenomenon of *screening*. The quark $q_0$ is said to be screened by the newly produced $\bar{q}_1$ because $q_0$ now exerts no effect on the quark $q_1$. Therefore, after taking into account this additional interaction due to the produced antiquark $\bar{q}_1$, the magnitude of the force experienced by $q_1$ is $\kappa$, and not $2\kappa$. Thus, when we take into account the fact that the production of $q_1$ is accompanied by the production of $\bar{q}_1$, the potential experienced by the produced quark $q_1$ is given not by Eq. (5.4), but by

$$V(z) = -\kappa z \quad (\text{ additional antiquark } \bar{q}_1 \text{ interaction included }). \tag{5.5}$$

This reduction of the potential due to the additional interaction was first pointed out by Glendenning and Matsui [4].

By treating the color charged pair $q_0$ and $\bar{q}_0$ as static external sources of the color electric field, we can identify the potential $V(z)$ as the time-like component $A_0(z)$ of an Abelian gauge field with vanishing space-like components $\mathbf{A}$. Equation (5.5) gives the potential that a quark will experience inside the flux tube. Outside the flux tube, the color electric field is zero, and so the potential is a constant. Let us set this constant at zero. Then the complete spatial dependence of the potential experienced by a quark is given by

$$A_0(z) = \begin{cases} 0 & \text{for } z \le 0 \text{ (Region I);} \\ -\kappa z & \text{for } 0 \le z \le L \text{ (Region II);} \\ -\kappa L & \text{for } L \le z \text{ (Region III).} \end{cases} \tag{5.6}$$

The potential $A_0(z)$ is shown as the solid curve in Fig. 5.3($a$).

Consider a quark with a rest mass $m$ and a transverse mass $m_T = \sqrt{m^2 + p_T^2}$ in the above static vector potential $A = (A_0, \mathbf{A})$. In Fig. 5.3($a$), the long-dashed curve shows $A_0(z) + m_T$, the potential $A_0$ plus the transverse mass, above which curve the positive energy continuum lies. The short-dashed curve shows $A_0(z) - m_T$, the potential $A_0$ minus the transverse mass, below which curve the negative energy continuum lies. According to Dirac's hole theory, the vacuum state is characterized by the configuration in which the single-particle states in the negative energy continuum are all occupied, and the states in the



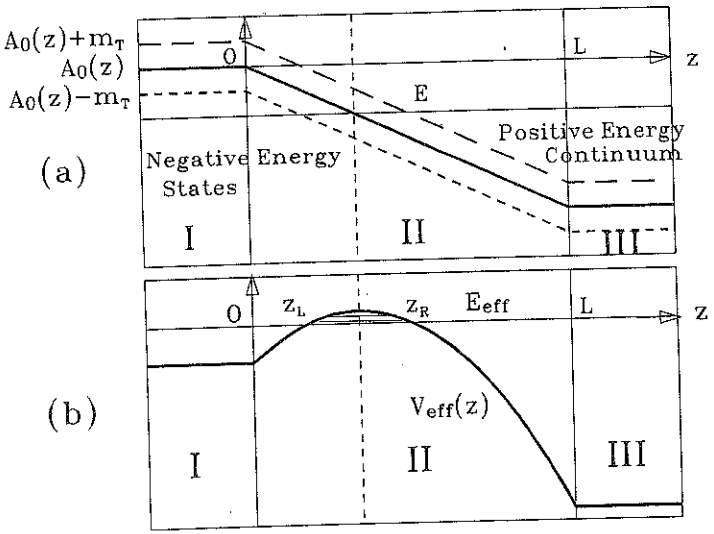


**Fig. 5.3**($a$) The potential $A_0(z)$ for the Dirac or the Klein-Gordon equation. ($b$) The corresponding $V_{\text{eff}}$ for the equivalent Schrödinger equation (5.13).

positive energy continuum are all empty. A quark from the negative energy sea in Region I can tunnel through a barrier and emerge as a quark in the positive energy continuum in Region III, leaving an empty hole in Region I and a quark in Region III. The hole has the characteristics of an antiquark. The tunneling process will lead to the production of a $q\bar{q}$ pair. How does one exhibit the nature of the tunneling mechanism explicitly?

To study the tunneling process, one should, in principle, use the Dirac equation to examine the behavior of a quark as a fermion in the external gauge potential $A$ of Eq. (5.6). This was carried out in Refs. [2] and [5]. As is well known, the Dirac equation gives rise to a Klein-Gordon-like equation with an extra term involving $\boldsymbol{\alpha} \cdot \nabla A_0$, where $\boldsymbol{\alpha}$ is the Dirac $\boldsymbol{\alpha}$ matrix [6]. This term comes from the spinor nature of the fermion wave function and gives only high-order effects in the tunneling probability; it does not affect the dominant term in the tunneling process. When this $\boldsymbol{\alpha} \cdot \nabla A_0$ term is neglected, one has a Klein-Gordon equation for a quark of mass $m$ in the vector potential of Eq. (5.6):

you may skim this page and the rest of this section till the next section "String Model" but notice the neat simple NR QM version of the problem, a little like Gamow-teller...note I'm pretty sure this is related to the "Instanton" stuff I briefly mentionned, including use of the WKB approximation

$$\left[(p-A)^2 - m^2\right]\psi = 0\,. \tag{5.7}$$

The potential does not depend on the transverse coordinates $x$ and $y$. (For a more realistic description of the transverse degrees of freedom, see Refs. [8]-[10]). The wave function in the transverse coordinates is just a plane wave with transverse momentum $\mathbf{p}_T = (p_x, p_y)$. The solution of the Klein-Gordon equation can be written in the form

$$\psi = \exp[i(p_x x + p_y y - Et)]f(z)\,, \tag{5.8}$$

where $E$ is the energy of the quark under consideration. Written out explicitly, the Klein-Gordon equation becomes

$$\left\{[E - A_0(z)]^2 - {p_z}^2 - m_T^2\right\}f(z) = 0\,. \tag{5.9}$$

For the tunneling process to occur, one observes from Fig. 5.3($a$) that the quark state under consideration must lie in the negative energy continuum in Region I and must lie in the positive energy continuum in Region III. Therefore, only those quarks with an energy $E$ lying within the range

$$-m_T \geq E \geq -\kappa L + m_T \tag{5.10}$$

can tunnel from Region I to Region III, leaving a quark particle in Region III and an empty hole in Region I, which has the characteristics of an antiparticle $\bar{q}$. There is thus the production of a particle-antiparticle $(q\bar{q})$ pair in the strong field.

From the above energy requirement, we see that the production of a $q\bar{q}$ pair will not occur if the separation $L$ between $q_0$ and $\bar{q}_0$ is not great enough to satisfy Eq. (5.10). The minimum separation $L_{\min}$ below which no pair will be produced can be inferred from Eq. (5.10) by equating its left-hand side with its right-hand side. We find

$$L_{\min} = 2m_T/\kappa\,. \tag{5.11}$$

If one takes the quark mass to be about 0.325 MeV, and the string tension $\kappa$ to be 1 GeV/fm, then the minimum length $L_{\min}$ required for the production of a $q\bar{q}$ pair is 0.7 fm for a particle with $p_T = 0$, but increases to 1 fm for a quark with a transverse momentum of 0.35 GeV.

To bring out the important concepts of tunneling through a barrier, it is simplest to write Eq. (5.9) in the form of the Schrödinger equation, for which techniques for its solution have been well developed.



Dividing Eq. (5.9) by $-2m_T$, we obtain the equivalent Schrödinger equation

$$\left\{\frac{p_z^2}{2m_T} + \left[\frac{m_T}{2} - \frac{[E - A_0(z)]^2}{2m_T}\right]\right\}f(z) = 0\,. \tag{5.12}$$

This can be cast in the form

$$\left[\frac{{p_z}^2}{2m_T} + V_{\text{eff}}(z) - E_{\text{eff}}\right]f(z) = 0\,, \tag{5.13}$$

which is the Schrödinger equation for a particle with a mass $m_T$ and an effective energy $E_{\text{eff}} = 0$, in an effective potential

$$V_{\text{eff}}(z) = \frac{m_T}{2} - \frac{[E - A_0(z)]^2}{2m_T}\,. \tag{5.14}$$

A simple inspection shows that the effective potential $V_{\text{eff}}(z)$ experienced by the particle of mass $m_T$ in the region between the plates (Region II) is quadratic in the spatial coordinate $z$. The quadratic dependence comes from the quadratic term $(p - A)^2$ in Eq. (5.7). In Region II, $V_{\text{eff}}(z)$ is in the form of an inverted parabola $m_T/2 - [E - A_0(z)]^2/2m_T$; this is joined onto a constant potential $m_T/2 - E^2/2m_T$ on the left of $z = 0$, and onto a constant potential $m_T/2 - (E + \kappa L)^2/2m_T$ on the right of $z = L$, as shown in Fig. 5.3($b$). The system under consideration corresponds to a single particle with a mass $m_T$ and an effective energy $E_{\text{eff}} = 0$. The particle must overcome a barrier to tunnel from Region I to Region III. The potential barrier is shown as the shaded region in Fig. 5.3($b$). The top of the barrier is located at the point at which $E = A_0(z)$ in Fig. 5.3($a$). It has a height of $m_T/2$. The left turning point is at

$$z_L = \frac{-E - m_T}{\kappa}\,,$$

and the right turning point is at

$$z_R = \frac{-E + m_T}{\kappa}\,.$$

From the equivalent Schrödinger equation (5.13), and the semiclassical WKB method [6], the penetrability for the system to go from one side of the barrier to the other side is given by

$$P = e^{-I}\,, \tag{5.15}$$

where

$$I = 2\int_{z_L}^{z_R} \sqrt{2m_{\rm eff}[V_{\rm eff}(z) - E_{\rm eff}]}dz\,,$$

and where $m_{\rm eff}$ is $m_T$. Then the integration of the above gives

$$I = 2\int_{z_L}^{z_R} \sqrt{m_T^2 - [E - A_0(z)]^2}dz = \frac{\pi m_T^2}{\kappa}\,. \tag{5.16}$$

Therefore, the penetrability for the quark to tunnel from Region I to Region III is

$$P = \exp\left\{-\frac{\pi m_T^2}{\kappa}\right\} = \exp\left\{-\frac{\pi(m^2+p_T^2)}{\kappa}\right\}. \tag{5.17}$$

This result indicates that the tunneling probability decreases with increasing particle mass. It increases with increasing string tension, which depends on the strength of the color electric field. The greater the strength of the color electric field, the greater will be the probability to produce a pair. The transverse momentum serves to provide an additional mass to the particle, and this effective mass must likewise tunnel through the barrier. It must work against the field with a strength $\kappa$; the result is a transverse momentum distribution of the form $\exp(-\pi p_T^2/\kappa)$.

## §5.2 Particle Production Rate

We are now in a position to estimate the rate of particle production per unit time per unit volume. We consider an energy $E$ in the interval given by Eq. (5.10) for which pair production can occur. The number of pairs produced in an elementary phase space volume, leading to the occurrence of positive energy particles at an energy $E$ within a time interval $\Delta t$, is equal to the number of quantum states in that phase space volume element multiplied by the penetrability $P$ given by Eq. (5.17).

The longitudinal length traveled by a particle with longitudinal momentum $p_z$ and energy $E = (m_T^2 + p_z^2)^{1/2}$, in time $\Delta t$, is $\Delta t$ multiplied by the longitudinal velocity $p_z/E$. The elementary phase space volume swept in time $\Delta t$ is therefore $\Delta x\Delta y[\Delta t\times(p_z/E)]dp_z d\mathbf{p}_T$. The number of quantum states in the phase space volume is

$$\Delta x\Delta y[\Delta t \times (p_z/E)]dp_z d\mathbf{p}_T/(2\pi)^3\,.$$



The number of positive energy outgoing particles which arise in this elementary phase space volume is

$$\Delta N = \Delta x\Delta y\Delta t\frac{p_z}{|E|}\frac{dp_z d\mathbf{p}_T}{(2\pi)^3}\exp\left\{-\frac{\pi(m^2+\mathbf{p}_T^2)}{\kappa}\right\}.$$

Therefore, we have

$$\frac{\Delta N}{\Delta x\Delta y\Delta t\Delta E d\mathbf{p}_T} = \frac{1}{(2\pi)^3}\exp\left\{-\frac{\pi(m^2+\mathbf{p}_T^2)}{\kappa}\right\}. \tag{5.18}$$

In the process of particle production due to the tunneling of a particle from the negative energy state, a positive energy particle is created, leaving a hole in the negative energy state which can be interpreted as an antiparticle moving in the direction opposite to the direction of motion of the created particle. The created particle is characterized by the energy $E$. For such a pair, one cannot, strictly speaking, identify any particular point as the location where the pair is produced, although this pair of particles begins to emerge into the classically allowed region at the points $z_L = (-E - m_T)/\kappa$ and $z_R = (-E + m_T)/\kappa$. Nevertheless, one can approximately associate the midpoint between $z_L$ and $z_R$,

$$z = -E/\kappa\,,$$

as the location in the vicinity of which the pair is created. With this association, the energy of the produced pair can be identified by the approximate location of production $z$, and the energy interval $\Delta E$ in which the particle states lie can be associated with a spatial interval $\Delta z$ at which the pair of particles are produced:

$$\Delta E = \kappa\Delta z\,. \tag{5.19}$$

From Eqs. (5.18-19), we obtain

$$\frac{\Delta N}{\Delta t\Delta x\Delta y\Delta z d\mathbf{p}_T} = \frac{\kappa}{(2\pi)^3}\exp\left\{-\frac{\pi(m^2+\mathbf{p}_T^2)}{\kappa}\right\}. \tag{5.20}$$

An integration over the transverse momentum leads to

$$\frac{\Delta N}{\Delta t\Delta x\Delta y\Delta z} = \frac{\kappa^2}{8\pi^3}\exp\left\{-\frac{\pi m^2}{\kappa}\right\}. \tag{5.21}$$

When we include the additional spin degree of freedom, we need to multiply the above result by the spin degeneracy factor $g_s$ which is

### §References for Chapter 6


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## 7. Classical String Model

In the last two chapters, particle production was considered as a tunneling process in a strong field, or as a dynamical excitation of an external polarizing field in two-dimensional QED. Although these basic considerations provide much insight into the dynamics of the production process, their applications to practical problems have been limited by the inability to describe many different produced mesons as stable states and to incorporate the transverse degrees of freedom. For phenomenological applications, there are some merits in the classical string model which describes mesons as string segments executing longitudinal expansion and contraction, and the process of particle production is represented by the fragmentation of a stretching string formed by a separating quark and antiquark pair. This type of model [1-3] has enjoyed a great deal of success and is useful for many applications. We shall describe the classical string model for mesons and for particle production.

### §7.1 The String Model of Hadrons

The notion that hadrons can be described as strings originated in the late 1960's from the study of the dual resonance model [4] which was later interpreted in terms of the scattering of strings [5-7]. In the dual resonance model, one calculates the amplitude for the elastic scattering of mesons with initial momenta $p_1$ and $p_2$ to end up as mesons with final momenta $p_3$ and $p_4$: $p_1 + p_2 \to p_3 + p_4$. In the lowest-order Feynman diagram, this can be done in two different ways, as shown in Fig. (7.1$a$) and (7.1$b$). In one way, as indicated in Fig. (7.1$a$), one allows $p_1$ and $p_2$ to interact to lead to a particle (or a resonance) $p_J$ which subsequently breaks up into $p_3$ and $p_4$. As the momentum of $p_J$ in Fig. (7.1$a$) is equal to the sum $p_1 + p_2$ and $(p_1 + p_2)^2 = s$, a diagram of the type (7.1$a$) is called an $s$-channel diagram. There is a diagram (7.1$a$) for each value of the spin $J$ of the intermediate particle $p_J$. The $s$-channel scattering amplitude is the sum of all the amplitudes coming from all the diagrams of the type in Fig. (7.1$a$) with different values of the spin $J$. Another way of calculating the amplitude describes the process as the exchange of a particle $p_J$ from the particle $p_1$ to $p_2$ as they propagate forward in time, leading to the final particles $p_3$ and $p_4$, as indicated by Fig. (7.1$b$). Because the momentum $p_J$ is equal to $p_1 - p_3$ and $(p_1 - p_3)^2 = t$,

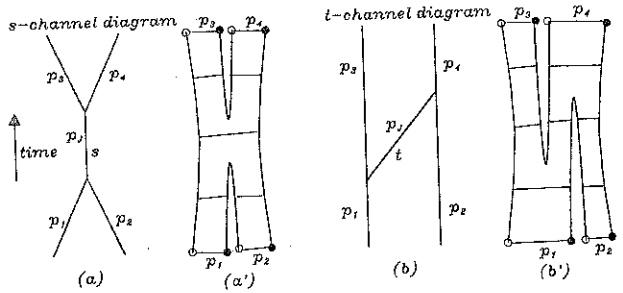



**Fig. 7.1** ($a$) The $s$-channel diagrams for elastic meson-meson scattering. ($a'$) The $s$-channel diagram for elastic meson-meson scattering when mesons are assumed to be strings. The configuration of the strings (mesons) at different proper times is represented by thin lines. The solid and the open circles represent a quark and an antiquark respectively. ($b$) The $t$-channel diagram for elastic meson-meson scattering. ($b'$) The $t$-channel diagram for elastic meson-meson scattering when mesons are assumed to be strings.

a diagram of the type in (7.1$b$) is called a $t$-channel diagram. There is a $t$-channel diagram (7.1$b$) for each value of the spin $J$ of the exchanged particle $p_J$. The $t$-channel scattering amplitude is the sum of all the amplitudes coming from all the $t$-channel diagrams with different values of $J$.

In the dual resonance model, it was found that the $s$-channel amplitude and the $t$-channel amplitude are the same. The $s$- and the $t$-channel diagrams give two alternative, or 'dual', descriptions of the same physics. This 'duality' property is the origin of the name 'dual resonance model'. A quantitative introduction to the dual resonance model will be presented in Chapter 8.

A very simple way the duality property can be realized is to assume that a meson is not a point object but an extended object, with the structure of a string connecting a quark and an antiquark as shown in Fig. (7.1$a'$) and Fig. (7.1$b'$). In terms of strings, the $s$-channel diagram Fig. (7.1$a$) for mesons as point particles is actually a configuration of the form in Fig. (7.1$a'$) if the mesons are strings. Similarly, the $t$-channel diagram in Fig. (7.1$b$) for point-like mesons becomes Fig. (7.1$b'$) if the mesons are described as strings. Because Figs. (7.1$a'$) and (7.1$b'$) have the same topology, they are just the same diagram in terms of quarks and antiquarks. The differences of the $s$-channel and $t$-channel diagrams in terms of point-like mesons are merely two manifestations of the same quark-antiquark diagram (Fig. (7.1$a'$) or



Fig. (7.1$b'$)) in two different Lorentz frames. The $s$-channel amplitude is therefore the same as the $t$-channel amplitude. It is in this way that the dual resonance model led to the suggestion of a string description of mesons.

this will provide some motivation for the next section too and is the beginning of understanding the pomeron

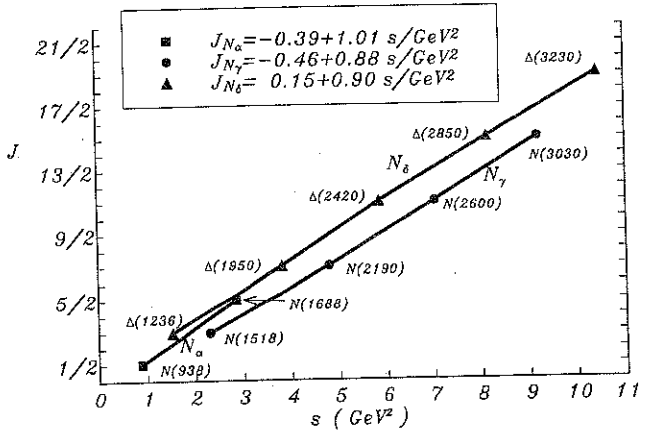



**Fig. 7.2** The relation between $J$ and $s(= M^2)$ for baryons (from Ref. 8).

A string picture of hadrons also gives the appropriate relationship between the mass $M$ of the hadron and its spin $J$. It was found empirically that hadron resonances with the same internal symmetry quantum numbers but different spins fall into straight lines when one plots $J$ as a function of $s(= M^2)$. These straight lines are called *Regge trajectories* (Fig. 7.2). That is, $J$ is related to $M$ by

$$J(M) = \alpha(0) + \alpha' M^2 \,, \tag{7.1a}$$

where $\alpha(0)$ is called the *Regge intercept* and $\alpha'$ the *Regge slope*. The Regge slopes for three different sets of baryons can be read off from the relation between $J$ and $s$ in Fig. 7.2. It appears that the Regge slope is approximately a universal constant with a value

$$\alpha' \approx 1 \ \ \mathrm{GeV}^{-2} \,. \tag{7.1b}$$

Such a relation between the spin and the mass can be obtained in the model of a spinning string. We can consider a string of length $2L$ rotating about the axis passing through its center $O$ (see Fig. 7.3) and

perpendicular to the rotating string. The end points of the string are attached to a massless quark (solid circle) and a massless antiquark (open circle) traveling with the speed of light in the direction perpendicular to the string. We assume a rigid-body rotation. The string segment $dx$ at the point $x$ has a speed $v = x/L$. If the string were at rest, the energy stored in the segment $dx$ would be $\kappa\, dx$, where $\kappa$ is the string tension coefficient. For the string in rotation, the energy content in the segment $dx$ is $\gamma\kappa dx$, where $\gamma = 1/\sqrt{1-v^2}$ is the gamma factor for this string segment in motion. The total energy content of the string is therefore

$$\begin{aligned} M =& 2\int_0^L \gamma\kappa dx \\ =& 2\int_0^L \frac{\kappa dx}{\sqrt{1-(x/L)^2}} \\ =& \pi\kappa L\,. \end{aligned}$$

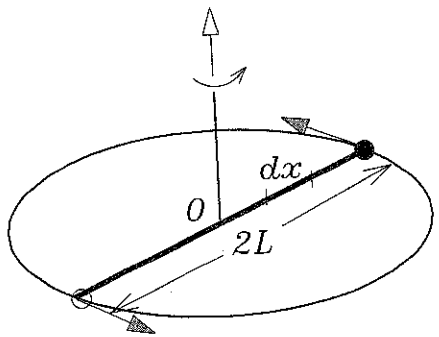



**Fig. 7.3** A rotating relativistic string.

On the other hand, the angular momentum content of the segment $dx$ is its energy content multiplied by $vx$. Hence, the total angular momentum of the string is

$$\begin{aligned} J =& 2\int_0^L \gamma\kappa v x dx \\ =& 2\int_0^L \frac{\kappa v x dx}{\sqrt{1-(x/L)^2}} \\ =& \kappa L^2\pi/2 \end{aligned}\,.$$



From these relationships, we have

$$J = \frac{1}{2\pi\kappa}M^2\,. \tag{7.2}$$

Thus, a string description of the structure of hadrons gives the proper relationship between the angular momentum and the mass as in Eq. (7.1a). In this picture, a comparison of Eqs. (7.1a) and (7.2) shows that the string tension $\kappa$ is related to the Regge slope $\alpha'$ by

$$\kappa = \frac{1}{2\pi\alpha'} \approx 1 \ \text{GeV/fm}\,. \tag{7.3}$$

It should be pointed out that, although strings may be used as an *effective* description of strong interaction phenomena in connection with the large-distance structure of hadrons, we now know that the string theory cannot be used as a *fundamental* model of the strong interaction. There is a general consensus that the fundamental theory for the strong interaction is the theory of quantum chromodynamics. The field theory of strings for strong interactions has the problem that it predicts a massless vector particle $\rho$ and a massless tensor particle. Any attempt to move these states to positive masses leads to other types of difficulties such as negative norms. The string theory of strong interaction also has difficulties in explaining many hard processes with large transverse momentum transfers observed in deep inelastic electron scattering experiments which led to the concept of partons in strong interactions. It has been suggested in the superstring theory that the field theory of strings may instead describe the structure of fundamental particles such as quarks, gluons, leptons, $W^{\pm}$, photons, and gravitons [9].

When we deal with the production of hadrons with a transverse momentum of a few hundred MeV/c, one is studying the strong interaction at a large distance scale. A phenomenological description in terms of strings provides much insight into the dynamics of the system. With the advent of the theory of QCD for strong interactions, and the realization that the color electric field between a quark and an antiquark, separated by a large distance, is constrained to reside mainly between them in the form of a flux tube [10,11], the string model of a meson is intuitively a reasonable concept. Much insight into the mechanism of particle production is provided by examining classical strings in one space and one time dimensions. They are useful as a simple representation of mesons, as well as highly excited $q\bar{q}$ systems. In this description, a meson at rest in its center-of-mass system is depicted as a string which has a quark and an antiquark at its two ends, the quark and the antiquark expanding and contracting in a yo-yo motion. A moving meson is described as a string with a translational motion superimposed on the intrinsic yo-yo motion. An

excited $q\bar{q}$ system is described as a string whose quark and antiquark at its two ends move apart with a large momentum. The excited string fragments into hadrons with new $q\bar{q}$ pairs created between the separating quark and antiquark. The classical string model is capable of describing many aspects of the high-energy particle production process.

## §7.2 The Yo-Yo Model of Stable Particles

We would like to begin our discussion of a simple string system which can be considered as a classical representation of a stable meson [1-3]. The description is simplest in the center-of-mass system in which the meson is at rest. The string is characterized by a quark and an antiquark located at the two ends of the string and interacting with a linear potential. It is convenient to take the quark and the antiquark to be massless so that their trajectories can be easily followed.

The Hamiltonian for this system of massless $q\bar{q}$ pair is

$$H(x_i, p_i) = |p_q| + |p_{\bar{q}}| + \kappa |x_q - x_{\bar{q}}| , \tag{7.4}$$

where $\kappa$ is the string tension, and $p_i$ and $x_i$ denote, respectively, the (longitudinal) momentum and the (longitudinal) spatial coordinate of the particle $i$, with $i = q$ and $\bar{q}$. The motion of the quark $q$ and the antiquark $\bar{q}$ can be determined from the Hamilton equations:

$$\dot{x}_i = \frac{\partial H}{\partial p_i} , \tag{7.5}$$

and

$$\dot{p}_i = -\frac{\partial H}{\partial x_i} . \tag{7.6}$$

From Eq. (7.5), we find that the velocity of the particle $i$ (in units of the speed of light) is governed by

$$\dot{x}_i = \text{sign}(p_i) . \tag{7.7}$$

Thus, the direction of motion of the particle $i$ is governed by the sign of its momentum. The spatial coordinate of the particle $i$ is

$$x_i(t) = x_i(t_0) + \text{sign}(p_i)(t - t_0) , \tag{7.8}$$

where $t_0$ is the most recent time when the quark and the antiquark meet together at $x_i = 0$. From the Hamilton equation (7.6), the rate of change of the momentum of the particle $i$ is determined by the sign of $x_i - x_{i'}$:

$$\dot{p}_i = -\text{sign}(x_i - x_{i'})\kappa , \tag{7.9}$$



where $i'$ is different from $i$. The rate of change of the magnitude of the momentum of the quark or the antiquark is governed by the string tension $\kappa$. The solution for the momentum of the $i$th particle is

$$p_i(t) = p_i(t_0) - \text{sign}(x_i - x_{i'})\kappa(t - t_0) . \tag{7.10}$$

We consider the initial situation where both the quark and the antiquark are located at the spatial origin at $t = 0$. Based on the above results, we have the following description for the motion of the quark and the antiquark which characterize the two ends of the string, as shown in Fig. (7.4).

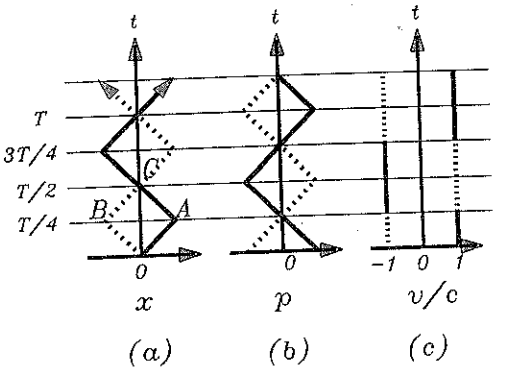



**Fig. 7.4** (*a*). The space-time trajectories of the quark (the solid line) and the antiquark (the dashed line) for a stable meson (the yo-yo state) at rest. The time $T$, given by Eq. (7.13), is the period of the yo-yo state in its rest frame. (*b*) The time variation of the momentum of the quark (the solid line) and the antiquark (the dashed line) for the yo-yo state. (*c*) The time variation of the velocity of the quark (the solid line) and the antiquark (the dashed line) for the yo-yo state.

Without loss of generality, we consider initially that the quark is moving to the right with an initial momentum $p_q(0) > 0$, and the antiquark is moving to the left with an initial momentum $p_{\bar{q}}(0) < 0$ and $p_q(0) = -p_{\bar{q}}(0)$. During the initial stage, $p_q(t) > 0$ and $p_{\bar{q}}(t) < 0$, the Hamilton equation (7.5) gives

$$\dot{x}_q = \text{sign}(p_q) = +1 \tag{7.11a}$$

and

$$\dot{x}_{\bar{q}} = \text{sign}(p_{\bar{q}}) = -1 . \tag{7.11b}$$

Thus, initially, the quark has a positive momentum and travels in the $+x$ direction, while the antiquark has a negative momentum and travels in the $-x$ direction, as shown in Figs. (7.4*a*) and (7.4*c*). Because these particles are massless, they travel with the speed of light.

In the initial stage, the motion of the quark and the antiquark pair leads to $x_q > x_{\bar{q}}$ (see Fig. (7.4*a*)). The Hamilton equation (7.6) for $\dot{p}_i$ then gives

$$\dot{p}_q = -\text{sign}(x_q - x_{\bar{q}})\kappa = -\kappa \tag{7.12a}$$

and

$$\dot{p}_{\bar{q}} = -\text{sign}(x_{\bar{q}} - x_q)\kappa = +\kappa\,. \tag{7.12b}$$

Therefore, as the quark and the antiquark separate from each other, the quark which has a positive momentum moves in the $+x$ direction, with its momentum diminishing linearly with time (see Fig. (7.4*b*)). It is pulled by the antiquark, by a force pointing in the $-x$ direction. As determined by Eq. (7.12a), the momentum of the quark is zero at the time $p_q(0)/\kappa$ which is one quarter of the period $T$ for the motion of the quark or the antiquark in this center-of-mass frame. Thus, the period $T$ is given by

$$T = 4p_q(0)/\kappa\,. \tag{7.13}$$

After $t > T/4$, the momentum of the quark will be negative and its direction of motion will be reversed. It will turn around and travel in the $-x$ direction toward the antiquark. It experiences an attractive pull from the antiquark. The magnitude of its momentum increases linearly with time, from a value of zero at $t = T/4$ to reach its maximum value at $x = 0$ and $t = T/2$. After $t > T/2$, the quark moves away from the antiquark and experiences a pull from the antiquark to slow down. The magnitude of its momentum decreases linearly with time, until the momentum becomes zero at $t = 3T/4$. The behavior of the antiquark is just a mirror image of that of the quark. The time variations of the momentum of the quark and the antiquark are shown in Fig. (7.4*b*).

Combining the motion of the quark and the antiquark, we can describe the dynamics of the string as follows. Initially at $t = 0$, the quark and the antiquark are located at point $O$ at $x = 0$ (Fig. (7.4*a*)). As time increases, the quark moves in the $+x$ direction, while the antiquark moves in the $-x$ direction. The string expands. The string reaches its maximum extension when the quark and the antiquark reach their respective turning points $A$ and $B$ at time $t = T/4$. Thereafter, the direction of motion of two particles reverses and the string contracts until the quark and the antiquark meet each other at point $C$, at $x = 0$ and $t = T/2$. This completes a half cycle of the



expansion and contraction motion. The expansion and contraction motion will continue and will repeat themselves. This type of motion is called a yo-yo motion. A string which undergoes a yo-yo motion is said to be in a *yo − yo state*.

In the center-of-mass system, the $(t, x)$ coordinates for the points $O, A, B$, and $C$ in Fig. 7.4 are given by

$$\begin{aligned} O &= (0,0) \\ A &= (T/4, T/4) \\ B &= (T/4, -T/4) \end{aligned} \tag{7.14}$$

and

$$C = (T/2, 0)\,.$$

With the results of Eqs. (7.8) and (7.10) given above, it is easy to show that the total energy of the system $H$ is a constant of motion, independent of time. We can evaluate the total energy $H$ at time $t = 0$ at which the separation between the quark and the antiquark is zero. The total energy is

$$H = 2|p_q(0)|\,. \tag{7.15}$$

We identify this energy $H$ as the rest mass $m$ of the stable meson. Since it is equal to the invariant mass $\sqrt{s}$ of the $q\bar{q}$ system at the moment when the quark $q$ and the antiquark $\bar{q}$ meet each other, we shall also call this quantity the invariant mass of the yo-yo system.

It is convenient to introduce the forward light-cone space-time coordinate $u$, and the backward light-cone space-time coordinate $v$. They are related to coordinates $t$ and $x$ by

$$(u, v) = (t + x, t - x)\,. \tag{7.16}$$

The $(u, v)$ coordinates of the points $O, A, B$, and $C$ are

$$\begin{aligned} O &= (\ \ 0,\ \ 0) \\ A &= (\ T/2,\ \ 0) \\ B &= (\ \ 0,\ T/2) \end{aligned} \tag{7.17}$$

and

$$C = (T/2, T/2)\,.$$

That is, $A$ lies on the axis $v = 0$ and $B$ lies on the axis of $u = 0$. In the $(u, v)$ coordinate system, the area $OACB$ enclosed by the quark and antiquark trajectories in a single half-cycle is

$$\begin{aligned} (\text{area } OACB) = OA \times OB &= T^2/4 \\ &= 4|p_q(0)|^2/\kappa^2 \\ &= s/\kappa^2\,. \end{aligned} \tag{7.18}$$

In subsequent expansions and contractions, the area enclosed by the quark and the antiquark trajectories in each half cycle is the same as that of $OACB$. This area in the $(u, v)$ coordinate system can be called an elementary area for the yo-yo state in question. From Eq. (7.18), we see that the elementary area enclosed by the trajectory of the quark and the antiquark lines is equal to the square of the invariant mass of the yo-yo state divided by $\kappa^2$. Therefore, in the classical string model, a meson is represented by a yo-yo state whose elementary area is equal to the square of the meson mass divided by $\kappa^2$.

you can probably skip this section

## §7.3 A Yo-Yo State in Motion

How do we describe a yo-yo state in translational motion? This is achieved by first following the yo-yo state in its center-of-mass system where the trajectories of the quark and the antiquark can be easily determined. This frame of reference is then boosted with a velocity $\beta$ relative to the laboratory frame. The points $O, A, B$, and $C$ for a yo-yo state at rest become $O, A', B'$, and $C'$ in the laboratory frame (see Fig. 7.5). The latter laboratory coordinates $(t, x)$ can be obtained from the coordinates of Eq. (7.14) by a Lorentz transformation. They are

$$\begin{aligned} O &= (\quad 0, \quad 0) \\ A' &= (\ \gamma(1+\beta)T/4, \quad \gamma(1+\beta)T/4) \\ B' &= (\ \gamma(1-\beta)T/4, \quad -\gamma(1-\beta)T/4) \end{aligned} \tag{7.19}$$

and

$$C' = (\gamma T/2, \quad \gamma\beta T/2)\,,$$

where $\gamma = 1/\sqrt{1-\beta^2}$. The velocity of the yo-yo in the laboratory system is determined by the ratio of the $x-$coordinate of $C'$ to the $t-$coordinate of $C'$, which is given, as expected, by

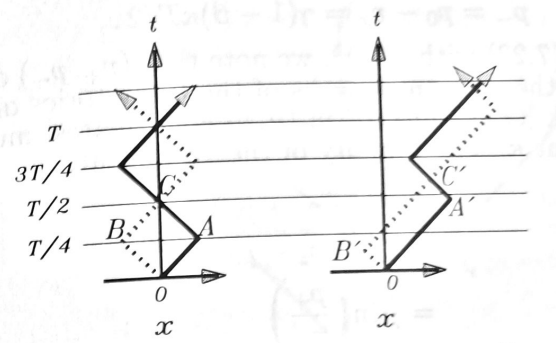
$$(\text{velocity of the yo} - \text{yo state}) = \frac{\gamma\beta T/2}{\gamma T/2} = \beta\,. \tag{7.20}$$

The trajectories for the boosted quark and the antiquark take on the form as shown in Fig. (7.5$b$).

In terms of the $(u, v)$ coordinates, the points $A'$ and $B'$ for the yo-yo in motion are

$$\begin{aligned} O &= (\quad 0, \quad 0) \\ A' &= (\quad \gamma(1+\beta)T/2, \quad 0) \\ B' &= (\quad 0, \quad \gamma(1-\beta)T/2) \end{aligned} \tag{7.21}$$






yo–yo at rest (a) yo–yo in motion (b)

**Fig. 7.5** ($a$). The space-time trajectories of the quark (the solid line) and the antiquark (the dashed line) for a yo-yo state at rest. ($b$) The space-time trajectories of the quark (the solid line) and the antiquark (the dashed line) for a yo-yo state in motion.

and

$$C' = (\gamma(1+\beta)T/2, \gamma(1-\beta)T/2)\,.$$

For the yo-yo state in motion, the elementary area $OA'C'B'$ enclosed by the quark and the antiquark trajectories in the $(u, v)$ coordinate system is

$$(\text{area } OA'C'B') = OA' \times OB' = T^2/4 = s/\kappa^2\,, \tag{7.22}$$

which is equal to the area $OACB$ for the yo-yo state at rest. Thus, the elementary area is an invariant quantity which does not depend on the motion of the yo-yo state. It is equal to the square of the invariant mass of the system divided by $\kappa^2$.

Strong interaction phenomena are characterized by a large number of stable particles and resonances. The ability to describe these stable particles as expanding and contracting strings is one of the advantages of the classical string model. The production of particles is represented in terms of the break-up of a string with a large invariant mass into smaller fragments with the masses of the stable mesons.

Knowing the velocity of the yo-yo state in the laboratory system, one can calculate the energy and the momentum of the yo-yo state as a single particle. These quantities give the forward and backward light-cone momenta $p_+$ and $p_-$ of the yo-yo state. We find

$$p_+ = p_0 + p_x = \gamma(1+\beta)\kappa T/2 \tag{7.23$a$}$$

and
$$p_- = p_0 - p_x = \gamma(1-\beta)\kappa T/2\,. \tag{7.23b}$$

Comparing Eqs. (7.23) with (7.21), we note that $(p_+, p_-)$ of the yo-yo state is given by the segment lengths of the trajectories of the quark and the antiquark [$OA'$ and $OB'$ in $(u,v)$ coordinates] multiplied by the string constant $\kappa$. The rapidity of the yo-yo state is

$$\begin{aligned} y &= \frac{1}{2}\ln\left(\frac{p_z+p_0}{p_z-p_0}\right) \\ &= \frac{1}{2}\ln\left(\frac{p_+}{p_-}\right) \\ &= \frac{1}{2}\ln\left(\frac{1+\beta}{1-\beta}\right), \end{aligned}$$

which is the same as the rapidity given by Eq. (1) of Exercise 2.4, for a single particle travelling with longitudinal velocity $\beta$.

skim this one again though

## §7.4 The Symmetric Lund Model

We shall examine here the symmetric Lund model [3] which gives the probability distribution function for the locations of the vertices at which the string is broken into separate pieces. The basic assumption of the symmetric Lund model is that the vertices at which the quark and antiquark pairs are produced lie approximately on a curve of constant proper time. This characteristics will lead to a rapidity distribution of the produced particles to be independent of the rapidity for an infinite-energy system (see Exercise 7.1) and is consistent with the experimental feature of a plateau shape for the rapidity distribution in finite-energy systems (Fig. 3.2). With additional simplifying assumptions on the functional dependence of the splitting function and the independence on the order of splitting, an explicit form of the distribution function for the locations of the vertices is then obtained.

To set up our framework, we consider a string system consisting of a pair of massless quark $q$ and antiquark $\bar{q}$ with large forward and backward light-cone momenta respectively (Fig. 7.6). At the space-time origin $(t,x) = (0,0)$, the quark $q$ of the string meets with, and begins to separate from, the antiquark $\bar{q}$ of the string. We shall work in the center-of-mass frame in which the initial momentum of the quark $p_q(0)$ is equal and opposite to the initial momentum of the antiquark $p_{\bar{q}}(0)$. The invariant mass of the system is $\sqrt{s} =$



slope of all lines = c

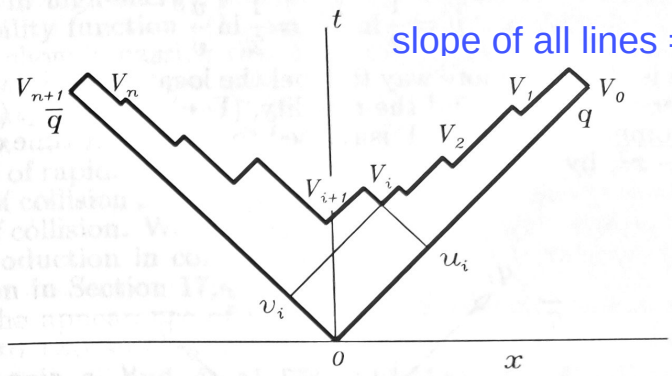



**Fig. 7.6** The space-time trajectories of quarks and antiquarks from a string fragmentation.

$2|p_q(0)|$. Without loss of generality, we can take $p_q(0) > 0$. As time proceeds, the massless quark $q$ traverses to the positive-$x$ direction, while the massless antiquark $\bar{q}$ traverses to the negative-$x$ direction. A string is formed joining the quark and the antiquark. When the $q\bar{q}$ pair separates, the string is stretched and new $q_i\bar{q}_i$ pairs will be spontaneously produced at various vertices $V_i$ (see Fig. 7.6). A quark $q_i$ produced at vertex $V_i$ will subsequently combine with an antiquark $\bar{q}_{i+1}$ from a neighboring vertex $V_{i+1}$ to form a yo-yo state, which can be considered to represent a hadron in the inside-outside cascade description of particle production, as shown previously in Fig. (6.4).

A vertex $V$ is specified by the light-cone space-time coordinates $(u,v) = (t+x, t-x)$. From the classical motion of a string fragment discussed in the last section, we note from Eqs. (7.23) and (7.21) that the forward and the backward light-cone momenta of a yo-yo state are directly proportional to the segment lengths of the trajectories of the quark and the antiquark in the $(u,v)$ coordinates. It is therefore convenient to label the location of the vertex $V$ alternatively by the forward and the backward light-cone momenta $(p_+, p_-)$ defined by

$$(p_+, p_-) \equiv (\kappa u, \kappa v)\,, \tag{7.24}$$

which differ from $(u,v)$ only by the multiplicative scale factor $\kappa$. These quantities can be written in terms of the square of the proper momentum $\Gamma$, and the rapidity $y$ as

$$\Gamma = p_+p_- = \kappa^2 uv = \kappa^2(t^2 - x^2)\,, \tag{7.25a}$$

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and

$$y = \frac{1}{2}\ln\frac{p_+}{p_-} = \frac{1}{2}\ln\frac{u}{v}\,. \tag{7.25b}$$

There is thus one more way to label the location of the vertex $V$ by the proper momentum and the rapidity, $(\Gamma, y)$. From Eq. (7.25a), the proper momentum variable $\Gamma$ is related to the proper time coordinate, $\tau = \sqrt{t^2 - x^2}$, by

$$\sqrt{\Gamma} = \kappa\tau\,. \tag{7.26}$$

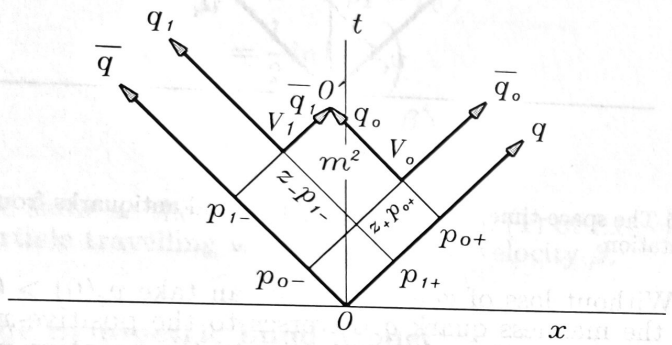



**Fig. 7.7** The space-time trajectories of quarks and antiquarks from a string fragmentation at vertices $V_0$ and $V_1$.

We consider the occurrence of two adjacent vertices $V_0$ and $V_1$ as shown in Fig. (7.7). The order of occurrence of the vertices depends on the frame of reference. One can boost the system to a frame of reference such that the vertex $V_0$ precedes $V_1$, or to another frame of reference in which the vertex $V_1$ precedes $V_0$.

We shall label the location of the vertex $V_0$ by $(\Gamma_0, y_0)$. The probability element for the occurrence of a vertex producing the pair $q_0$ and $\bar{q}_0$ at the point $V_0$, specified by the proper momentum $\Gamma_0$ and the rapidity $y_0$, is given by

$$dP_0 = \rho(\Gamma_0, y_0)d\Gamma_0 dy_0\,. \tag{7.27}$$

In particle production processes at very high energies, it is observed experimentally that the rapidity distribution of the produced particles $dN/dy$ has a plateau shape (Fig. 3.2). On the other hand, it is expected theoretically that if all the production vertices fall on a curve of a constant proper time $\tau_0$ (or proper momentum $\Gamma_0 = \kappa\tau_0$), the rapidity distribution of the produced particles will be independent of the rapidity, and $dN/dy$ will be a constant (see Exercise 7.1).



Therefore, in high-energy processes, it is reasonable to assume that the probability function $\rho(\Gamma_0, y_0)$ is a function of $\Gamma_0$ only.

We can show in passing that if all the production vertices fall on a curve of a constant proper time $\tau_0$, the produced particles are ordered in space-time (see Exercise 7.2). The ordering of produced particles means that in the center-of-mass system, particles with a greater magnitude of rapidity $|y|$ are produced at a distance farther away from the point of collision and at a time later than and more separated from the time of collision. We shall come back to discuss these properties of particle production in connection with pion interferometry in $e^+$-$e^-$ annihilation in Section 17.4 of Chapter 17.

After the appearance of the vertex $V_0$, the string is further fragmented into two smaller strings with the production of the quark-antiquark pair $q_1$ and $\bar{q}_1$ at the vertex $V_1 = (p_{1+}, p_{1-})$. In the configuration of Fig. (7.7), the set of vertices $V_0$, and $V_1$ occur in the sequence from right to left. Before the quark-antiquark pair is produced at the vertex $V_1$, the coordinates $(\Gamma_0, y_0)$ or the light-cone momentum corrdinates $(p_{0+}, p_{0-})$ of vertex $V_0$ are presumed known. Thus, the coordinates of $V_1$ can be specified by the forward light-cone momentum fraction $z_+$

$$z_+ = (p_{0+} - p_{1+})/p_{0+}\,, \tag{7.28}$$

and the square of the invariant mass of the produced hadron:

$$M_{01}^2 = (p_{0+} - p_{1+})(p_{1-} - p_{0-})\,.$$

We can use the quantities $z_+$ and $M_{01}^2$ to specify the location $(p_{0+}, p_{0-})$ of vertex $V_1$.

We introduce the probability distribution $f_{01}$ so that the probability element for the occurrence of this vertex at $V_1$ is

$$dP_1 = f_{01}(z_+, M_{01}^2)g(M_{01}^2)dz_+ dM_{01}^2\,, \tag{7.29}$$

where the ordering of the subscript 01 in $f_{01}$ and $M_{01}$ indicates the first occurrence of vertex $V_0$, to be followed by the occurrence of the vertex $V_1$. In general, $f_{01}$ depends on $z_+, M_{01}^2$, and $\Gamma_0$ [12]. The assumption that $f$ depends only on $z_+$ and $M_{01}^2$ allows one to obtain an explicit form of the distribution by solving well-defined equations [3].

The invariant mass $M_{01}$ formed between the vertices $V_0$ and $V_1$ is given by a distribution $g(M_{01}^2)$ which depends on the dynamics of the problem. We can consider the case where $M_{01}^2$ is determined from some other considerations to have a certain set of values $m_\lambda^2$, with probabilities $c_\lambda$ for the occurrence of the mass $m_\lambda$:

$$g(M_{01}^2) = \sum_\lambda c_\lambda \delta(M_{01}^2 - m_\lambda^2)\,. \tag{7.30}$$

JETSET IS PART OF PYTHIA



For example, in the Lund model, as implemented in the computer program JETSET [13], one finds from a given probability distribution what particle will be produced next. From a transverse momentum distribution, the transverse momentum and the transverse mass of the produced particle is determined. The transverse mass is then identified as the mass $m$ of the particle produced by the occurrence of that vertex.

Combining Eqs. (7.27) and (7.29), the joint probability element for the occurrence of both vertices $V_0$ and $V_1$ is

$$dP = dP_0 \cdot dP_1 = \rho(\Gamma_0) f_{01}(z_+, M_{01}^2) g(M_{01}^2) d\Gamma_0 dy_0 dz_+ dM_{01}^2 . \quad (7.31)$$

On the other hand, the occurrence of these two vertices in Fig. 7.7 can proceed by the occurrence of the vertex $V_1$ first, and the vertex $V_0$ subsequently. Following the preceding arguments as before and interchanging the labels 0 and 1, the combined probability element for this sequence of events is

$$dP' = dP_1' \cdot dP_0' = \rho(\Gamma_1) f_{10}(z_-, M_{10}^2) g(M_{10}^2) d\Gamma_1 dy_1 dz_- dM_{10}^2 , \quad (7.32)$$

where $(\Gamma_1, y_1)$ are respectively the square of the proper momentum and the rapidity of $V_1$, and

$$z_- = (p_{1-} - p_{0-})/p_{1-} . \quad (7.33)$$

The quantity $M_{01}$ is equal to $M_{10}$, as the invariant mass is independent of the order of the occurrence of the two vertices. The symmetric Lund model makes the observation that if the vertices are indistinguishable, the different ordering of the occurrences of the two vertices represent the same physical process in different Lorentz frames, and the joint probability should be symmetrical with respect to the ordering of the vertices $V_0$ and $V_1$. The joint probability should fulfill the following "left-right symmetry" condition:

$$dP = dP' . \quad (7.34)$$

From this condition, Andersson *et al.* obtained the splitting function [3]

$$f(z_+, m^2) = N z_+^{-1} (1 - z_+)^a e^{-bm^2/z_+} \quad (7.35)$$

and

$$\rho(\Gamma) = C \Gamma^a e^{-b\Gamma} , \quad (7.36)$$

where the constant $N$ and $C$ are normalization constants, and we have used the notation $m$ for $M_{10}$, (see Exercise 7.3).



If the vertices are distributed according to (7.36), the average proper time at which the vertices occur is then

$$< \tau >= \frac{\Gamma(a + 1 + \frac{1}{2})}{\kappa \sqrt{b} \Gamma(a + 1)} , \quad (7.37)$$

where $\Gamma$ in the above equation is the gamma function (not to be confused with the $\Gamma$'s in Eq. (7.36)). The standard deviation of the proper formation time for the distribution (7.36) is

$$\sigma_\tau = \sqrt{\frac{(a+1)}{\kappa^2 b} - < \tau >^2} . \quad (7.38)$$

Thus, the values of the parameters $a$ and $b$ give information about the vertex occurrence time and the particle formation time.

starting here is an exercise you can skip

•[ Exercise 7.1
Show that if all the pair-production vertices of a fragmenting string fall on the curve of the proper time $\tau = \tau_0$, the rapidity distribution of the produced particles is a constant given by [12]

$$\frac{dN}{dy} = \frac{\kappa \tau_0}{m_T} .$$

◊Solution:
We consider vertices $V_{i-1}$ and $V_i$. The light-cone space-time coordinates of the vertices are $V_{i-1} = (u_{i-1}, v_{i-1})$ and $V_i = (u_i, v_i)$ (see Fig. 7.6). A particle $i$ is formed by combining the antiquark from vertex $V_{i-1}$ with the quark from vertex $V_i$. This particle is a yo-yo state which has a forward light-cone momentum $p_{i+} = \kappa(u_{i-1} - u_i)$ and a backward light-cone momentum $p_{i-} = \kappa(v_i - v_{i-1})$. The rapidity of this yo-yo particle $i$ is given according to Eq. (7.25b) by

$$y_i = \frac{1}{2} \ln\left(\frac{p_{i+}}{p_{i-}}\right) = \frac{1}{2} \ln\left(\frac{u_{i-1} - u_i}{v_i - v_{i-1}}\right) . \quad (1)$$

Similarly, the rapidity of the adjacent yo-yo particle $i+1$ formed by combining the antiquark from vertex $V_i$ with the quark from vertex $V_{i+1}$ is

$$y_{i+1} = \frac{1}{2} \ln\left(\frac{u_i - u_{i+1}}{v_{i+1} - v_i}\right) .$$

There is the addition of one produced particle in the interval between $y_i$ and $y_{i+1}$. Therefore, in the neighborhood of the vertex $V_i$, the rapidity density $\Delta N/\Delta y$ is

$$\frac{\Delta N}{\Delta y} = \frac{1}{y_{i+1} - y_i} = 2 \Big/ \left[ \ln\left(\frac{u_i - u_{i+1}}{v_{i+1} - v_i}\right) - \ln\left(\frac{u_{i-1} - u_i}{v_i - v_{i-1}}\right) \right] .$$

it goes on for a couple pages, I've not included...
However its an interesting execise, it's saying that the rapidity distribution of hadrons in the jet frame is flat which means its like ~how I described the y-distributions along the BEAM direction when we learned about rapidity -- (excluding the "beam spikes" ) -- universal behavior!

Because $|\sinh y|$ is a monotonically increasing function of $|y|$, we deduce that particles with a greater magnitude of rapidity $|y|$ are produced at a greater longitudinal distance $|x|$ from the point of collision. Furthermore, because $\cosh y$ is a monotonically increasing function of $|y|$, particles with a greater magnitude of rapidity $|y|$ are produced at a time $t$ later and more separated from the time of collision.. ]•

•[ Exercise 7.3
Prove that Eqs. (7.35) and (7.36) satisfy the symmetry condition (7.34).

◇Solution:
The validity of Eqs. (7.35) and (7.36) can be readily verified by substituting them into Eqs. (7.31) and (7.32). A change of variables and some simple geometrical considerations from Fig. 7.7 give the following relations for the independent variables:

$$\Gamma_0 = M_{01}^2(1-z_-)/(z_+z_-)\,, \tag{1}$$

$$\Gamma_1 = M_{01}^2(1-z_+)/(z_+z_-)\,, \tag{2}$$

and

$$\frac{dz_+dM_{01}^2}{z_+} = dp_{1+}dp_{1-} = d\Gamma_1 dy_1\,, \tag{3}$$

$$\frac{dz_-dM_{10}^2}{z_-} = dp_{0+}dp_{0-} = d\Gamma_0 dy_0\,. \tag{4}$$

After substituting Eqs. (1) and (3) into (7.31), we get

$$dP = NCM_{01}^{2a}\exp\{-b\frac{M_{01}^2}{z_+z_-}\}\left[\frac{(1-z_-)}{z_+}\right]^a\left[\frac{(1-z_+)}{z_-}\right]^a d\Gamma_0 dy_0 d\Gamma_1 dy_1 g(M_{10}^2)\,. \tag{5}$$

After substituting Eqs. (2) and (4) into (7.32), we obtain

$$dP' = NCM_{01}^{2a}\exp\{-b\frac{M_{01}^2}{z_+z_-}\}\left[\frac{(1-z_-)}{z_+}\right]^a\left[\frac{(1-z_+)}{z_-}\right]^a d\Gamma_0 dy_0 d\Gamma_1 dy_1 g(M_{10}^2)$$
$$= dP\,.$$

Therefore, Eqs. (7.35) and (7.36) satisfy the symmetry condition (7.34).

We note that $M_{01}^2/z_+z_-$ in Eq. (5) is $\kappa^2$ times the area formed by the rectangle with the diagonal $OO'$ (Fig. 7.7). The probability for the occurrence of the vertices is proportional to the exponential of this area. ]•

start here only till end of 7.4,

In practical Monte Carlo calculations with the Lund model, an iterative scheme is introduced [13] with an additional assumption. It is stipulated that a subsystem fragments independently, and the same splitting function $f(z,m^2)$ is used iteratively to generate all the vertices, starting from the known right-most vertex $V_0$ and the left-most vertex $V_{n+1}$ (see Fig. 7.6). The assumption of the same splitting

"independent frag. of each jet"



function for all fragmentations is the same as that in the outside-inside cascade picture of multiperipheral parton fragmentation, as was recognized earlier by Artru [14].

It should be pointed out that when one uses the splitting function $f(z_+)$ of Eq. (7.35) iteratively, starting with the left-most (or the right-most) vertex for which $\tau = 0$, the proper time for the first few vertices will not distribute themselves according to $\rho(\Gamma)$ of Eq. (7.36). Only by using $f(z_+)$ of Eq. (7.35) iteratively many times will one obtain vertices whose proper times coordinates will be distributed according to $\rho(\Gamma)$ of Eq. (7.36) and whose average vertices occurrence proper times are given by Eq. (7.37). Other types of fragmentation functions similar to Eqs. (7.35) and (7.36) have been proposed [2,12] in the literature.

stop here.

## §7.5 Lund Model for Nuclear Collisions

In the Lund model for hadron-hadron collisions [15,16], it is assumed that the hadrons are not transversely excited but are only longitudinally stretched. The collision of a beam hadron $b$ with a target hadron $a$ will result in two excited hadrons $b'$ and $a'$:

$$b + a \to b' + a'\,. \tag{7.39}$$

The longitudinally excited hadrons $b'$ and $a'$ subsequently decay. They are the sources of particle production. The decay of these stretched hadrons is analyzed in the same way as in the fragmentation of a string stretched between a $q$ and a $\bar{q}$ which has been discussed in the last section.

It is convenient to represent the energy-momentum of a hadron by the light-cone momenta. We introduce

$$(b_+, b_-) = (b_0 + b_z, b_0 - b_z)\,, \tag{7.40}$$

where the forward light-cone momentum $b_+$ is the sum of its energy and its longitudinal momentum, while the backward light-cone momentum $b_-$ is the difference (Eq. (2.9)). Without loss of generality, we can take the beam particle $b$ to be moving in the positive $z$ direction. The initial light-cone momenta of the beam particle $b$ are

$$b = (b_+, b_-) = (b_+, \frac{m_b^2}{b_+})\,, \tag{7.41}$$

where $b_+ \gg b_-$ . The initial light-cone momenta of the target particle $a$ are

$$a = (a_+, a_-) = (\frac{m_a^2}{a_-}, a_-)\,, \tag{7.42}$$

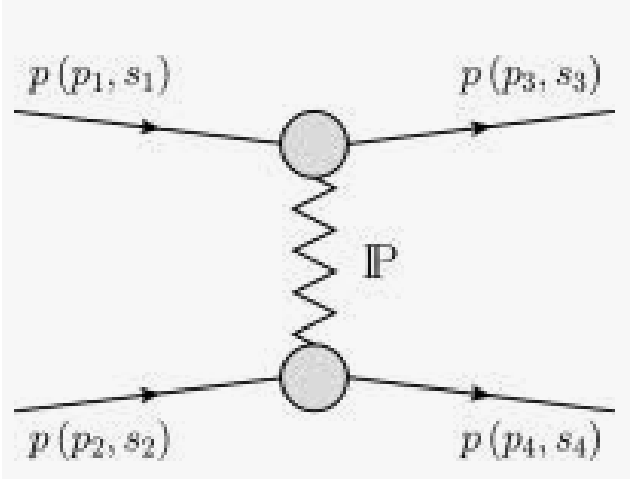
# Dual Parton Model

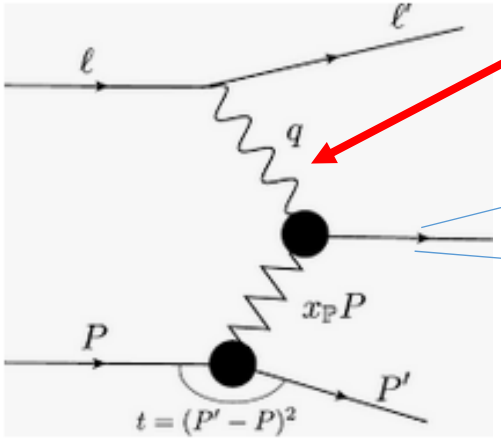
- Now the Dual Parton Model (next reading from Wong). This is the main part which explains Reggeon phenomenology and the first introduction to the Pomeron.
  - These are more like global properties/classifications, a little like an alternative to the Eightfold way before it was integrated into the quark model.
  - At this level, there aren't too many details e.g. no aspects of the mechanisms such as the diffractive diagrams shown previously in class (see next slide—I include again the slide I had on this when we discussed this in the first few weeks—**read it again.**)
- Although much more has been learned, interest/progress has been sporadic, starting only in the last ~20 years—starting with a resurgence especially with HERA. The EIC will probably yield a more complete understanding finally--we are only at the beginning of seeing how the Pomeron can be explained with perturbative tools such as factorization/structure functions.
- One aspect known for a long time is that because the photon in the process (see the red on the bottom right of next slide) is vector particle, the dominant production will be of VECTOR mesons. This is known as Vector Meson Dominance VMD.

# Case Study: Other Particles - Pomeron

Review

- Pomeron / Odderon /Regge-ons - Diffractive Physics - very odd "states"
- These are multigluon "states" that ~only act as force mediators of a sort, somewhat like pion is an effective force mediator for nucleons
- Usually these are not considered among the glueballs themselves e.g. predicted by the SM/Lattice QCD– more of an empirical data explainer
  - so like virtual force mediators particles only
- An early (<=70's) general classification of hadrons in scattering grouped them in a classifier type called Regge-ons → a theory that successfully explained the mysteriously rising .x.s. energy dependence through exchange of a Reggeon not observed elsewhere that had "vacuum quantum numbers" e.g. spin O
  - This x.s. rise is a non-perturbative QCD phenomenon
- More importantly these processes are "elastic" or pseudo elastic → despite high energy, all incoming hadrons are left intact by the scattering.
  - But ~usually, additional particles are produced
- The pomeron attains seemingly very real nature in some cases, because some of these processes involve e.g. jet production where it is thought that the quarks emerge from the breakup of the pomerons themselves.




Photon

breakup?- but not of incoming particles

most often though just a single vector meson

# Quick Digression: The Color Dipole Framework

- **IMPORTANT ASIDE:** Closely related to VMD is a family of perturbative theories known as the **Color Dipole Model.**
- In this model, which is an alternative to standard factorization in calcs of even regular DIS, the photon in a very similar way to the diffractive diagram, interacts with the incoming proton as a qqbar pair "fluctuation".
- A VERY LARGE amount of theoretical work quietly exists for this dipole model. In this model the dipole cross section sort of takes the place of the parton cross section in factorization, but oppositely, in this framework, it is this CROSS SECTION which is "global fit" non-perturbative part, while the hadronic structure part, analogous to the [usually non-perturbative] structure functions in regular factorization, are what can be and are perturbatively calculated.
  - It's a favorite side-gig of a lot of medium energy nuclear theorists?!?!
- Some references
  - http://home.thep.lu.se/~leif/ariadne/
  - https://www.mpi-hd.mpg.de/imprs-ptfs/pics/QCD-HD.pdf
  - (Kopeliovich is certainly its current "biggest" proponent & user – **"Color Transparency"** is one of its off-shoots)
  - https://inspirehep.net/literature/26426 ???

## 8. Dual Parton Model

In the last few chapters, we discussed different descriptions of the particle production process. We shall give here an alternative description of particle production based on the dual resonance model, which describes many characteristics of soft hadron physics. We shall briefly review the dual resonance model and the concepts of a Reggeon and a Pomeron, at an elementary level.

### §8.1 Dual Resonance Model

In Section 7.1, we gave a simple qualitative discussion of the dual resonance model in order to introduce the concept of the string model of hadrons. We shall present here a more quantitative discussion of the dual resonance model [1,2]. We recall that the basic assumption of the dual resonance model is that hadrons interact through the formation of intermediate states (resonances) as described by Fig. 8.1*a*, which is called an *s-channel scattering process*. They can also interact through the exchange of particles (or resonances) as depicted in Fig. 8.1*b*, which is called a *t-channel scattering process*. The variables $s = (p_1 + p_2)^2$ and $t = (p_1 - p_3)^2$ are the usual Mandelstam variables.

The intermediate particle $p_J$ in Fig. 8.1*a* or the exchanged particle $p_J$ in Fig. 8.1*b* can possess a spin. There are problems when this intermediate particle or exchange particle possesses a large spin. In a $t$-channel process (Fig. 8.1*b*), if the exchange particle has a spin $J$, the scattering amplitude $A$ at high energies grows as $s^J$ (see Supplement 8.1). The scattering amplitude and the total cross section grow without bounds if there is only the exchange of a single particle with a high spin $J$ at very high energies.

Similarly, in the $s$-channel process of Fig. 8.1*a*, if the intermediate state $p_J$ in a hadron-hadron scattering is a single particle (or a resonance) with a spin $J$, the scattering amplitude will likewise grow as $t^J$, which diverges for large values of $J$ and $t$. These types of asymptotic behavior for large $s$ and $t$ are not desirable.

Remarkably, there is a way to write down a scattering amplitude $A(s,t)$ which is asymptotically convergent for large values of $s$ (and fixed $t$) and represents the sum of amplitudes for the exchange of a whole set of particles with all possible spins. This is the Veneziano



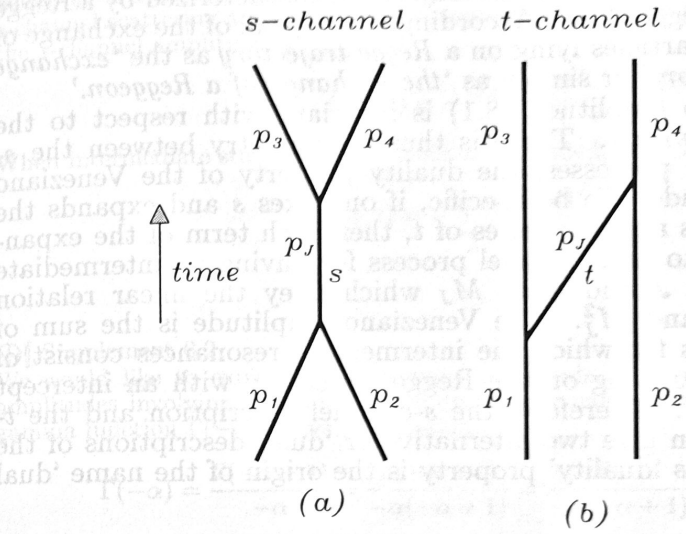



**Fig. 8.1** (*a*) An $s$-channel diagram for elastic meson-meson scattering. (*b*) A $t$-channel diagram for elastic meson-meson scattering.

scattering amplitude [1] given by

$$A(s,t) = g^2 \frac{\Gamma(-\alpha(s))\Gamma(-\alpha(t))}{\Gamma(-\alpha(s) - \alpha(t))}, \tag{8.1}$$

where $\alpha(x)$ is taken to depend linearly on $x$ as

$$\alpha(x) = \alpha_0 + \alpha' x \tag{8.2}$$

and $\Gamma$ is the gamma function.

This amplitude has many remarkable properties (see Supplement 8.2). If one keeps $t$ to be fixed and expands the amplitude (8.1) as a power series of $s$, then each term of the expansion corresponds to a $t$-channel Feynman diagram (8.1*b*) for the exchange of a particle with a spin $J$ and a mass $M_J$, which are related to each other by

$$J = \alpha_0 + \alpha' M_J^2 . \tag{8.3}$$

The set of particles whose spin $J$ and mass $M_J$ obey the relation (8.3) is said to lie on a *Regge trajectory* with a *Regge intercept* $\alpha_0$ and a *Regge slope* $\alpha'$. The amplitude (8.1) then represents the sum of amplitudes for the exchange of the set of particles lying on the Regge trajectory with intercept $\alpha_0$ and slope $\alpha'$. For brevity of notation, we often simply use the term 'the Regge trajectory' to represent the

whole set of particles lying on the trajectory, characterized by a Regge intercept and a Regge slope. Accordingly, we speak of the exchange of the whole set of particles lying on a *Regge trajectory* as the '*exchange of a Regge trajectory*' or simply as '*the exchange of a Reggeon*.'

The Veneziano amplitude (8.1) is invariant with respect to the interchange of $s$ with $t$. There is thus a symmetry between the $s$- and the $t$-channel processes, the duality property of the Veneziano scattering amplitude. To be specific, if one fixes $s$ and expands the amplitude (8.1) as a power series of $t$, then each term of the expansion corresponds to an $s$-channel process for having an intermediate resonance of spin $J$ and mass $M_J$ which obey the linear relation (8.3) between $J$ and $M_J^2$. The Veneziano amplitude is the sum of all the amplitudes for which the intermediate resonances consist of the set of particles lying on the Regge trajectory with an intercept $\alpha_0$ and a slope $\alpha'$. Therefore, the $s$-channel description and the $t$-channel description give two alternative or 'dual' descriptions of the same physics. This 'duality' property is the origin of the name 'dual resonance model.'

⊕⟦ Supplement 8.1 skip supplement 8.1

We would like to study here the Veneziano scattering amplitude for the $t$-channel diagram (8.1$b$) in perturbation theory, when the exchange particle possesses a spin $J$. The spin $J$ field is specified by $\sigma_{\mu_1\mu_2\ldots\mu_J}$, and the simplest coupling of the spin $J$ field to the particle field $\phi$ is

$$-g'_J\phi\partial^{\mu_1}\partial^{\mu_2}\ldots\partial^{\mu_J}\phi\sigma_{\mu_1\mu_2\ldots\mu_J}\,, \tag{1}$$

where $g'_J$ is a coupling constant which may depend on $J$. The scattering amplitude for the process $p_1+p_2\to p_3+p_4$ is obtained by taking the matrix element of the product of an operator of the form (1) associated with the vertex joining $p_1$ and $p_3$ to $p_J$ and another operator of the same form associated with the vertex joining $p_2$ and $p_4$ to $p_J$ (Fig. 8.1$b$). If there were no derivative operators in Eq. (1), the scattering amplitude would be simply $g_J^2/(t-M_J^2)$ [3], where $M_J$ is the rest mass of the exchanged particle with spin $J$. For each of the two vertices, each operator $\partial_\mu$ brings down a factor of the momentum $-ip_{1\mu}$ or $-ip_{2\mu}$. Hence for high energies, the scattering amplitude is proportional to $s^J$,

$$A_J(s,t)\propto-\frac{(g'_J)^2(-s)^J}{t-M_J^2}\,. \tag{2}$$

When one includes the exchanges of particles with all possible spins, we have

$$A(s,t)=-\sum_J\frac{g_J^2(-s)^J}{t-M_J^2}\,, \tag{3}$$



where $g_J$ is related to $g'_J$ by a constant. In a similar way, one can consider the $s$-channel scattering amplitude. If one considers an intermediate state with spin $J$, the $s$-channel amplitude is

$$A_J(s,t)=-\frac{(g'_J)^2(-t)^J}{s-M_J^2}\,.$$

When intermediate states of all $J$ values are included, the total amplitude is

$$A(s,t)=-\sum_J\frac{g_J^2(-t)^J}{s-M_J^2}\,. \tag{4}$$ ⟧⊕

⊕⟦ Supplement 8.2 and 8.2

We would like to expand the Veneziano amplitude (8.1) as a sum of $t$-channel amplitudes involving the poles of the exchange particles. We first note that the gamma function $\Gamma(-\alpha)$ obeys the relation

$$\Gamma(-\alpha)=\frac{\Gamma(-\alpha+1)}{-\alpha}=\frac{\Gamma(-\alpha+2)}{-\alpha(-\alpha+1)}=\frac{\Gamma(-\alpha+J+1)}{-\alpha(-\alpha+1)\ldots(-\alpha+J)}\,.$$

In the neighborhood of $-\alpha\simeq-J$ where $J$ is an integer, the gamma function can be represented by

$$\begin{aligned}\Gamma(-\alpha)&=\frac{\Gamma(1)}{(-J)(-J+1)\ldots(-\alpha+J)}\\&=\frac{(-1)^J}{J!\,(-\alpha+J)}\,.\end{aligned} \tag{1}$$

Thus, $\Gamma(-\alpha)$ has poles at $-\alpha=0,-1,-2,\ldots,-J,\ldots$.

We consider now the Veneziano amplitude (8.1)

$$A(s,t)=g^2\frac{\Gamma(-\alpha(s))\Gamma(-\alpha(t))}{\Gamma(-\alpha(s)-\alpha(t))}\,.$$

For fixed $s$ and when $-\alpha(t)$ approaches $-J$, then according to Eq. (1)

$$\begin{aligned}A(s,t)&=g^2\frac{\Gamma(-\alpha(s))}{\Gamma(-\alpha(s)-J)}\frac{(-1)^J}{J!\,[-\alpha(t)+J]}\\&=g^2[-\alpha(s)-1][-\alpha(s)-2]\ldots[-\alpha(s)-J]\frac{(-1)^J}{J!\,[-\alpha(t)+J]}\\&=g^2\frac{[\alpha(s)+1][\alpha(s)+2]\ldots[\alpha(s)+J]}{J!\,[-\alpha(t)+J]}\,.\end{aligned}$$

If we include all the singularities of $-\alpha(t)=0,-1,-2,\ldots-J,\ldots$, we have

$$A(s,t)=\sum_{J=0}^{\infty}g^2\frac{[\alpha(s)+1][\alpha(s)+2]\ldots[\alpha(s)+J]}{J!\,[-\alpha(t)+J]}\,.$$

When the function $\alpha$ is taken to be a linear form,

$$\alpha(x) = \alpha_0 + \alpha' x\,,$$

then the Veneziano amplitude for large $s$ is

$$A(s,t) = \sum_{J=0}^{\infty} g^2 \frac{(\alpha' s)^J}{J!\,[-\alpha_0 - \alpha' t + J]}\,,$$

which can be written as

$$A(s,t) = -\sum_{J=0}^{\infty} \frac{g^2(\alpha')^{J-1}}{J!} \frac{s^J}{t - \frac{J-\alpha_0}{\alpha'}}\,. \tag{2}$$

We can now compare this amplitude with the amplitude (3) in Supplement 8.1 for the exchange of a set of particles with different spins $J$. We see that the amplitude $A(s,t)$ in Eq. (2) is a sum of amplitudes for the exchange of particles with spin $J = 0, 1, 2, ..., \infty$ and mass $M_J$ given by

$$M_J^2 = \frac{J - \alpha_0}{\alpha'}\,. \tag{3}$$

The amplitude (8.1) then represents the sum of amplitudes for the exchange of the set of particles whose spin $J$ and mass $M_J$ are related to each other by

$$J = \alpha_0 + \alpha' M_J^2.$$

starts here again:

The Veneziano amplitude $A(s,t)$ is symmetric with respect to the interchange of $s$ and $t$. A complete crossing symmetry requires the amplitude to be invariant under the interchange of $s$, $t$, and $u$, where $u = (p_1 - p_4)^2$. We would like to use a scattering amplitude that has such a complete *crossing symmetry* which is given by

$$A_4 = A(s,t) + A(t,u) + A(u,s). \tag{8.4}$$

It is easy to show that for large values of $s$ and a fixed value of $t$, the crossing-symmetric amplitude $A_4$ behaves as

$$A_4 = -g^2\pi e^{-\alpha(t)} \frac{(1 + e^{-i\pi\alpha(t)})}{\Gamma(1+\alpha(t))} \frac{(\alpha' s)^{\alpha(t)}}{\sin\pi\alpha(t)} \tag{8.5}$$

(see Supplement 8.3).

It is instructive to compare the behavior of the above scattering amplitude with the behavior of the scattering amplitude for the exchange of a particle with spin $J$ as given by Eq. (2) of Supplement



8.1. One notes from the power of $s$ that the scattering amplitude $A_4$ behaves as if the hadrons had exchanged a *single* particle with a spin $J$, which needs not be an integer, and the spin $J$ is related to $t$ by

$$J = \alpha(t) = \alpha_0 + \alpha' t\,. \tag{8.6}$$

It is thus interesting that an *infinite* sum of the amplitudes for the exchange of a set of particles lying on a Regge trajectory can be effectively described as the exchange of a *single* fictitious particle with an effective spin $J = \alpha(t)$.

The optical theorem states that the imaginary part of the scattering amplitude at zero degree is related to the total cross section by [4]

$$\mathcal{I}m\ A_4\big|_{t=0} = \lambda(s, m^2, m^2)\sigma_{\rm tot}, \tag{8.7}$$

where $\lambda(s, a^2, b^2)$ is given by Eq. (4.5c). From Eq. (8.5), the imaginary part of $A_4$ is

$$\mathcal{I}m\ A_4 = g^2\pi e^{-\alpha(t)} \frac{(\alpha' s)^{\alpha(t)}}{\Gamma(1+\alpha(t))}\,. \tag{8.8}$$

From Eqs. (8.7) and (8.8), the total cross section for a hadron-hadron collision in the dual resonance model then depends on $s$ according to

$$\sigma_{\rm tot} \propto s^{\alpha_0 - 1}\,. \tag{8.9}$$

Eq 8.9 is a key signature of the Reggeon/Pomeron behavior

Supplement 8.3

We would like to study the behavior of the crossing-symmetric amplitude $A_4$ as defined by Eq. (8.4), for large values of $s$ and a fixed value of $t$. We first examine the amplitude $A(s,t)$, which is the first term in Eq. (8.4). Using the formula [5]

Skip Supplement 8.3

$$\Gamma(-\alpha) = \frac{-\pi}{\Gamma(1+\alpha)\sin\pi\alpha}\,, \tag{1}$$

we can write

$$\frac{\Gamma(-\alpha(s))}{\Gamma(-\alpha(s) - \alpha(t))} = \left[\frac{\Gamma(1+\alpha(s)+\alpha(t))}{\Gamma(1+\alpha(s))}\right]\left[\frac{\sin\pi(\alpha(s)+\alpha(t))}{\sin\pi\alpha(s)}\right]. \tag{2}$$

Upon using Eq. (1) and Eq. (2), the scattering amplitude (8.1) becomes

$$A(s,t) = g^2 \frac{-\pi}{\Gamma(1+\alpha(t))\sin\pi\alpha(t)}\left[\frac{\Gamma(1+\alpha(s)+\alpha(t))}{\Gamma(1+\alpha(s))}\right]\left[\frac{\sin\pi(\alpha(s)+\alpha(t))}{\sin\pi\alpha(s)}\right]. \tag{3}$$

We shall use Stirling's formula [5]

$$\Gamma(x) \simeq \sqrt{2\pi}x^{x+\frac{1}{2}}e^{-x},$$

which gives an approximate representation of $\Gamma(x)$ for large values of $|x|$ away from the negative $x$-axis. Applying Stirling's formula to the first factor in (2), we obtain

$$\frac{\Gamma(1+\alpha(s)+\alpha(t))}{\Gamma(1+\alpha(s))} \simeq \frac{[1+\alpha(s)+\alpha(t)]^{1+\alpha(s)+\alpha(t)+\frac{1}{2}} e^{-1-\alpha(s)-\alpha(t)}}{[1+\alpha(s)]^{1+\alpha(s)+\frac{1}{2}} e^{-1-\alpha(s)}}$$
$$\simeq [\alpha(s)]^{\alpha(t)} e^{-\alpha(t)} . \tag{4}$$

The second factor in Eq. (2) is

$$\frac{\sin\pi(\alpha(s)+\alpha(t))}{\sin\pi\alpha(s)} = \cos\pi\alpha(t) + \cot\pi\alpha(s)\sin\pi\alpha(t) . \tag{5}$$

To simplify this factor, we note that when $s$ is large, this factor is a rapidly varying function of $s$. From Eq. (3), we see that $A(s,t)$ passes through many poles whenever $\alpha(s)$ is an integer and it passes through zeros whenever $\alpha(s)+\alpha(t)$ is an integer. One can average over these poles and zeros by giving a positive imaginary part to $s$, corresponding to providing a width to the resonances. One expects that as $s$ increases, the width of the resonances also increases. So, this imaginary part of $s$ is large compared to the separation between the poles (although it is still small compared to the real part of $s$). By giving an imaginary part to $s$, the function $\alpha(s)$, which is linear in $s$ with real coefficients, will have a real part $\alpha_R(s)$ and an imaginary part $\alpha_I(s)$. The large widths of the resonances correspond to the situation $\pi\alpha_I(s) \gg 1$. We can evaluate the limit $\pi\alpha_I(s) \gg 1$,

$$\lim_{\pi\alpha_I(s)\gg 1} \cot\pi\alpha(s) = i \lim_{\pi\alpha_I(s)\gg 1} \frac{e^{i\pi(\alpha_R(s)+i\alpha_I(s))} + e^{-i\pi(\alpha_R(s)+i\alpha_I(s))}}{e^{i\pi(\alpha_R(s)+i\alpha_I(s))} - e^{-i\pi(\alpha_R(s)+i\alpha_I(s))}}$$
$$= -i .$$

We then have

$$\lim_{\pi\alpha_I(s)\gg 1} \frac{\sin\pi(\alpha(s)+\alpha(t))}{\sin\pi\alpha(s)} = e^{-i\pi\alpha(t)} . \tag{6}$$

Eqs. (2), (4), (5) and (6) lead to

$$\frac{\Gamma(-\alpha(s))}{\Gamma(-\alpha(s)-\alpha(t))} = (\alpha_0+\alpha' s)^{\alpha(t)} e^{-\alpha(t)} e^{-i\pi\alpha(t)} .$$

Using the above equation and Eq. (1), we find from Eq. (8.1) the amplitude $A(s,t)$ for large $s$ and fixed $t$ as given by

$$A(s,t) \simeq -g^2 \frac{\pi(\alpha' s)^{\alpha(t)} e^{-\alpha(t)} e^{-i\pi\alpha(t)}}{\Gamma(1+\alpha(t))\sin\pi\alpha(t)} . \tag{7}$$

We consider next the amplitude $A(t,u)$, which is the second term in Eq. (8.4),

$$A(t,u) = g^2 \frac{\Gamma(-\alpha(t))\Gamma(-\alpha(u))}{\Gamma(-\alpha(t)-\alpha(u))} .$$



We note that $\alpha(s), \alpha(t)$, and $\alpha(u)$ are related by

$$\alpha(s)+\alpha(t)+\alpha(u) = 3\alpha_0 + \alpha'(s+t+u)$$
$$= 3\alpha_0 + \alpha'(4m^2) ,$$

where $m$ is the rest mass of the colliding (equal) hadrons. Using this relation, we can express $\alpha(u)$ in terms of $\alpha(s)$ and $\alpha(t)$:

$$\alpha(u) = -\alpha(s) - \alpha(t) + 3\alpha_0 + 4\alpha' m^2 ,$$

and

$$\alpha(t)+\alpha(u) = -\alpha(s) + 3\alpha_0 + 4\alpha' m^2 .$$

The amplitude $A(t,u)$ then becomes

$$A(t,u) = g^2 \frac{\Gamma(\alpha(s)+\alpha(t)-3\alpha_0-4\alpha' m^2)\Gamma(-\alpha(t))}{\Gamma(\alpha(s)-3\alpha_0-4\alpha' m^2)} .$$

Applying Stirling's formula and using Eq. (1), we find

$$A(t,u) = -g^2\pi e^{-\alpha(t)} \frac{1}{\Gamma(1+\alpha(t))} \frac{(\alpha' s)^{\alpha(t)}}{\sin\pi\alpha(t)} . \tag{8}$$

We consider finally the amplitude $A(u,s)$, which is the last term in Eq. (8.4). Writing $\alpha(u)$ again in terms of $\alpha(s)$ and $\alpha(t)$, we have

$$A(u,s) = g^2 \frac{\Gamma(-\alpha(u))\Gamma(-\alpha(s))}{\Gamma(-\alpha(u)-\alpha(s))}$$
$$= -g^2 \frac{\Gamma(\alpha(s)+\alpha(t)-3\alpha_0-4\alpha' m^2)\pi}{\Gamma(\alpha(t)-3\alpha_0-4\alpha' m^2)\Gamma(1+\alpha(s))\sin\pi\alpha(s)}$$
$$\simeq -g^2 \frac{(\alpha(s))^{\alpha(t)-1-3\alpha_0-4\alpha' m^2}}{\Gamma(\alpha(t)-3\alpha_0-4\alpha' m^2)} \frac{2\pi i}{e^{i\pi(\alpha_R(s)+i\alpha_I(s))} - e^{-i\pi(\alpha_R(s)+i\alpha_I(s))}} .$$

This amplitude $A(u,s)$ vanishes when $\pi\alpha_I(s) \gg 1$.

Summing up all the terms, the crossing-symmetric amplitude $A_4$ for large $s$ is given by

$$A_4 = -g^2\pi e^{-\alpha(t)} \frac{(1+e^{-i\pi\alpha(t)})}{\Gamma(1+\alpha(t))} \frac{(\alpha' s)^{\alpha(t)}}{\sin\pi\alpha(t)} ,$$

which is Eq. (8.5). ]⊕

starts here again

One can use the scattering amplitude (8.5) to analyze experimental scattering data for elastic $\pi N$ reactions for large $s$ and fixed $t$. In principle, there can be many contributions coming from the exchange of many different Regge trajectories. Among the trajectories, the *leading trajectory* is the one which gives the dominant contribution.

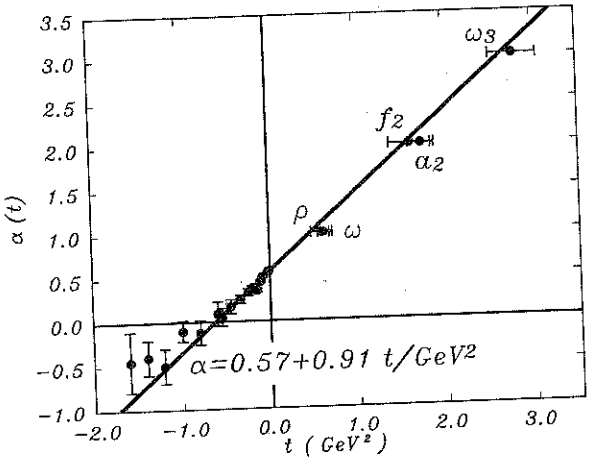



**Fig. 8.2** The function $\alpha(t)$ from elastic $\pi N$ data (for negative $t$) and from resonance masses for positive $t$, from Ref. [6].

to the scattering amplitude. From Eq. (8.5), one observes that the trajectory with the largest value of $\mathcal{R}e\ \alpha(t)$ will be dominant asymptotically; the leading trajectory is the one with the largest value of $\mathcal{R}e\ \alpha(t)$.

From the experimental data, one can extract the quantity $\alpha(t)$ for the leading trajectory for a fixed value of $t$, which is negative for elastic $\pi N$ reactions. We show in Fig. 8.2 the data points from such an analysis. We also show $\alpha(t)$ as a function of $t$ for positive values of $t$, which gives the relation between the mass of a resonance and its spin $J = \alpha$. It is remarkable that $\alpha(t)$ extracted from elastic $\pi N$ data joins on smoothly to $\alpha(t)$ as determined from the masses of resonances for positive values of $t$. Thus, the leading Regge trajectory from the elastic data coincides with the $\rho - \omega$ meson trajectory from meson spectra. The intercept of the leading Regge trajectory is $\alpha_R(0) \sim 0.5$ and the slope parameter $\alpha'_R \sim 1\ \mathrm{GeV}^{-2}$, where the subscript $R$ indicates that these quantities are for the leading Regge trajectory. The Regge slope for mesons is, within the experimental uncertainty, the same as the Regge slope for the baryons, as seen by comparison of Figs. 8.2 and 7.2. These comparisons indicate the approximate validity of the dual resonance model as a description of some aspects of hadron physics.



## §8.2 Pomeron Exchange

The concept of a *Pomeron* arises from the consideration of many special features of hadronic reactions at very high energies [7]. It was found experimentally that for hadron-hadron reactions at high energies

(i) inelastic processes with no quantum number flow appear to dominate the reactions in the fragmentation regions,

(ii) the forward scattering amplitudes are approximately purely imaginary,

and (iii) the total cross section approaches nearly a constant value at very high energies.

Property (i) suggests that the exchange of an object with the quantum numbers of the vacuum can be a useful concept for forward reactions. Features (ii) and (iii) can be understood if the exchanged object is a Regge trajectory having a Regge intercept $\alpha_0 = 1$ [7].

It is easy to show that the exchange of such a Regge trajectory will lead to the characteristics (ii) and (iii). From the scattering amplitude $A_4$ in (8.5), the real part of $A_4$ is given by

$$\mathcal{R}e\ A_4 = -g^2\pi e^{-\alpha(t)}\frac{(1+\cos\pi\alpha(t))}{\Gamma(1+\alpha(t))}\frac{(\alpha' s)^{\alpha(t)}}{\sin\pi\alpha(t)}\,. \tag{8.10}$$

Forward scattering is characterized by $t \to 0$. Therefore, if the Regge intercept $\alpha_0$ is equal to unity, then

$$\mathcal{R}e\ A_4\big|_{t\to 0} = -\lim_{\alpha(t)\to 1} g^2\pi e^{-\alpha(t)}\frac{(1+\cos\pi\alpha(t))}{\Gamma(1+\alpha(t))}\frac{(\alpha' s)^{\alpha(t)}}{\sin\pi\alpha(t)} \to 0\,.$$

On the other hand, the imaginary part of $A_4$, as given by Eq. (8.8) is

$$\mathcal{I}m\ A_4\big|_{t\to 0} = g^2\pi e^{-1}\alpha' s\,,$$

which is a nonzero quantity. The scattering amplitude for the exchange of a Regge trajectory with an intercept $\alpha_0 = 1$ is approximately a purely imaginary quantity, in agreement with property (ii).

From Eq. (8.9), the total cross section is proportional to $s^{\alpha_0 - 1}$. The exchange of a Regge trajectory with $\alpha_0 = 1$ gives a total cross section that is independent of energy, in agreement with property (iii).

this whole last page is quite important. but that's good for now...we will now skip a few pages....

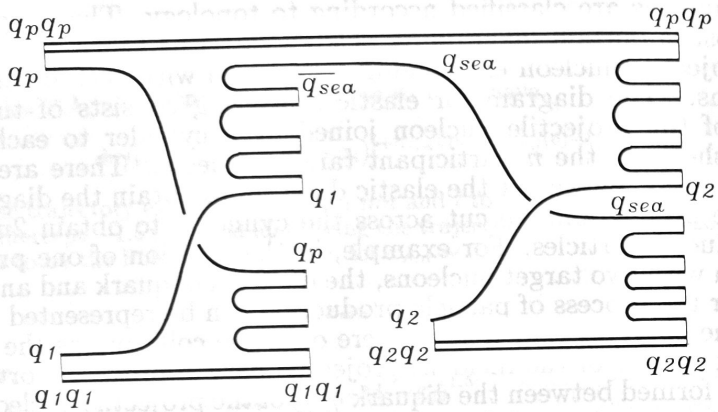



**Fig. 8.6** The diagram for the collision of one projectile nucleon with two target nucleons in the dual-parton model.

start here again... but you only need to skim pp 150-151

particles can be obtained by convolution. The dual-parton model has been applied successfully to analyze a large body of experimental data in hadron-nucleus and nucleus-nucleus collisions [13-16].

It is worth mentioning some other variations and modifications of the dual-parton model and concepts. In the multistring model of Werner (with the computer code VENUS)[17], the dominant diagram for a nucleon-nucleon collision includes the same diagram as in Fig. 8.5b and 8.6. In addition, it also includes the probability for finding an antiquark in a nucleon and the collision of this antiquark with the other nucleon. For nucleus-nucleus collisions, there are thus additional contributions from the collision of the antiquark with nucleons of the other nucleus. In contrast to the dual-parton model, for which the momentum distribution of the partons are inferred from the dual resonance model, the momentum distribution function of quarks and antiquarks in the multistring model of Werner are taken to be the same as those determined by deep-inelastic collision experiments, represented in terms of the functions and parameters as given by Duke and Owens [18]. The momentum distribution of the diquark is then determined by momentum conservation. The partons form strings by interacting with the partons of other nucleons. The fragmentation of these strings is described, as in the dual-parton model, by using fragmentation functions which are a generalization of the Field-Feynman

this section is only meant to show you a few pages later that the DPM results can be related directly to the fragmentation functions D(z)



distribution functions [19]. For example, in reference [17], for the fragmentation of a pion and a baryon from a quark, the fragmentation functions are

$$xD_{\pi/q}(x) = (1-x)\,,$$

$$xD_{b/q}(x) = x^{\alpha}(1-x)^3\,.$$

For the fragmentation of a pion and a baryon from a diquark, the fragmentation functions are

$$xD_{\pi/q}(x) = (1-x)^3\,,$$

$$xD_{b/q}(x) = x^{\alpha}(1-x)\,,$$

where $\alpha$ is a parameter. A later version of the model (VENUS 3) [20] uses a different fragmentation scheme, which is similar to the string fragmentation scheme of the Lund model (Chapter 7), treating the quark and the diquark pair in a string as the two ends of a relativistic string. The probability for the breaking of the string at the space-time point $x_+, x_-$ is given by the probability function

$$dP \propto e^{-\text{const}\times x_+ x_-}\,,$$

which was first suggested by Artru and Mennessier [21]. The model has been applied successfully to analyze a large set of experimental data in nucleus-nucleus collisions [17,20].

A different multiple-collision model, the multi-chain model, was proposed by Kinoshita, Minaka, and Sumiyoshi [22]. In this model for a baryon-nucleus collision, incident baryons make collisions with many nucleons in the target nucleus. Each collision leads to the formation of a chain which later evolves into produced particles. The leading cluster degrades its momentum as the collision process proceeds. How the leading cluster loses energy is taken to be an unknown parametrized function $P(x)$ to be determined by experiment. A convenient parametrization of the probability function $P(x)$ is taken to be

$$P(x) = \alpha x^{\alpha-1}\,,$$

where $x$ is the light-cone variable of the leading cluster. Other methods of partitioning the light-cone momentum have also been attempted [22]. The momentum degradation of the leading cluster is assumed to be sequential so that the momentum distribution after $n$ collisions is the folding of the momentum distribution after $n-1$ collisions and the probability function $P(x)$. After each collision, each chain also acquires a momentum fraction, and it decays into

obviously see the book for more details.

# Lastly the (MIT) Bag model of Hadrons

- Now one very well known model of the non-perturbative (NOT as just valence quarks) description of the hadronic bound state structure.

- I include some slides from this talk…
  - https://slideplayer.com/slide/15362831/

- …before Wong's short section on it.

- Wong's treatment is focused on understanding the formation of the QGP but it has other applications.

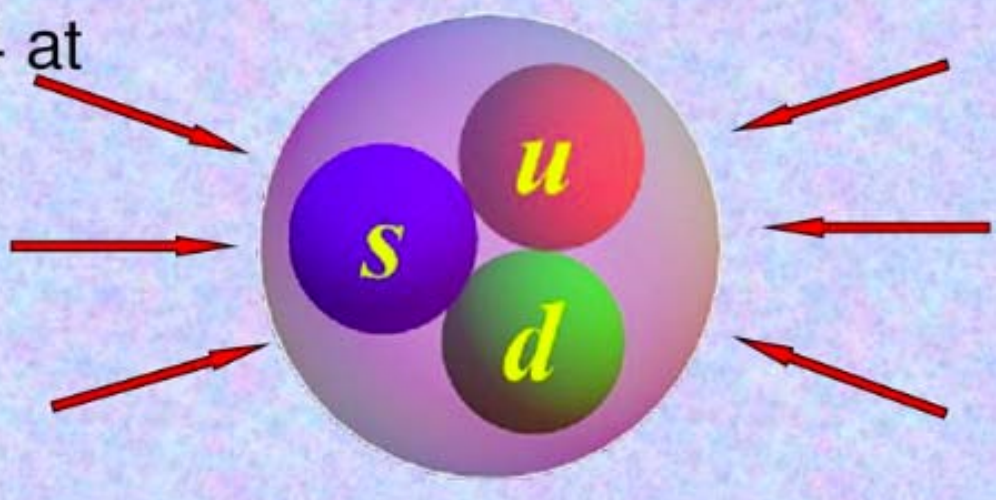
# The Color Charge & Bag Model

## Outline

- Color in Quantum Chromodynamics (QCD)
  - Color Charges
  - Quark confinement
- M.I.T. Bag Model
  - Assumptions
  - Bag Model of Quark Confinement

# M.I.T. Bag Model

- Developed in 1974 at Massachusetts Institute of Technology (MIT)
- It models spatial confinement only




- Quarks are forced by a fixed external pressure to move only inside a given spatial region
- Quarks occupy single particle orbitals
- The shape of the bag is spherical, if all the quarks are in ground state

# M.I.T Bag Model

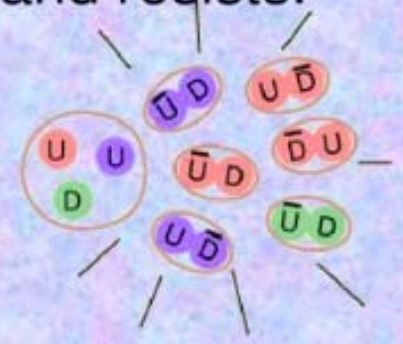
- *Inside* the bag, quarks are allowed to move quasi-free.
- An appropriate boundary condition at the *bag surface* guarantees that no quark can leave the bag
- This implies that there are no quarks outside the bag

# Bag Model of Quark Confinement

In dealing with the nature of quark confinement, one visualization is that of an elastic bag which allows the quark to move freely around, as long as you don't try to pull them further apart. But if you try to pull a quark out, the bag stretches and resists.

The quarks of a proton are free to move within the proton volume

If you try to pull one of the quarks out, the energy required is on the order of 1 GeV per fermi, like stretching an elastic bag.




The energy required to produce a separation far exceeds the pair production energy of a quark-antiquark pair, so instead of pulling out an isolated quark, you produce mesons as the produced quark-antiquark pairs combine.

# M.I.T. Bag Model

A potential function which has been successfully used to describe some quark systems is of the form:

$$V = \frac{-k_1}{r} + k_2 r$$

$k_1$ = strength of coulomb attraction of the quarks.

$k_2$ = strength of the color force attraction, about 1 GeV/fm.

- The boundary condition generates discrete energy eigenvalues.

$$\varepsilon_n = \frac{x_n}{R}$$

R - radius of the Bag

$x_1$=2.04

$$E_{kin}(R) = N_q \frac{x_n}{R}$$

$N_q$ = # of quarks inside the bag

$$E_{pot}(R) = \frac{4}{3}\pi R^3 B$$

B – bag constant that reflects the bag pressure

# M.I.T. Bag Model

- Minimizing E(R), one gets the equilibrium radius of the system

$$R_n = \left( \frac{N_q x_n}{4\pi B} \right)^{1/4}$$

$$E_n = \frac{4}{3} \left( 4\pi B N_q^3 x_n^3 \right)^{1/4}$$

Fixing the only parameter of the model B, by fitting the mass of the nucleon to 938MeV we have *first order* predictions

quark-antiquark bound states, and baryons can be considered as three-quark bound states [8]. Up to now, it is found that all the hadron states which can be observed in isolation are *color singlet* states which are completely antisymmetric with respect to the exchange of any two quarks of the hadron. Experimentally, no single quark, which is described by a color triplet state, has ever been isolated. Therefore, it is held that only hadrons in the color singlet state can be isolated and observed. The absence of the observation of a single quark in isolation also suggests that the interaction between quarks and gluons must be strong on large distance scales.

In the other extreme, much insight into the nature of the interaction of quarks and gluons on short distance scales was provided by deep-inelastic scattering experiments. In these experiments, an incident electron interacts with a quark within a hadron and is accompanied by a transfer of momentum from the electron to the quark. The measurement of the momentum of the electron before and after the collision allows a probe of the momentum distribution of the quarks (or partons) inside the nucleon. It was found that with very large momentum transfers, the quarks inside the hadron behave as if they were almost free, as demonstrated by the success of Bjorken scaling [9] and the parton model [10].

*A non-Abelian gauge theory* can describe a system which is weakly interacting on short distance scales but very strongly interacting on a large distance scale [11]. The non-Abelian gauge theory is essentially unique in this respect [12]. In a non-Abelian gauge field theory, the gauge field operators do not commute. This is in contrast to the Abelian gauge field theories for electromagnetic interactions, in which they do commute. It was found that non-Abelian gauge field theories possess the property of '*asymptotic freedom*' [11]. The effective strength of the interaction between quarks and gluons depends on the conditions of their interaction. The QCD coupling constant $\alpha$ is related to the scale of the momentum transfer $q$ by the relation [11]

$$\alpha(q^2) = \frac{\alpha_0}{1 + \alpha_0 \frac{(33-2n_f)}{12\pi} \ln(\frac{-q^2}{\mu^2})}, \tag{9.1}$$

where $\alpha_0$ is the coupling constant for the momentum transfer $\mu$ and $n_f$ is the number of flavors. When the distance scale of the interaction is small, as for example when one probes the high momentum component of the distribution of the quarks, the coupling constant is small. This is the case of 'asymptotic freedom'. For that case, since the coupling constant is small, a perturbative treatment is a good description of the process and the parton model [8-10] is a useful concept. Much knowledge of the interaction of quarks and gluons has been derived from such studies. On the other hand, when the distance scale is large, as for example in the study of the structure of the ground state of a hadron, then the interaction strength is large. A perturbative treatment, based on an expansion in powers of the coupling constant, is no longer applicable. The constituent quarks are subject to confining forces and a nonperturbative treatment is needed.



start here

## §9.2 Bag Model of a Hadron

A very useful method to study nonperturbative quark-gluon systems is to examine the system on a discrete lattice of space and time. The formulation of QCD on a lattice is known as the QCD lattice gauge theory which we shall discuss in the next chapter. Theoretical results from lattice gauge theory indicate that when the distance scale is comparable to the size of a hadron, quarks interact with an effective interaction which goes approximately linearly with the spatial distance. Experimentally, no single quark has been isolated. This leads to the concept that the large scale behavior of quarks in hadrons is characterized by their confinement inside the hadrons. An intuitive way to visualize confinement is the picture of a linear tube between a quark and an antiquark (or a diquark) in which the color electric flux is restricted (see Chapters 5 and 7). As the separation between the quark and the antiquark becomes large, it becomes energetically more favorable to produce another quark-antiquark pair at a point along the tube such that the produced quark is connected to the antiquark while the produced antiquark is connected to the quark. To isolate a quark by separating it from its antiquark partner would then be an impossible task. The confinement of quarks when one considers the large distance behavior of QCD is sometimes referred to as the '*infrared slavery*' of quarks. The term 'infrared' comes from analogy to optics where the wavelengths in the infrared region are greater than those in the optical spectrum. It is used here to denote the circumstances involving a large spatial distance, or small momentum transfer. Likewise, it is customary to use the term 'ultraviolet' to denote the circumstances involving a small distance scale, or a large momentum transfer.

With quarks being confined inside a hadron, a useful phenomenological description of quarks in hadrons is provided by the *bag model* [13,14]. While there are many different versions of the model, the MIT bag model [13] contains the essential characteristics of the phenomenology of quark confinement; we shall use it to understand the

circumstances of how quarks can become deconfined in new phases of quark matter.

In the MIT bag model, quarks are treated as massless particles inside a bag of finite dimension, and are infinitely massive outside the bag. Confinement in the model is the result of the balance of the *bag pressure* $B$, which is directed inward, and the stress arising from the kinetic energy of the quarks. Here, the bag pressure $B$ is a phenomenological quantity introduced to take into account the nonperturbative effects of QCD. If the quarks are confined in the bag, the gluons should also be confined in the bag. In this description, the total color charge of the matter inside the bag must be colorless, by virtue of the Gauss's Law. As there are three different types of color, the bag model would imply that the allowable hadronic bags should include colorless $qqq$ and $q\bar{q}$ states.

⊕[ Supplement 9.1

We can get an estimate of the magnitude of the bag pressure by considering massless free fermions in a spherical cavity of radius $R$. The Dirac equation for a massless fermion in the cavity is

$$\gamma \cdot p\psi = 0. \tag{1}$$

We use the Dirac representation of the $\gamma$ matrices,

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix},$$

and

$$\boldsymbol{\gamma} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix},$$

where $I$ is a 2×2 unit matrix and $\boldsymbol{\sigma}$ are the Pauli matrices. We write the four-component wave function for the massless fermion $\psi$ as

$$\psi = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix},$$

where $\psi_+$ and $\psi_-$ are two dimensional spinors. Equation (1) becomes

$$\begin{pmatrix} p^0 & -\boldsymbol{\sigma}\cdot\mathbf{p} \\ +\boldsymbol{\sigma}\cdot\mathbf{p} & -p^0 \end{pmatrix} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = 0. \tag{2}$$

Writing the above matrix equation into two equations and eliminating the $\psi_-$ component, we obtain

$$[\mathbf{p}^2 - (p^0)^2]\psi_+ = 0.$$

The lowest energy solution for the above equation is the $s_{1/2}$ state given by

$$\psi_+(\mathbf{r},t) = \mathcal{N} e^{-ip^0 t} j_0(p^0 r)\chi_+,$$



where $j_0$ is the spherical Bessel function of order zero, $\chi_+$ is a two dimensional spinor, and $\mathcal{N}$ is a normalization constant. From Eq. (2), the other component is

$$\psi_-(\mathbf{r},t) = \mathcal{N} e^{-ip^0 t}\, \boldsymbol{\sigma}\cdot\hat{\mathbf{r}}\, j_1(p^0 r)\, \chi_-.$$

The confinement of the quarks is equivalent to the requirement that the normal component of the vector current $J_\mu = \bar{\psi}\gamma_\mu\psi$ vanishes at the surface. This condition is the same as the requirement that the scalar density $\bar{\psi}\psi$ of the quark vanishes at the bag surface $r = R$ (see p. 412 of Ref. [8]). This leads to

$$\bar{\psi}\psi\Big|_{r=R} = [j_0(p^0R)]^2 - \boldsymbol{\sigma}\cdot\hat{\mathbf{r}}\ \ \boldsymbol{\sigma}\cdot\hat{\mathbf{r}}[j_1(p^0R)]^2 = 0$$

or

$$[j_0(p^0R)]^2 - [j_1(p^0R)]^2 = 0.$$

From the tabulated values of the spherical Bessel functions [15], this equation is satisfied for

$$p^0R = 2.04, \quad \text{or} \quad p^0 = \frac{2.04}{R}.$$

For a system of $N$ quarks in a bag, the total kinetic energy of the confined quarks is inversely proportional to $R$. It decreases with an increase in the bag radius. In the bag model, the nonperturbative effects of confinement are represented phenomenologically by the presence of a bag pressure directed from the region outside the bag toward the region inside the bag. The energy density of the 'vacuum' inside the bag is higher than that outside the bag, the difference being the bag pressure $B$. The energy of a system of $N$ confined quarks in a bag of radius $R$ is

$$E = \frac{2.04N}{R} + \frac{4\pi}{3}R^3B.$$

We observe that the tendency to increase the radius due to the kinetic energy of the quarks is counterbalanced by this inward pressure $B$ directed from the region outside the bag towards the region inside the bag. The equilibrium radius of the system is located at the radius $R$ determined by $dE/dR = 0$, which leads to a bag pressure constant $B$ related to the radius by

$$B^{1/4} = \left(\frac{2.04N}{4\pi}\right)^{1/4}\frac{1}{R}.$$

If we take the confinement radius to be 0.8 fm for a 3 quark system in a baryon, we obtain an estimate of the bag pressure constant

$$B^{1/4} = 206 \text{ MeV}.$$

(We have chosen to use the units $\hbar = c = 1$. To include $\hbar$ and $c$ explicitly, the bag pressure constant should be $B^{1/4} = 206\text{MeV}/(\hbar c)^{3/4}$.) The value of the bag pressure $B^{1/4}$ is in the range between 145 MeV (Ref. 16) and 235 MeV (Ref. 17). ]⊕

## §9.3 Quark-Gluon Plasma

We have outlined the essence of the bag model in the last section. More refined descriptions taking into account chiral symmetry and perturbative gluon interactions have been quite successful in describing many experimental data, such as hadron masses and magnetic moments [18]. For our purposes, it is useful to use it as a heuristic model to discuss matter under extreme conditions.

We can interpret the bag model as indicating that the essential effect of nonperturbative QCD is to give rise to an inward bag pressure of magnitude $B$. For a hadron in which the quarks can be considered to be confined to be in the lowest $s_{1/2}$ state inside the bag, the inward bag pressure is balanced by the quantum stress [19] arising from the wave function of the quarks. This balance of opposing pressures allows a simple intuitive description of the competing forces leading to a stable system. It also provides a simple intuitive understanding of why new phases of quark matter are expected.

It is clear that if the pressure of the quark matter inside the bag is increased, there will be a point when the pressure directing outward is greater than the inward bag pressure. When that happens, the bag pressure cannot balance the outward quark matter pressure and the bag cannot confine the quark matter contained inside. A new phase of matter containing the quarks and gluons in an unconfined state is then possible. It is this situation which leads to the possible existence of different phases of the quark matter. The main condition for a new phase of quark matter is the occurrence of a large pressure exceeding the bag pressure $B$.

A large pressure of quark matter arises 1) when the temperature of the matter is high, and/or 2) when the baryon number density is large. New phases of quark matter are then expected [20]. We shall discuss these two extreme cases separately.

### §9.3.1 Quark-Gluon Plasma at High Temperatures

We consider the case of a quark-gluon system in thermal equilibrium at a high temperature $T$, within a volume $V$. For simplicity, we examine the case where quarks and gluons are idealized to be noninteracting and massless and there is no net baryon number. The number of quarks and the number of antiquarks in the system are equal.

The partial pressures arising from the quarks, antiquarks, and the gluons can be obtained from standard textbooks [21]. For completeness, we shall describe the steps in Supplement 9.2 below.



Adding the contributions from quarks, antiquarks and gluons together (see Eqs. (3a), (3b), and (8) in Supplement 9.2), we have the total pressure of an ideal quark-gluon plasma given by

$$P = g_{\text{total}} \frac{\pi^2}{90} T^4, \tag{9.2}$$

where

$$g_{\text{total}} = \left[ g_g + \frac{7}{8} \times (g_q + g_{\bar{q}}) \right], \tag{9.3}$$

where $g_g, g_q$, and $q_{\bar{q}}$ are respectively the degeneracy number of the gluons, the quarks and the antiquarks. If the quarks and gluons are confined in a finite volume, the total pressure should include additional contributions from the quantum kinetic energies of the particles, which is inversely proportional to the radius of the confined volume. We are interested in the case when the quark matter may be deconfined in a large volume, and we can neglect this contribution.

To evaluate the degeneracy numbers, we note that there are 8 gluons, each having two possible polarizations. Therefore, we have

$$g_g = 8 \times 2 \,. \tag{9.4}$$

The degeneracy number $g_q$ of the quarks depends on the number of flavors under consideration. In order to find the critical temperature for a transition into a quark-gluon plasma with two flavors, we shall take the flavor degeneracy to be two. The degeneracy numbers $g_q$ and $g_{\bar{q}}$ are

$$g_q = g_{\bar{q}} = N_c \, N_s \, N_f \,. \tag{9.5}$$

where $N_c(= 3)$ is the number of colors, $N_s(= 2)$ is the number of spins, and $N_f(= 2 \text{ or } 3)$ is the number of flavors. From Eq. (9.3), the total number of degrees of freedom $g_{\text{total}}$ is 37. The pressure of a quark-gluon plasma at a temperature $T$ is therefore

$$P = 37 \frac{\pi^2}{90} T^4, \tag{9.6}$$

and the energy density of the quark-gluon matter at a temperature $T$ is

$$\epsilon = 37 \frac{\pi^2}{30} T^4, \tag{9.7}$$

which gives an energy density of 2.54 GeV/fm$^3$ at a temperature of 200 MeV.

From Eq. (9.6), the *critical temperature* at which the quark-gluon pressure is equal to the bag pressure $B$ is given by

$$T_c = (\frac{90}{37\pi^2})^{1/4} B^{1/4}. \tag{9.8}$$

Thus, for $B^{1/4} = 206$ MeV as estimated in Section 9.2, we have $T_c \sim 144$ MeV. If the quark matter in a bag are heated up to a high temperature greater than the critical temperature, the quark matter inside the bag will have a pressure which is greater than that of the bag pressure. When this happens, the bag will not be able to hold the quark matter in the bag and the quark matter will be deconfined. The deconfined phase of the quark matter is given the general name '*quark-gluon plasma*'. Thus, a quark-gluon plasma may arise when the temperature of the quark matter is very high.

⊕⟦ Supplement 9.2 ignore spplement 9.2

We would like to obtain the pressure arising from a relativistic massless quark gas at temperature $T$. We first determine the energy density of the quark gas. The phase space volume of quarks in a spatial volume $V$ with momentum $p$ in the momentum interval $dp$ is $4\pi p^2 dpV$. As each state occupies a phase space volume of $(2\pi\hbar)^3$, the number of states characterized by a momentum $p$ in the interval $dp$ is $4\pi p^2 dpV/(2\pi)^3$. (In our notation, we use $\hbar = 1$.) At the specified temperature $T$, not all of the states are occupied. The occupation probability for the state with a momentum $p$ is given by the Fermi-Dirac distribution appropriate for the temperature $T$. The number of quarks in a volume $V$ with momentum $p$ within the interval $dp$ is

$$dN_q = \frac{g_q V 4\pi p^2 dp}{(2\pi)^3} \left\{ \frac{1}{1 + e^{(p-\mu_q)/T}} \right\}$$

where the factor in the curly brackets is the Fermi-Dirac distribution, $\mu_q$ is the quark Fermi energy (or chemical potential), and $g_q (= N_c N_s N_f)$ is the degeneracy of quarks.

Given the quark chemical potential $\mu_q$, we can obtain the density of the antiquarks. The presence of antiquarks corresponds to the absence of quarks in the negative energy states. The number density of antiquarks is therefore

$$\begin{aligned} n_{\bar{q}}(\mu) &= \frac{g_q}{(2\pi)^3} \int_{-\infty}^{0} 4\pi p_0^2 dp_0 \left[ 1 - \frac{1}{1 + e^{(p_0-\mu_q)/T}} \right] \\ &= \frac{g_q}{(2\pi)^3} \int_{0}^{\infty} 4\pi p_0^2 dp_0 \frac{1}{1 + e^{(p_0+\mu_q)/T}}. \end{aligned}$$

Thus, for our case when the number density of the quarks is the same as that of the antiquark, we have $\mu_q = 0$.

After setting $\mu_q = 0$, the energy of the massless quarks in the system of volume



$V$ and temperature $T$ is

$$\begin{aligned} E_q &= \frac{g_q V}{2\pi^2} \int_0^\infty \frac{p^3 dp}{1 + e^{p/T}} \\ &= \frac{g_q V}{2\pi^2} T^4 \int_0^\infty \frac{z^3 dz}{1 + e^z} \\ &= \frac{g_q V}{2\pi^2} T^4 \int_0^\infty z^3 dz e^{-z} \sum_{n=0}^{\infty} (-1)^n e^{-nz} \\ &= \frac{g_q V}{2\pi^2} T^4 \Gamma(4) \sum_{n=0}^{\infty} (-1)^n \frac{1}{(n+1)^4}, \end{aligned}$$

where $\Gamma$ is the gamma function. It is easy to show that

$$\begin{aligned} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(n+1)^4} &= \sum_{m=1,3,5..} \frac{1}{m^4} - \sum_{m=2,4,6..} \frac{1}{m^4} \\ &= \sum_{m=1,2,3,..} \frac{1}{m^4} - 2 \sum_{m=2,4,6..} \frac{1}{m^4} \\ &= \sum_{m=1,2,3,..} \frac{1}{m^4} - 2 \sum_{m=1,2,3..} \frac{1}{(2m)} \\ &= (1 - 2^{-3}) \zeta(4), \end{aligned}$$

where $\zeta(x)$ is the Riemann zeta function defined by [15]

$$\zeta(x) = \sum_{m=1,2,3,..} \frac{1}{m^x}.$$

The function $\zeta(4)$ has the value $\pi^4/90$ [15]. The energy of the syster is therefore

$$E_q = \frac{7}{8} g_q V \frac{\pi^2}{30} T^4.$$

For massless fermions and bosons, it can be shown [21] that the press to the energy density $E/V$ by

$$P = \frac{1}{3} \frac{E}{V}.$$

The pressure due to quarks is

$$P_q = \frac{7}{8} g_q \frac{\pi^2}{90} T^4.$$

Similarly, the pressure due to antiquarks is

skipped a few pages, just skim this one, ...the punch line (the plot below)



corresponding to a quark Fermi momentum $\mu_q$ with the value

$$\mu_{u,d} = 434 \text{ MeV}. \tag{9.16b}$$

These values for the quark-gluon plasma should be compared with the nucleon number density $n_B = 0.14/fm^3$ and a nucleon Fermi momentum of 251 MeV for the normal nuclear matter at equilibrium. Thus, the critical baryon density is about 5 times the normal nuclear matter density. When the density of baryons exceeds this density, the baryon bag pressure is not strong enough to withstand the pressure due to the degeneracy of quarks, and the confinement of quarks within individual baryon bags will not be possible, leading to formation of a state of deconfined quarks.

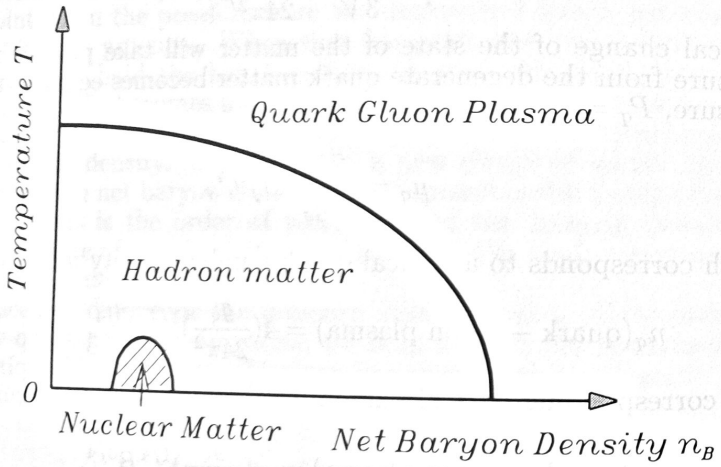



Fig. 9.1 Phase diagram in the plane of the temperature and the net baryon density. The region of nuclear matter is indicated by the shaded area.

We have discussed the possible occurrence of deconfined phases of quark matter in two extreme situations. The first situation occurs when the temperature is very high and the net baryon density (the density of baryons minus the density of the antibaryons) is zero. We estimate that the critical temperature is about 140 MeV. The second situation occurs when the temperature is zero and the baryon density is about 5 times the equilibrium nuclear matter density. For a system in between these two limits, there is a pressure arising from the thermal motion of the particles, there is also a pressure arising from



the degeneracy of the fermion gas. The total pressure is the sum of the two contributions. Thus, for a system whose temperature and net baryon density is not zero, the critical temperature at which the quark matter becomes deconfined, will lie between the limits of $T = 0$ for a degenerate quark gas, and the other limit of $T_c$ for a pure plasma with no net baryon density. The phase diagram one can construct, in the space of temperature and baryon density can be shown schematically as in Fig. 9.1. A important objective of relativistic heavy ion physics is to explore the phase diagram of quark matter in various regions of temperature and baryon density so as to confirm the existence of the new phase of quark matter.

It should be pointed out that we have used the bag model to provide a heuristic understanding of the possibility of the occurrence of a new phase of quark matter. The bag model has nevertheless limitations. There have been many predictions of the existence of exotic states [13] using the bag model. These exotic states are multiquark and multi-gluon color singlet bag states which do not occur in the simplest quark models, of $q\bar{q}$ and $qqq$. Many of these are quark-gluon systems with a large number of quarks (greater than 3). Experimental observation of such exotic multiquark systems is still rather elusive. To gain a better understanding of the possibility of a new phase of quark matter, it is necessary to examine the quark matter system within the framework of the lattice gauge theory. This will be followed in the next chapter.

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